North American Bat Monitoring Program in British Columbia

Data Summary and Activity Trend Analyses: 2016 - 2025

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1 Executive Summary

The North American Bat Monitoring (NABat) program is a multi-agency initiative administered by the US Geological Survey (USGS). In 2024, the North by Northwest (NNW) Bat Hub was formed by combining the former Alberta Hub and the BC and Southeast Alaska (BC-SE_AK) Hub. The NNW Bat Hub leads the NABat program within British Columbia, with key support from provincial and state representatives and WCSC. In 2025, 57 grid cells throughout British Columbia were monitored. Results indicate that bat populations across British Columbia remain stable, with some key species showing small positive trends, including Little Brown Myotis and Eastern Red Bat. Given growing threats to bat populations in BC, continued monitoring is essential to detect any significant changes.

2 Land Acknowledgement

Biodiversity Pathways respectfully acknowledges that our work takes place on Treaty 8 and Douglas Treaties Territories as well as the traditional and unceded territories of First Nations and Métis Peoples across all regions of British Columbia, whose histories, languages, and cultures are deeply connected to the biodiversity we monitor. We acknowledge the traditional teachings of the lands that we work on, and that reciprocal, meaningful, and respectful relationships with Indigenous peoples make our work possible. We are deeply grateful for their stewardship of these lands, and we are committed to supporting Indigenous-led monitoring programs, while learning Indigenous ways of knowing, being, and doing.

3 Introduction

3.1 Overview of NABat and the NNW Bat Hub

North American Bat Monitoring Program (NABat) is a large-scale coordinated effort to monitor bat species to address gaps in knowledge and lack of long-term studies across North America (Loeb et al. 2015).The program is administered by the US Geological Survey (USGS) and implemented by the North by Northwest Bat (NNW) Hub in British Columbia, Alberta and S.E Alaska.

NNW Bat Hub was established in 2024 with collaboration from Government of British Columbia, Government of Alberta, and Government of Alaska. The hub was formed by combining the previous Alberta Hub and the BC and southeast Alaska (BC-SE_AK) Hub. The implementation of the program within British Columbia has key support from Wildlife Conservation Society Canada (WCSC), government staff, other wildlife biologists, Indigenous Nations, community members and naturalists across the province.

The centralization of efforts for the hub will expand exciting programs to coordinate more efficiently across jurisdictions, ensuring consistent and long-term bat monitoring through standardized data collection, and streamlined bioacoustic data management. It also seeks to enhance engagement with NABat, foster collaboration among stakeholders, and produce annual reports on regional monitoring.

3.2 Threats to bat populations in BC

Bat populations in British Columbia face a wide range of threats including forestry practices, anthropogenic expansion, and climate change among others. A comprehensive review of tese threats and their relative impacts on individual bat species can be found in the BC Bat Action Plan (Team 2024). Of these threats, White-Nose Syndrome and wind energy development have emerged as two of the highest-priority concerns.

3.2.1 White-Nose Syndrome

White-nose Syndrome (WNS) is a deadly fungal disease caused by the fungus *Pseudogymnoascus destructans* (*Pd*) that affects hibernating bats. As of 2026, WNS has been confirmed in nine Canadian provinces including Alberta, and *Pd* has been detected within British Columbia at two locations (Figure 1: CWHC-RCSF (2024)). Continued monitoring of populations in BC is critical as *Pd* and WNS continue to spread throughout the province.

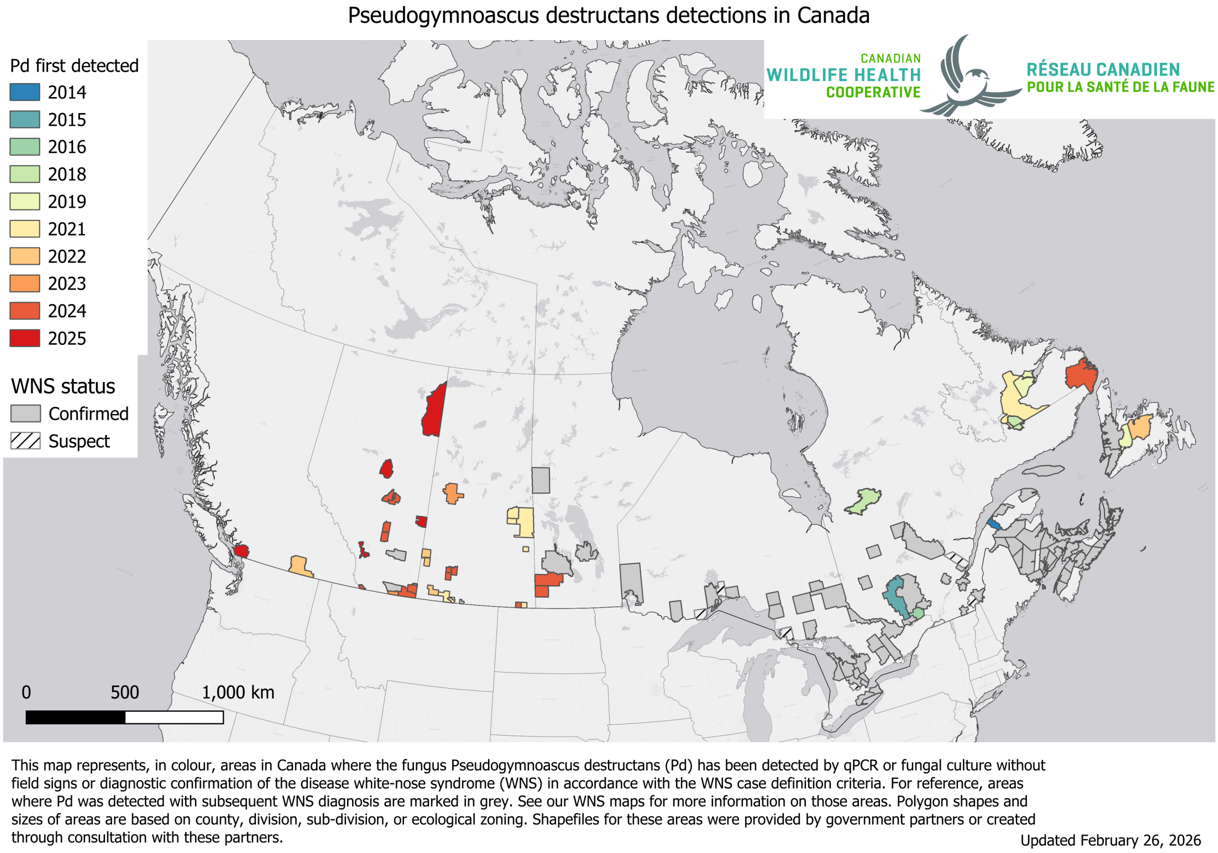


Figure 1: White-nose syndrome (WNS) and Pseudogymnoascus destructans (Pd) occurrence in Canada. Source: CWHC-RCSF (2026)

3.2.2 Wind Energy Development

Wind energy farms has been linked to high mortality rates across all bat species in Canada, with long distance migratory species accounting for 73% of recorded fatalities (Zimmerling and Francis 2016). In British Columbia alone, one estimate predicts an annual mortality rate of 680 bats per wind turbine (Zimmerling and Francis 2016). There are currently close to 300 turbines active in the province across 10 different projects with the most turbines found in the Peace region (Figure 2). In late 2024, B.C announced nine new wind projects which would require the installation of over 300 new wind turbines across the province (Energy and Climate Solutions 2024). As of 2023, COSEWIC designated all three of B.C.'s long distance migratory species as endangered due in part to fatalities at wind turbine facilities (Status of Endangered Wildlife in Canada 2023).

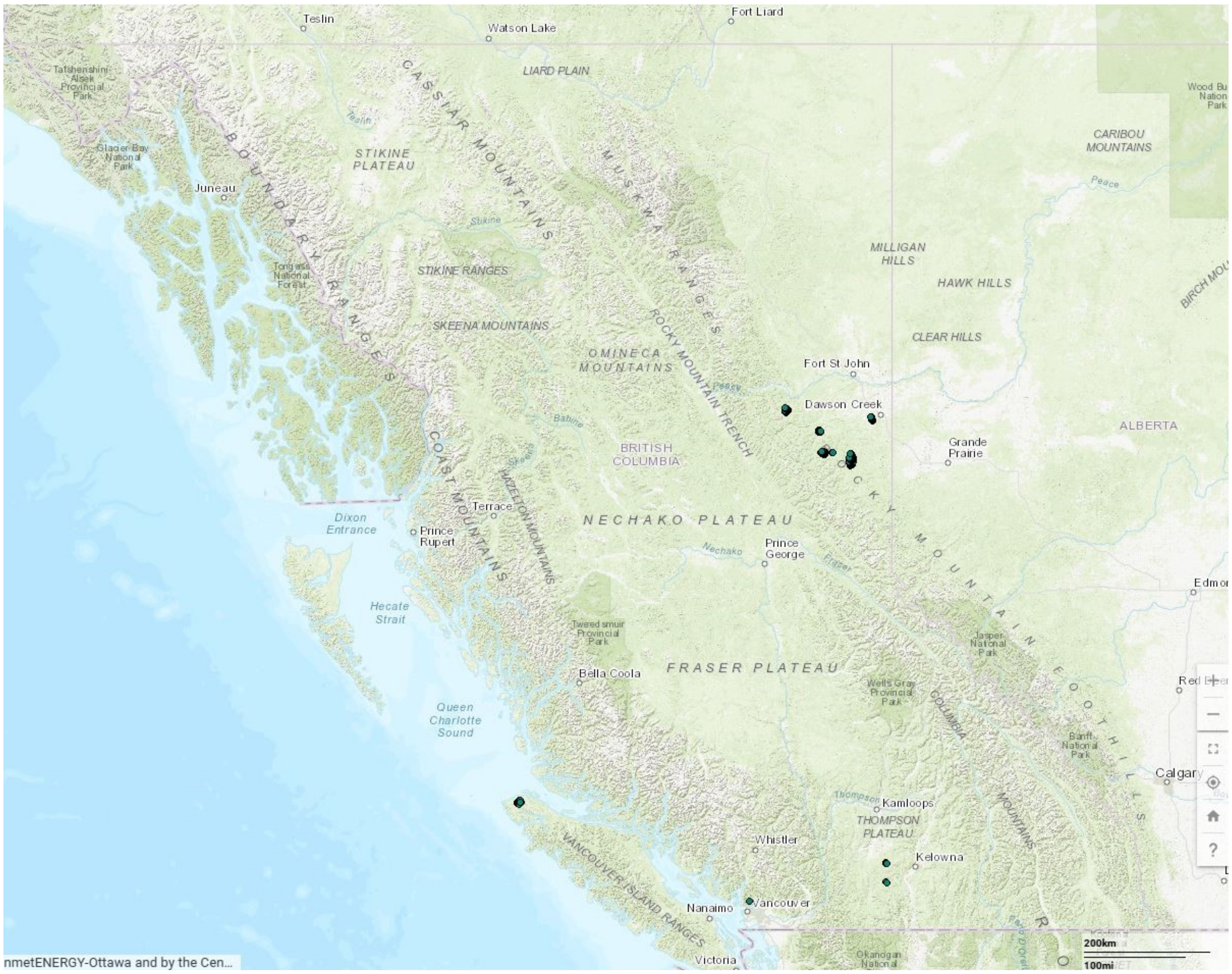


Figure 2: Current wind turbines in British Columbia based on the Canadian Wind Turbine Database (Source: https://search.open.canada.ca/openmap/79fdad93-9025-49ad-ba16-c26d718cc070)

3.3 Acoustic Monitoring of Bats

Acoustic recording is a cost-effective method for monitoring bat populations across large geographic and temporal scales. However, several limitations affect the conclusions that can be drawn from these data:

1. **Imperfect species identification** — Unlike birds, bats rely on echolocation rather than song, and species with similar ecology and physiology often produce overlapping calls. While detectors are deployed to maximize recordings of diagnostic Search Phase calls, species identifiability will always vary depending on the local assemblage.
2. **Variable detectability** — Bat species are not equally detectable acoustically. Larger, wide-ranging species produce louder calls that travel farther, while smaller species with quieter calls or specialized habitat use are less reliably recorded. As a result, trend estimates cannot be produced for all species using acoustic methods alone.
3. **Data variability** — Many factors influence recording quality and bat activity, including wind speed, humidity, and local habitat. Additionally, stationary detectors cannot account for individual recapture, meaning acoustic data can only inform relative activity and occupancy.
4. **Processing limitations** — Auto-ID software enables efficient processing of large datasets but has known high misclassification rates for bats, making manual verification necessary to ensure accurate results.

3.4 Update in Statistical Models

Bat monitoring data can be analyzed using different statistical approaches, each with distinct assumptions and outputs. Activity trend models examine changes in relative acoustic activity over time, providing a straightforward measure of whether bat detections are increasing, decreasing, or remaining stable across the monitoring network.

Dynamic occupancy models take a different approach, estimating the probability that a species occupies a given site, while explicitly accounting for imperfect detection. Rather than treating a non-detection as an absence, these models distinguish between a species being truly absent versus simply going undetected — an important consideration given the variable detectability of bats discussed above. They also track colonization and extinction probabilities across sites over time, providing insight into range shifts and local population dynamics.

In 2025, both model types were run to examine differences in results and assess the complementary information each approach provides.

4 Methods

4.1 Data Collection

The NABat monitoring program was established in BC in 2016 by WCSC and has sampled 65 grid cells in total. Cells were selected using the GRTSID system (Loeb et al. 2015), based on accessibility, presence of bat habitat, suitable recorder sites, and where possible, a road suitable for a driving transect (30–45 km, relatively straight).

Of 63 currently active cells, 56 were monitored in 2025 for a total of 635 site nights (Table 1); the remaining 7 were not monitored due to limited local capacity. The network is well-distributed across the province, with a slight bias toward southern BC and population centres ([Figure 3]).

Stationary Surveys

Two to four stationary detectors were deployed across at least two quadrants per grid cell for a minimum of seven nights. Sites are selected to maximize species detection and recording quality across different habitat types (see [Rae and Lausen (2024)] for full site selection criteria). Deployment occurs between mid-May and mid-July, after bats have emerged from hibernation but before young bats are volant. Sites and timing are kept consistent between years to minimize variability in conditions affecting bat activity.

Transect Surveys

Where possible, driving transects were conducted following Loeb et al. (2015). Two replicates of a 30–35 km route are driven at 30–35 km/h with a rooftop detector, using straight roads that minimize switchbacks. Transects are conducted concurrently with stationary deployments.

Table 1: Total number active grid cells and number of site nights sampled summarized by calendar year in British Columbia. Site nights are defined as the cumulative number of nights during which recorders were operational

Year	Number of Grid Cells Sampled	Site Nights
2016	22	203
2017	27	353
2018	43	314
2019	49	446
2020	49	446
2021	49	534
2022	56	614
2023	50	578
2024	56	635
2025	56	739

4.1.1 Equipment

A wide range of equipment is used across sampling locations in BC. Stationary units are primarily Wildlife Acoustics SM4Bat-FS or SM2Bat units (Maynard, MA), alongside an increasing number of Titley Scientific Swift and Express units (Brisbane, AU). Mobile transects are conducted using either the Wildlife Acoustics EchoMeter Touch2Pro or Titley Scientific Anabat units (SD1/2 zero-cross or Walkabout models).

2025 represents a transition year for equipment standardization. All SM2Bat units are being replaced with Titley Scientific Chorus units, and Anabat units are being phased out in favour of the

EchoMeter Touch2Pro for mobile transects. To account for detection differences between outgoing and incoming equipment, paired deployments were conducted at a number of grid cells this year, both stationary (SM2Bat and Chorus) and mobile (Anabat SD1 and EchoMeter Touch2Pro). These paired deployments will provide correction factors to ensure data comparability across the transition period.

4.1.2 Model Covariates

Numerous environmental factors influence bat activity across British Columbia. Species-specific covariates for stationary models were established by Rae and Lausen (2022) using AIC-based stepwise selection (Table 2).

Table 2: Top-performing stationary models selected via AIC-based stepwise selection. Covariates are described as, Moon = moon illumination × proportion of night above horizon; Clutter = distance (m) to nearest object; Nearby Water = detector within 100m of water;RH = mean nightly relative humidity (%); Degree Days = cumulative degree days above 10°C since Jan 1. Source Rae and Lausen (2024)

Species	Species Code	Top Performing Model Covariates
Pallid Bat	ANPA	Nightly Mean Temperature
Townsend's Big-eared Bat	COTO	Water Nearby,Nightly Min Temperature, Degree Days
Big Brown Bat	EPFU	Nightly Mean Wind Speed,Moon*Distance to Clutter,Nightly Min Temperature
Spotted Bat	EUMA	Nightly Mean Temperature
Eastern Red Bat	LABO	Nightly Mean Wind Speed*Distance to Clutter,Water Nearby,Nightly Mean Temperature
Hoary Bat	LACI	Moon*Distance to Clutter,Degree Days,Nightly Mean Temperature
Silver-haired Bat	LANO	Nightly Mean RH,Nightly Mean Wind Speed*Distance to Clutter,Water Nearby,Nightly Min Temperature
California Myotis	MYCA	Nightly Mean Wind Speed*Distance to Clutter,Water Nearby,Nightly Mean Temperature
Western Small-footed Myotis	MYCI	Nightly Mean Wind Speed*Distance to Clutter,Water Nearby,Nightly Mean Temperature
Long-eared Myotis	MYEV	Nightly Mean Wind Speed*Distance to Clutter,Water Nearby,Degree Days,Nightly Min Temperature
Little Brown Myotis	MYLU	Nightly Mean Wind Speed*Distance to Clutter,Water Nearby,Nightly Mean Temperature

Northern Myotis	MYSE	Nightly Mean RH,Moon*Distance to Clutter,Water Nearby
Fringed Myotis	MYTH	Degree Days
Long-legged Myotis	MYVO	Nightly Mean Wind Speed*Distance to Clutter,Water Nearby,Nightly Min Temperature
Yuma Myotis	MYYU	Nightly Mean Wind Speed*Distance to Clutter,Water Nearby,Degree Days

Transect models followed the same selection approach, testing all combinations of nightly mean temperature, relative humidity, wind speed, and moon illumination, with the most parsimonious model for each species selected based on lowest AIC (Table 3).

Table 3: Top-performing transect models selected via AIC-based stepwise selection. Covariates are described as, Moon = moon illumination × proportion of night above horizon;RH = mean nightly relative humidity (%).

Species	Species Code	Top Performing Model Covariates
Big Brown Bat	EPFU	Nightly Mean Wind Speed
Hoary Bat	LACI	Nightly Mean Temperature,Nightly Mean Wind Speed
Silver-haired Bat	LANO	Nightly Mean Wind Speed,Moon
California Myotis	MYCA	Nightly Mean Temperature,Nightly Mean Wind Speed,Moon
Western Small-footed Myotis	MYCI	Nightly Mean Wind Speed
Long-eared Myotis	MYEV	Nightly Mean Wind Speed
Little Brown Myotis	MYLU	Nightly Mean Wind Speed,Moon
Long-legged Myotis	MYVO	Nightly Mean RH,Nightly Mean Wind Speed
Yuma Myotis	MYYU	Nightly Mean Wind Speed,Moon

Dynamic occupancy model covariates were based on those established by Rae and Lausen (2022), with additional exploration of factors influencing colonization and extinction. These included distance to roads and distance to harvest cutblocks, derived from the unpublished Human Footprint Index (HFI) layer currently being developed by the Geospatial Centre within Biodiversity Pathways.

Where possible, temperature and relative humidity are recorded in each grid cell using onsite logging devices; otherwise, data are sourced from the nearest climate station. Additional covariates include nightly mean wind speed (km/h), cumulative precipitation since January 1 (mm), degree days above 10°C since January 1, and moon illumination (proportion illuminated × proportion of night above horizon). Site-level covariates — including distance to clutter, percent clutter, distance to water, and water feature type — are recorded by grid leaders during deployment.

Since 2022, insect activity has been measured along transect routes using vehicle-mounted sticky traps. Collected samples are identified to order level by WCSC and will be evaluated as a potential covariate of bat activity.

4.2 Acoustic Processing

In 2025 acoustic analysis mostly followed the standards set in previous years (Rae and Lausen 2022, 2024). This included running all files through two automatic classifiers, Kaleidoscope's Bats of North America 5.4.0 classifier and Sonobat 3.0's Northwestern British Columbia classifier, to remove files that do not contain bat pulses and giving a preliminary bat species identification.

Manual vetting follows Reichert et al. (2018) and analysis standards developed by the NNW Bat Hub and WCSC. Consistent with 2024, approximately 50% of all recordings were manually verified; for full details on this process, refer to the 2024 BC NABat Report.

4.3 Statistical Analyses

4.3.1 Activity Trends

Statistical analyses followed closely what was established by WCSC and Dr. Carl Schwarz in 2021 (Rae and Lausen 2022). This includes conducting species-specific trend analyses examining activity patterns in the data collected to date using the (negative binomial GLMM) models and the covariates identified in Table 2.

Species were excluded from the analysis if model fit was not appropriate due to issues including under-dispersion or standard errors that were more than 3 times the estimate. Model results were also excluded if there was less than 150 recordings across all sample years. We conducted our statistical analyses on provincial and regional scales. For the regional analyses, we assigned all sampled grid cells into four geographic regions (Southwestern, Southcentral, Southeastern, and Northern), each roughly corresponding to different species sets (Figure 3). Since transects are only conducted in roughly 50% of our grid cells on a typical year and each transect records fewer bats per night than stationary monitoring, we have elected not to conduct regional-scale transect trend analyses due to low sample sizes.

4.3.1.1 Use of Automated Identification Results in Stationary Trend Analyses

Stationary trend analyses used AutoID classifications from Kaleidoscope Pro and Sonobat 3.0. Manual verification was applied only to flag non-bat noise; where manual review suggested a different species than the AutoID label, the original AutoID classification was retained.

This approach was first identified as necessary in 2023 (Rae and Lausen 2022), when analyses of manually verified data revealed a consistent positive trend across all species, this was interpreted as an artifact of reviewer learning over time rather than a true biological signal. As verifier experience increased across years, calls were increasingly assigned to specific species rather than left unidentified, producing an apparent increase in detections independent of actual bat activity. Because this observer effect is difficult to control for statistically, AutoID classifications are now used as the standard for all activity trend analyses, as automated classifiers apply consistent criteria across all years. While AutoID is subject to some misclassification, the error rate is consistent across years and does not confound detection of genuine temporal trends.

Trend analyses are presented only for species confirmed present through manual verification and for which models fully converged.

4.3.2 Dynamic Occupancy

This analysis used nightly acoustic detection/non-detection data from passive bat detectors deployed at GRTS-selected sites in British Columbia (BC) as part of the NABat long-term monitoring program. Data were filtered to the 2017–2025 period, and analysis used an every-other-night subsampling protocol to reduce temporal autocorrelation between consecutive nights, using max 4 visits per site-year. Sites had to have a minimum of 2 visits within a given year and data from at least 2 years total. Models were run separately for all 17 bat species.

Fitted multi-season (dynamic) occupancy models (MacKenzie et al. 2003) to estimate trends in occupancy, colonization, and persistence while accounting for imperfect detection. Initial occupancy ( ) was modeled as a function of elevation and water nearby, with a random effect for region. Colonization ( ) and persistence ( ) were modeled with time-varying intercepts and distance to road as a covariate. Detection probability (p) was modeled as a function of percent clutter, nightly mean temperature, and Julian date.

Elevation for each site was extracted from AWS terrain tiles (~30 m resolution, SRTM-based) using the elevatr package in R, distance to road and distance to harvest was extracted using the Biodiversity Pathways Human Footprint Index (not published yet). All continuous covariates were standardized by centering on their mean and dividing by their standard deviation; missing covariate values were imputed with the global mean prior to standardization.

Models were fit in a Bayesian framework using JAGS via the jagsUI package in R, with 30,000 iterations, 15,000 burn-in, thinning by 5, and 3 chains run in parallel, yielding 9,000 posterior samples for inference. Priors on all regression coefficients were normally distributed (mean = 0, precision = 0.368, equivalent to SD 1.65), and the region-level random effect standard deviation received a uniform(0, 5) prior. Model convergence was assessed using Rhat diagnostics. Results are interpreted using posterior means and 95% Bayesian credible intervals. An overall occupancy trend metric ( ) was calculated as the ratio of estimated occupancy in the final year to occupancy in the initial year; values of > 1 with 95 CIs entirely above 1 indicate an increase in occupancy, values < 1 with 95 CIs entirely below 1 indicate a decline, and values with CIs overlapping 1 indicate uncertain or stable trends.

5 Results

2025 marked the tenth year of coordinated NABat sampling done in BC. A total of 56 grid cells were sampled with the support of over 60 individuals including contractors, volunteers and staff members from BC Parks, BC WLRS, and BC Ministry of Forests. Grid leaders have also collected a total of 16 transect insect sticky trap samples in 2025.

5.1 Species Detection

In 2025, all species previously documented in BC were detected. The manually verified detections varied among grid cell and regions (Figure 3). The most widely identified species as well as the most geographically distributed bats that were confirmed through manual ID were the Little Brown Bat followed by the Silver-haired Bat. Any notable species detection are addressed in the discussion

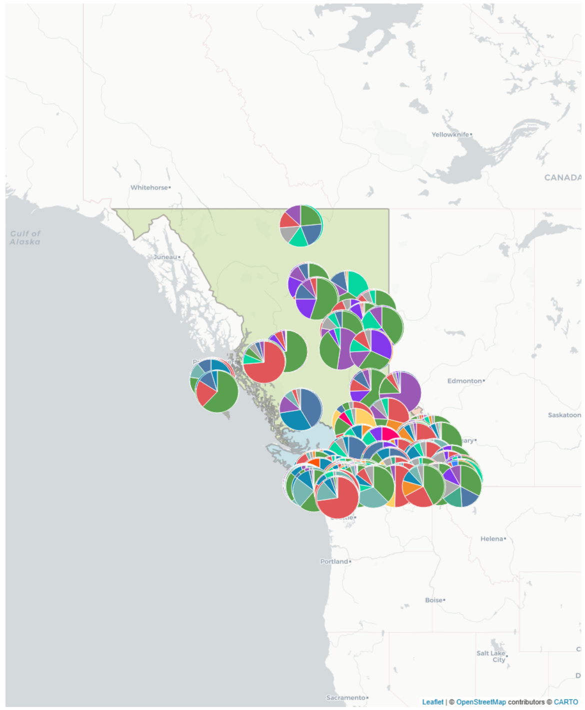


Figure 3: NABat survey efforts in 2025 within British Columbia. Analysis regions were defined by Wildlife Conservation Society Canada (WCSC) based on species ranges maps to group areas with similar species compositions.

5.2 Activity Trend Models

Stationary data was used to estimate activity trends for 11 of BC's 15 resident species, while transect data was sufficient to estimate trends for 9 species (Table 4). Of the stationary models, seven species showed statistically significant positive trends ($p < 0.05$):

- Townsend's Big-eared Bat (0.12 ± 0.06)
- Big Brown Bat (0.12 ± 0.08)
- Spotted Bat (0.15 ± 0.15)
- Silver-haired Bat (0.12 ± 0.05)
- California Myotis (0.08 ± 0.04)
- Little Brown Mytois(0.06 ± 0.04).
- Long-legged Myotis (0.08 ± 0.04).

No significant negative trends were detected in the stationary models. The remaining species with sufficient data had non-significant trend estimates.

Transect models yielded results for fewer species, with only Little Brown Myotis showing a significant positive trend (0.16 ± 0.08). Trend estimates for the remaining species with transect data were small and non-significant.

Table 4: Provincial stationary and transect trend estimates from 2017-2024 NABat monitoring in BC adjusted for covariates. Trends estimated on a logarithmic scale, thus a trend estimate of 0.05 corresponds to an annual change of 5%. Significant results (p value < 0.05) are highlighted in bold.

	Stationary Models		Transect Models	
Species	Trend Estimate ± 95%CI	P-value	Trend Estimate ± 95%CI	P-value
Pallid Bat				
Townsend's Big-eared Bat	**0.12 ± 0.06**	**<0.001**		
Big Brown Bat	**0.12 ± 0.08**	**0.001**	0.06 ± 0.09	0.150
Spotted Bat	**0.15 ± 0.15**	**0.049**		
Eastern Red Bat				
Hoary Bat	0.21 ± 0.33	0.225	0.01 ± 0.13	0.906
Silver-haired Bat	**0.12 ± 0.05**	**<0.001**	0.04 ± 0.07	0.311
California Myotis	**0.08 ± 0.04**	**<0.001**	0.06 ± 0.12	0.331
Western Small-footed Myotis			0.22 ± 0.26	0.103
Long-eared Myotis	0.02 ± 0.06	0.530	0.04 ± 0.12	0.488
Little Brown Myotis	**0.06 ± 0.04**	**0.008**	0.16 ± 0.08	
Northern Myotis				
Fringed Myotis	-0.05 ± 0.09	0.274		
Long-legged Myotis	**0.08 ± 0.04**	**<0.001**	0.12 ± 0.14	0.094
Yuma Myotis	0.03 ± 0.08	0.505	0.04 ± 0.15	0.582

Regional trend analyses revealed significant results ($p < 0.05$) across all regions with the most found in the South Coastal region (Table 5). Most significant results were positive, but significant negative trends were found in:

- Hoary Bat in the South coastal Region (**-0.60 ± 0.58**)
- Yuma Myotis in the Northern Region (**-0.22 ± 0.22**).

Table 5: Regional trend estimates from 2017-2024 NABat monitoring in BC adjusted for covariates. Trends estimated on a logarithmic scale, thus a trend estimate of 0.05 corresponds to an annual change of 5%. Significant results (p value < 0.05) are highlighted in bold.

Species	Kootenay Region		Northern Region	
	Estimate ± 95%CI	P-value	Estimate ± 95%CI	P-value
Pallid Bat				
Townsend's Big-eared Bat	0.08 ± 0.10	0.118		
Big Brown Bat	0.02 ± 0.12	0.707	0.02 ± 0.24	0.881
Spotted Bat				
Eastern Red Bat				
Hoary Bat	0.54 ± 0.59	0.073	1.05 ± 1.19	0.084
Silver-haired Bat	**0.17 ± 0.08**	**<0.001**	0.01 ± 0.14	0.882
California Myotis	0.03 ± 0.07	0.342	0.01 ± 0.11	0.829
Western Small-footed Myotis				
Long-eared Myotis	-0.02 ± 0.10	0.734	0.01 ± 0.15	0.934
Little Brown Myotis	**0.11 ± 0.06**	**0.001**	-0.05 ± 0.09	0.308
Northern Myotis				
Fringed Myotis	-0.02 ± 0.14	0.828		
Long-legged Myotis	0.06 ± 0.06	0.088	-0.03 ± 0.11	0.617
Yuma Myotis	-0.09 ± 0.15	0.222	**-0.22 ± 0.22**	**0.043**

	Okanagan Region		South Coastal Region	
Species	Estimate ± 95%CI	P-value	Estimate ± 95%CI	P-value
Pallid Bat				
Townsend's Big-eared Bat	0.13 ± 0.14	0.067	**0.20 ± 0.11**	**<0.001**
Big Brown Bat	0.08 ± 0.14	0.279	**0.21 ± 0.13**	**0.001**
Spotted Bat	0.07 ± 0.16	0.375	0.23 ± 0.49	0.364
Eastern Red Bat				
Hoary Bat	-0.28 ± 0.63	0.379	**-0.60 ± 0.58**	**0.044**
Silver-haired Bat	0.05 ± 0.10	0.292	0.07 ± 0.10	0.180
California Myotis	0.06 ± 0.09	0.216	**0.10 ± 0.07**	**0.004**
Western Small-footed Myotis				
Long-eared Myotis	0.03 ± 0.13	0.670	0.08 ± 0.10	0.123
Little Brown Myotis	0.04 ± 0.10	0.389	**0.11 ± 0.08**	**0.008**
Northern Myotis				
Fringed Myotis	-0.09 ± 0.13	0.195	0.16 ± 0.31	0.295
Long-legged Myotis	**0.12 ± 0.11**	**0.034**	**0.09 ± 0.07**	**0.011**
Yuma Myotis	-0.10 ± 0.16	0.229	**0.23 ± 0.12**	**<0.001**

5.3 Dynamic Occupancy Models

Across the 17 bat species modeled over the 9 years, no species showed a clear decline in occupancy in BC. Several species showed increases, and the rest were stable or showed too much uncertainty to draw conclusions regarding trends (E.g., TABR, PAHE).

Species showing clear increases in occupancy (BC models): MYLU. MYEV, MYVO, MYVU, MYCA, MYCI, LABO. LABO showed the strongest increase (0.22 to 0.59). MYLU was the most widespread species overall going from ~89% to 97% occupancy. Increases generally driven by colonization rates exceeding extinction rates–bats showing up at sites faster than they are leaving occupied ones.

Species with stable or uncertain trends: EPFU, LACI, LANO, EUMA, COTO, MYTH. ANPA is a special case– showing a massive lambda value in the BC-only model (and the BC + AK to some extent), but as per the note below, this is mainly because of an artifact of going from near zero occupancy to still-low occupancy, so the ratio is uncertain, reflected by the wide credible intervals.

Poorly estimated species: PAHE and TABR had extremely low detection probabilities (0.5-0.7%), leading to poor model convergence and very wide credible intervals-interpret with caution or just state they are data-deficient (models with that low of detection rates are basically meaningless).

Detection: Temperature was the most consistent detection covariate–warmer nights equate to higher detection for most species. There were some clutter effects on detection probability that were species-specific. Julian date effects were mixed across species.

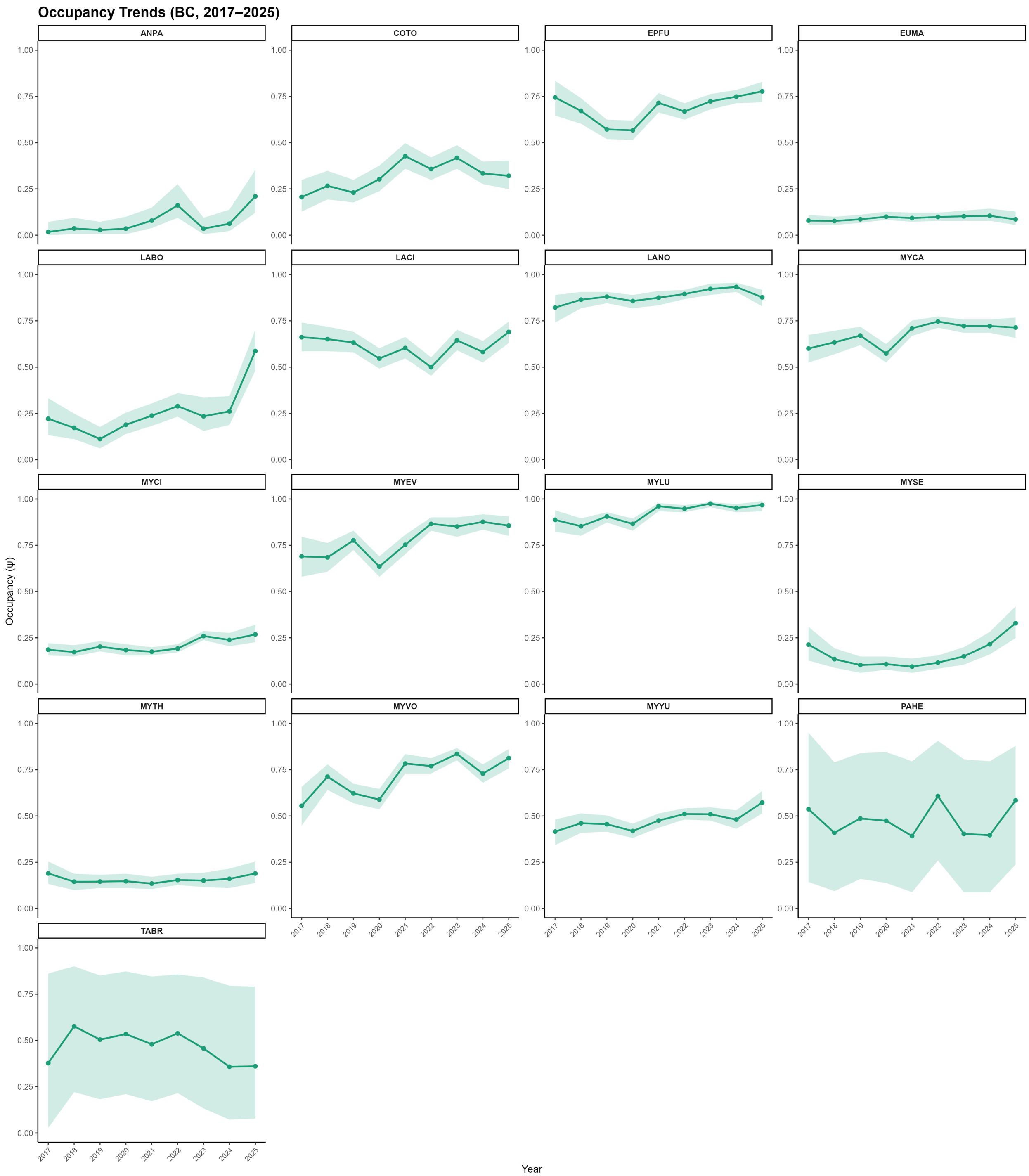


Figure 4: Dynamic occupancy estimates ( ) for 17 bat species monitored across British Columbia from 2017–2025. Points represent annual mean occupancy estimates and shaded areas indicate 95% credible intervals.

6 Discussion

6.1 Species Presence

Species ranges across the project period (2016–2025) continue to appear broadly stable, with no grid cells showing conspicuous or sustained absences for previously detected species.

6.2 Trend analyses

6.2.1 Activity Trends

Overall, results suggest that bat populations remain broadly stable across British Columbia, with no species showing widespread negative trends.

Several species showed statistically significant positive trends in both province-wide and South Coastal regional models. However, these results should be interpreted with caution. Caution is warranted when evaluating the magnitude of the trend estimates. Several models exhibited underdispersion, which, while not unexpected given the structural characteristics of acoustic bat activity data, means that confidence intervals and effect sizes should be interpreted carefully rather than taken at face value.

The significant negative trend detected for Hoary Bat in the South Coastal Region warrants attention and should be monitored closely in coming years to determine whether this represents a genuine localized decline or reflects sampling variability.

More broadly, as threats to bat populations continue to grow, it may be worth exploring complementary analytical approaches alongside activity-based trend models. Acoustic activity metrics carry inherent noise, and detecting meaningful population signals can be challenging, particularly for species with low detection rates or patchy distributions. We recommend exploring occupancy modelling as a complementary approach in future analyses, as it may provide a clearer population signal by reducing the noise inherent in activity-based trend estimates from stationary acoustic data.

6.2.2 Dynamic Occupancy

The dynamic occupancy estimates suggest that the bat community in BC appears to be generally stable to increasing over 2017-2025, with no species showing evidence of decline. Species showing strongest increases are Myotis species and LABO. Two species can't be reliably assessed (PAHE, TABR) due to very low detection rates. Distance to road has an effect on colonization and persistence dynamics for several species, with sites closer to roads tending to have higher turnover.

6.3 Challenges and Lessons Learned

6.3.0.1 Timing of Data Retrieval and Submission

Field crews across British Columbia continue to reliably collect recordings for their grid cells. However, organizing and submitting these recordings, along with associated metadata, remains a challenge. Delays in data submission from grid leaders extend processing and analysis times, making it difficult to complete reports by the end of the fiscal year. The transition in project management this year further complicated data submission. We anticipate these challenges will be resolved in future years as teams adjust to the new process. To improve efficiency, we plan to increase reminders to grid leaders and provide additional support to address ongoing challenges in data organization and submission.

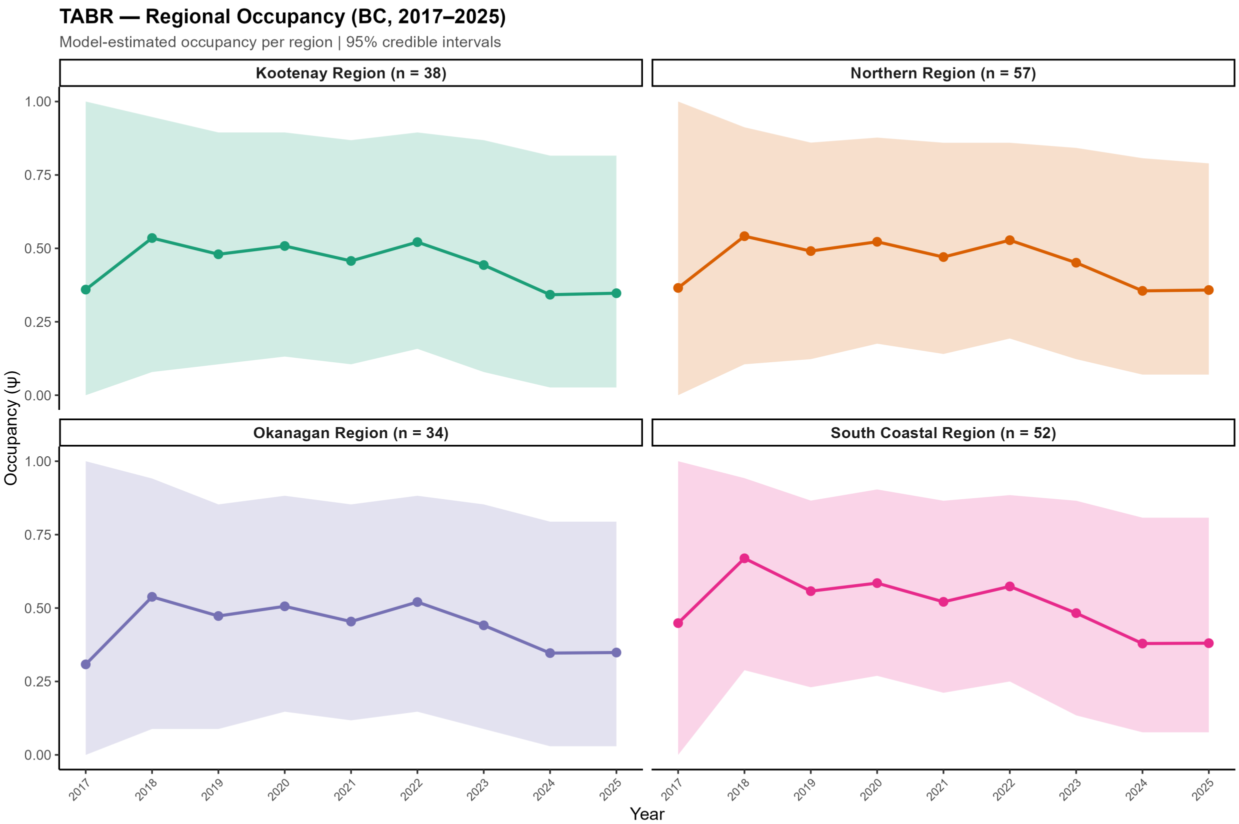
6.4 Looking Forward

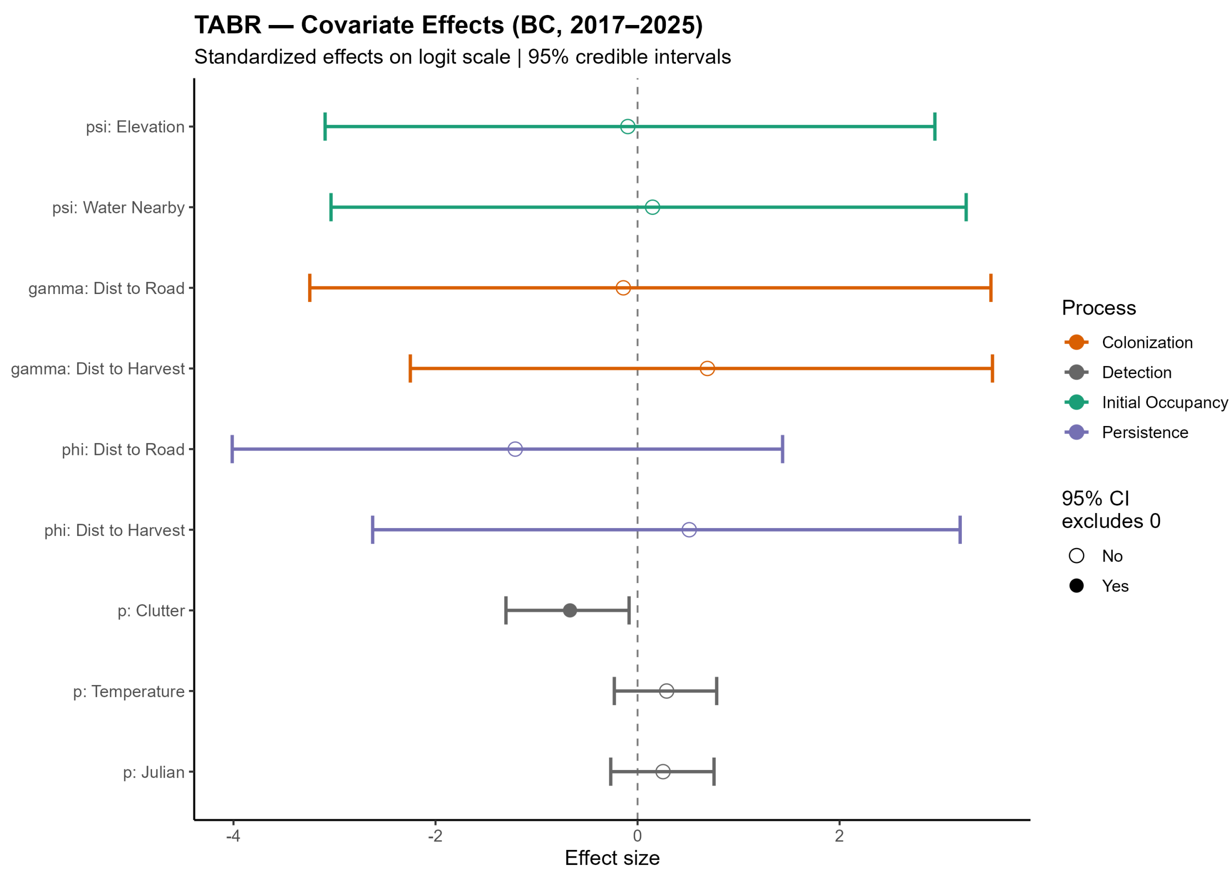
The BC NABat program has yielded a large and robust acoustic dataset that provides solid baseline information on distribution and relative abundance of 12 of 15 resident BC bat species. The data that has been collected so far are an invaluable reference point for BC's bat distributions and activity that improve with each additional year of monitoring. Key considerations for the future of the program include:

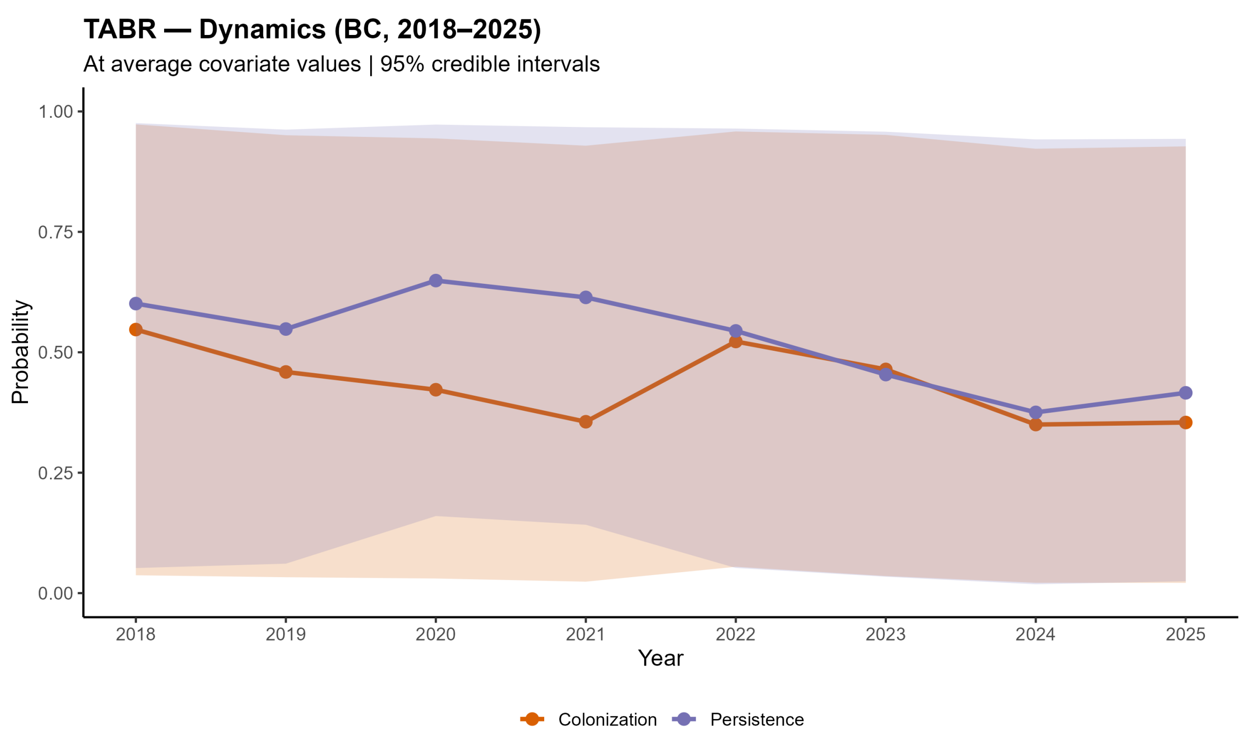
- **Manual Verification Efficiency.** This was the first year that the BC monitoring program reduced the proportion of manual vetting carried out in the recordings to ~50% based on the findings from Stratton et al. (2022). Careful consideration of these results compared to previous years will be considered as a guide for future models and analyses.
- **Transect Sampling.** Transects are a relatively high labour/effort source of data than stationary detectors that require participants to work late into the night. They also often do not collect many recordings of bats in BC relative to the stationary detectors. However, transects are still the only source of reliable estimates of relative abundance. While continuing to support current transect sampling efforts, the NNW Bat Hub will work along side other partners to also explore other methods of creating reliable abundance estimates that are less labour intensive.
- **Adding new grid cells.** Efforts will continue to be made to monitor as many of the same grid cells as possible, as the power of the dataset improves with each additional year. Moreover, looking for new opportunities to add new grid cells especially in under sampled regions of the province can help fill knowledge gaps for species distributions. New cells could also be added within ranges of species that are not widespread but easily monitored using acoustics. For example, Fringed Myotis and Spotted Bats are highly suited to acoustic surveillance, but only occur within a small portion of the grid cells being monitored currently

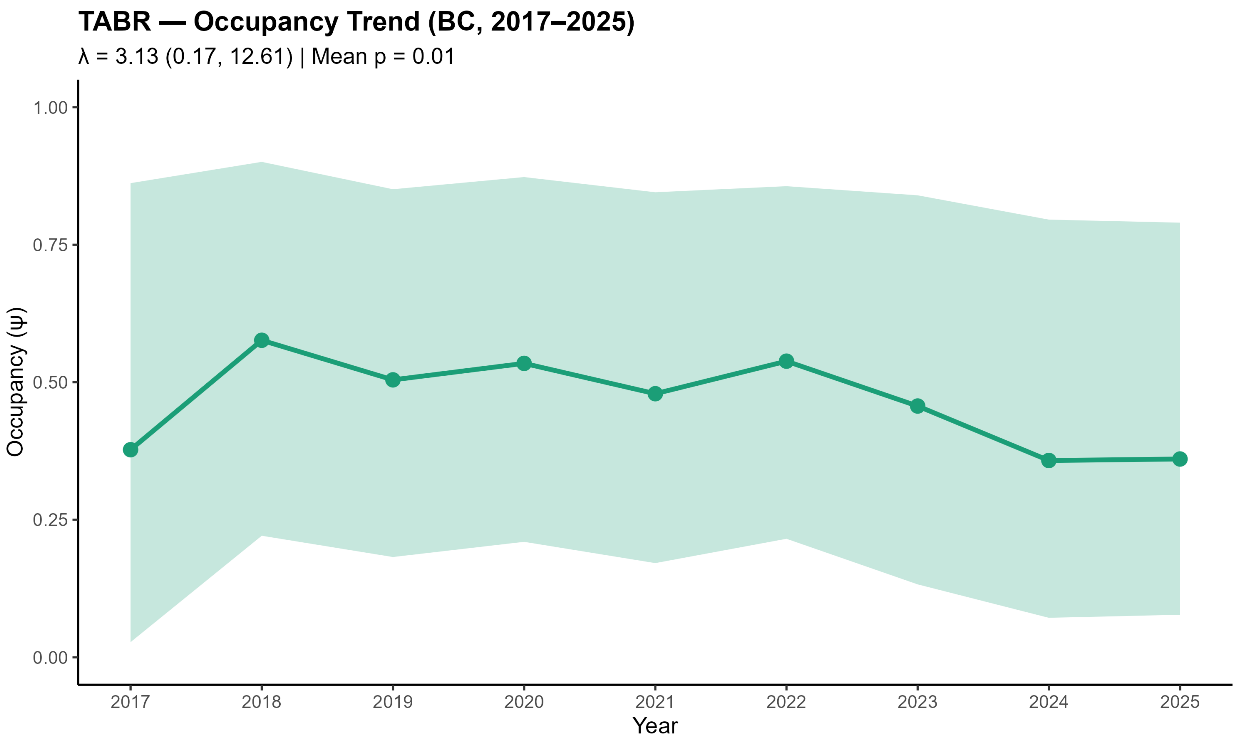
7 Appendix A

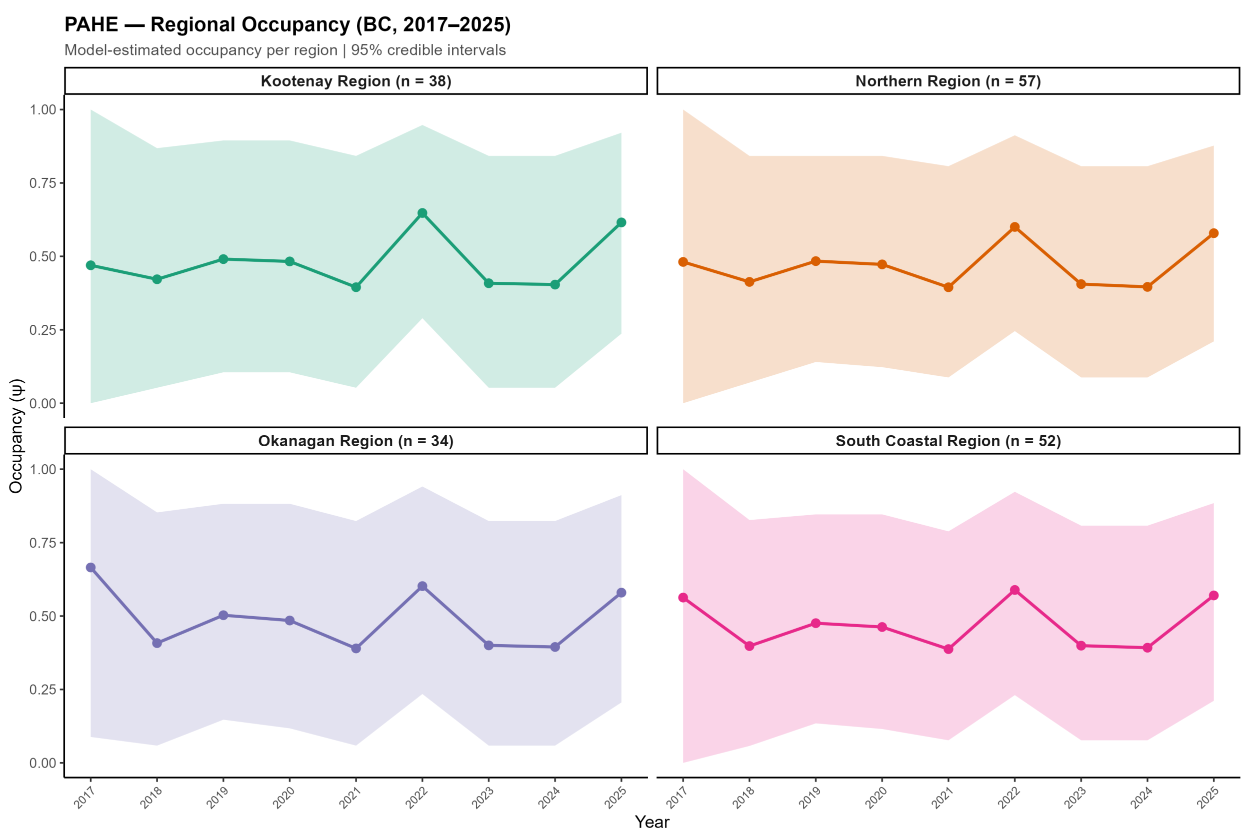
Species and region specific dynamic occupancy results

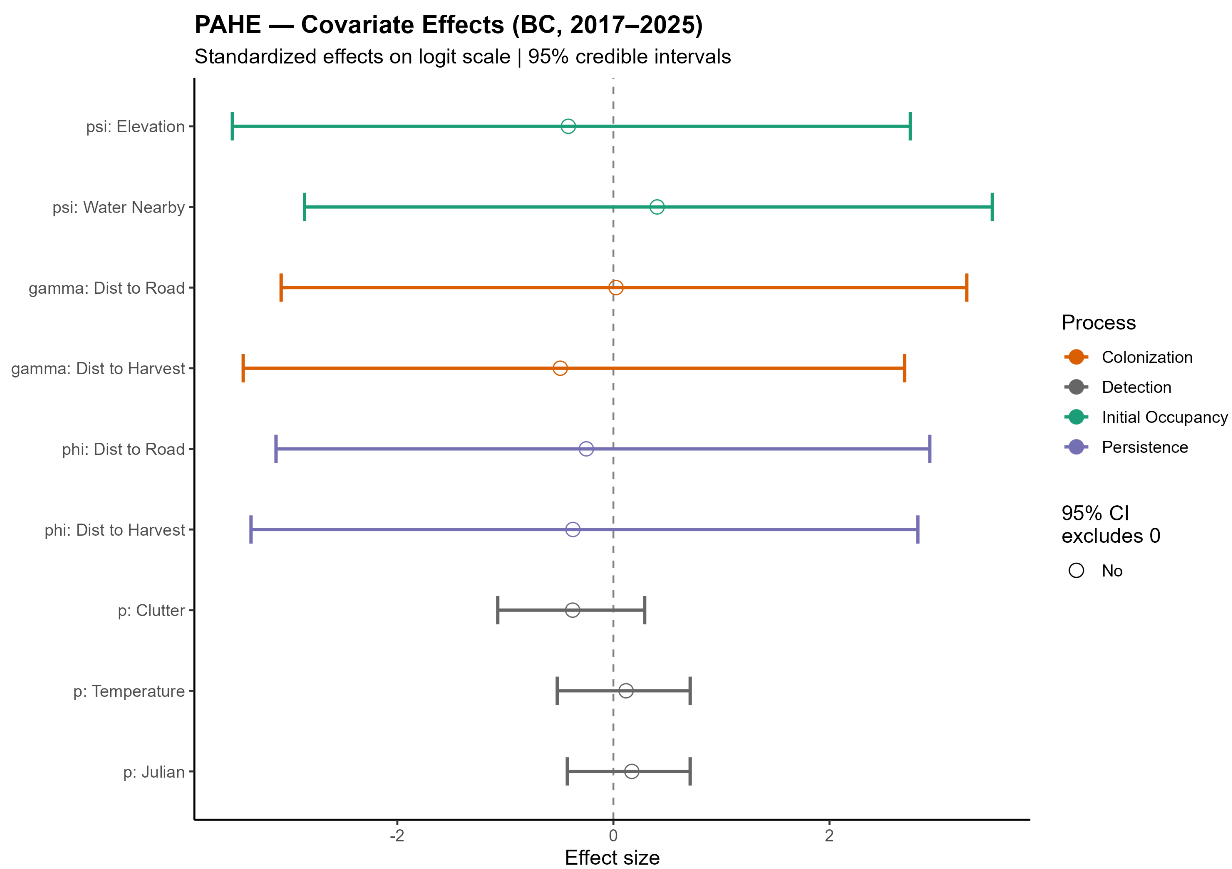



TABR — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-4
-2
0
2
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI
excludes 0
No
Yes

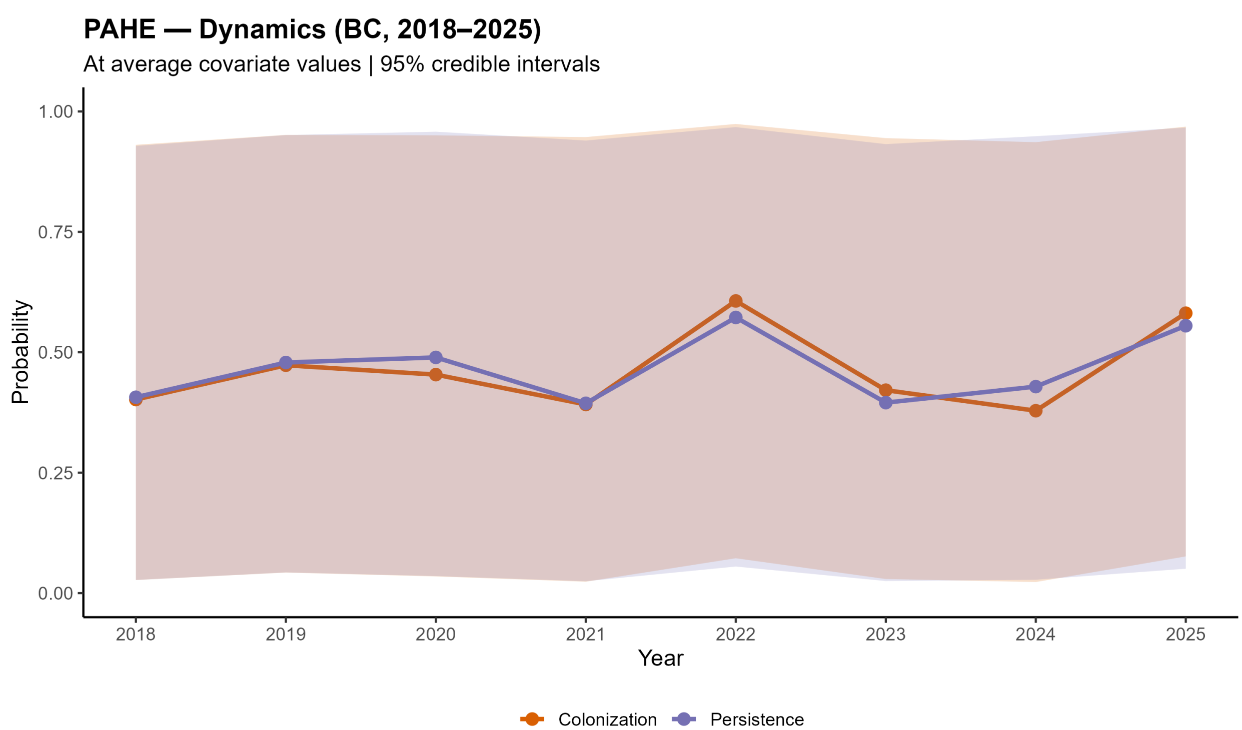

TABR — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence


TABR — Occupancy Trend (BC, 2017–2025)
λ = 3.13 (0.17, 12.61) | Mean p = 0.01
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

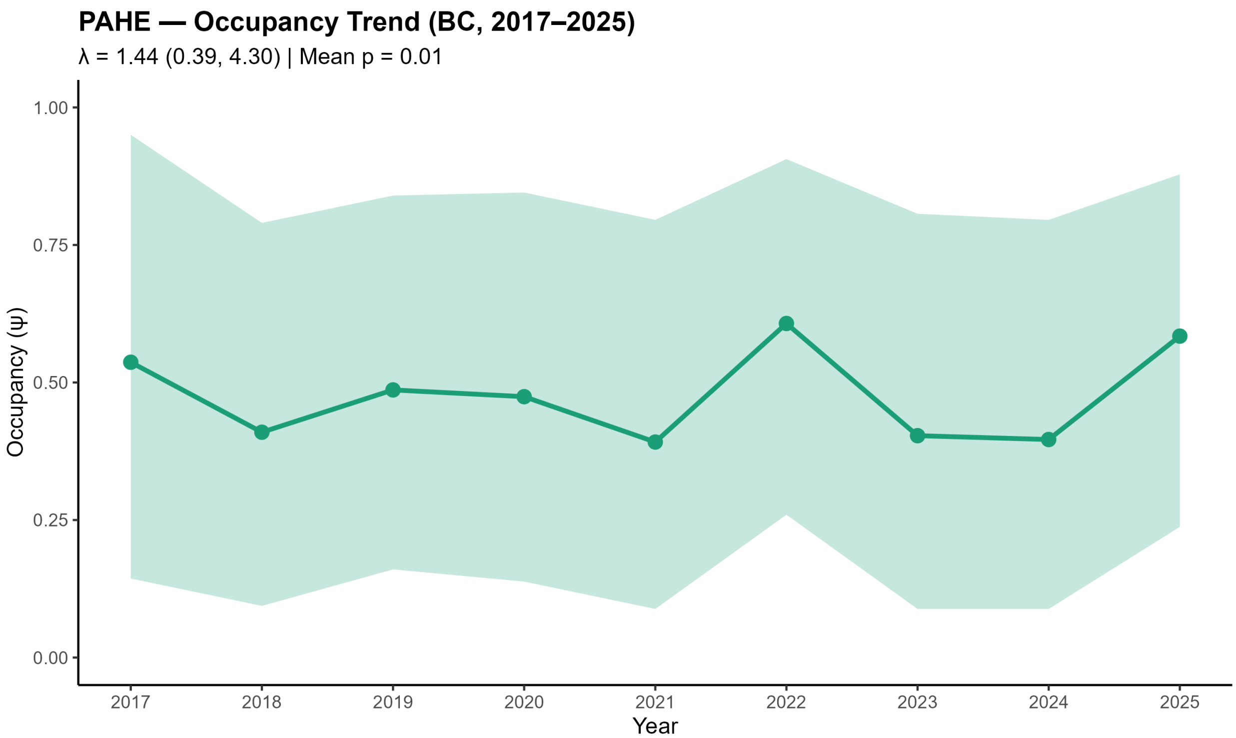

PAHE — Regional Occupancy (BC, 2017–2025)
Model-estimated occupancy per region | 95% credible intervals
Kootenay Region (n = 38)
Northern Region (n = 57)
Okanagan Region (n = 34)
South Coastal Region (n = 52)
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

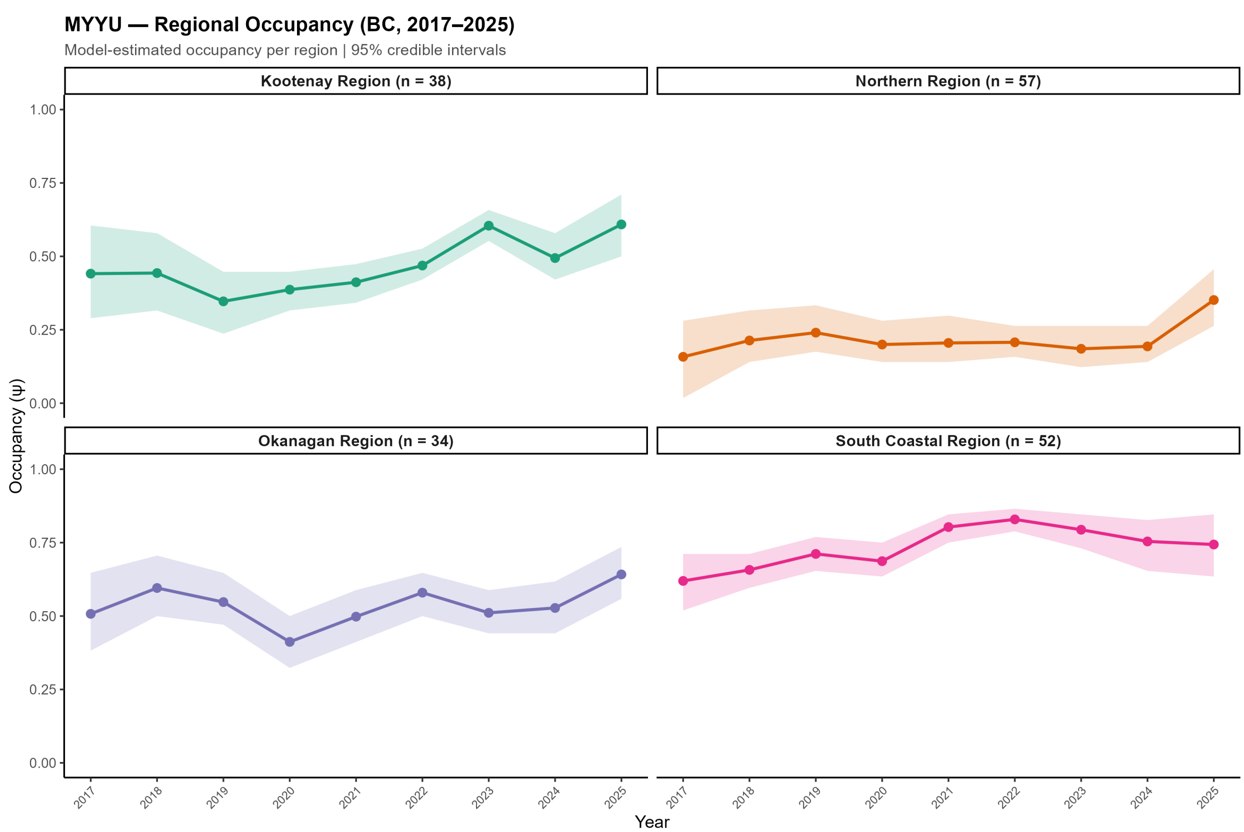


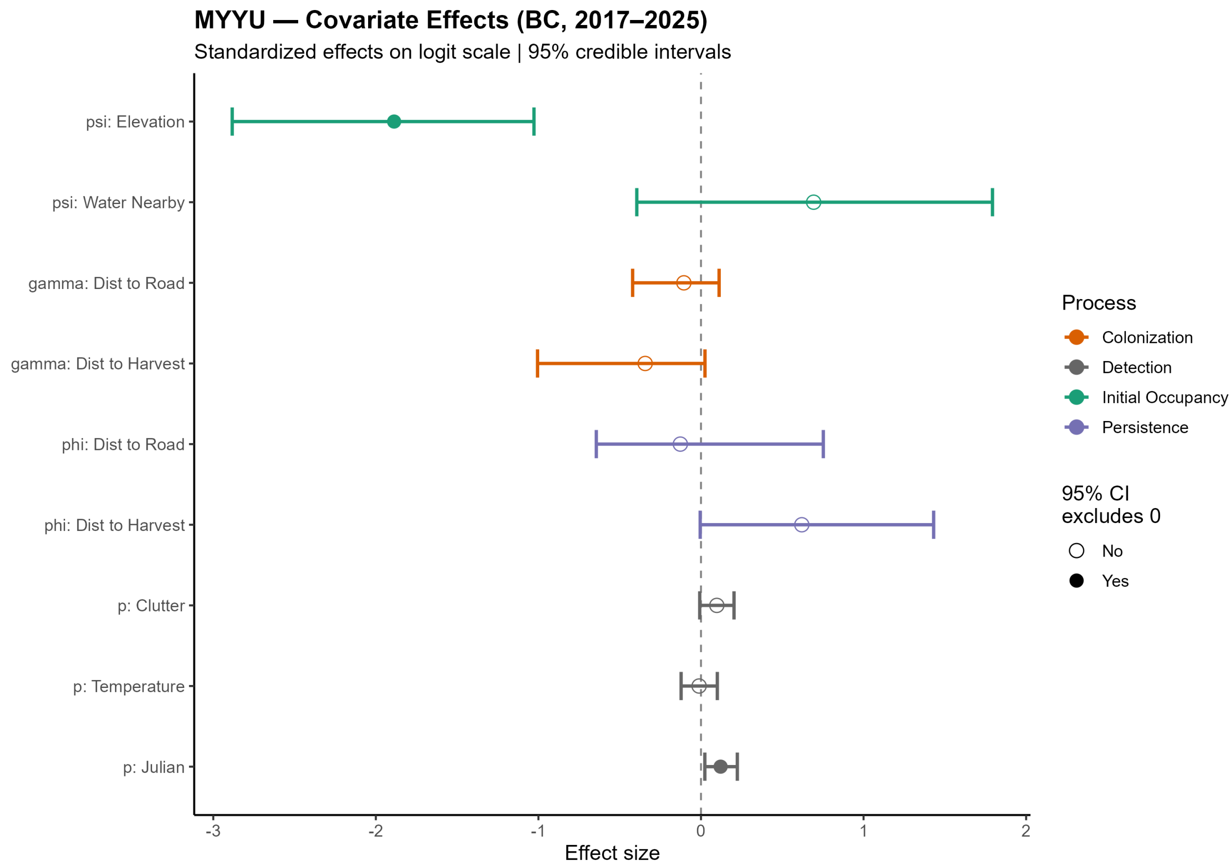
PAHE — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-2
0
2
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI excludes 0
No

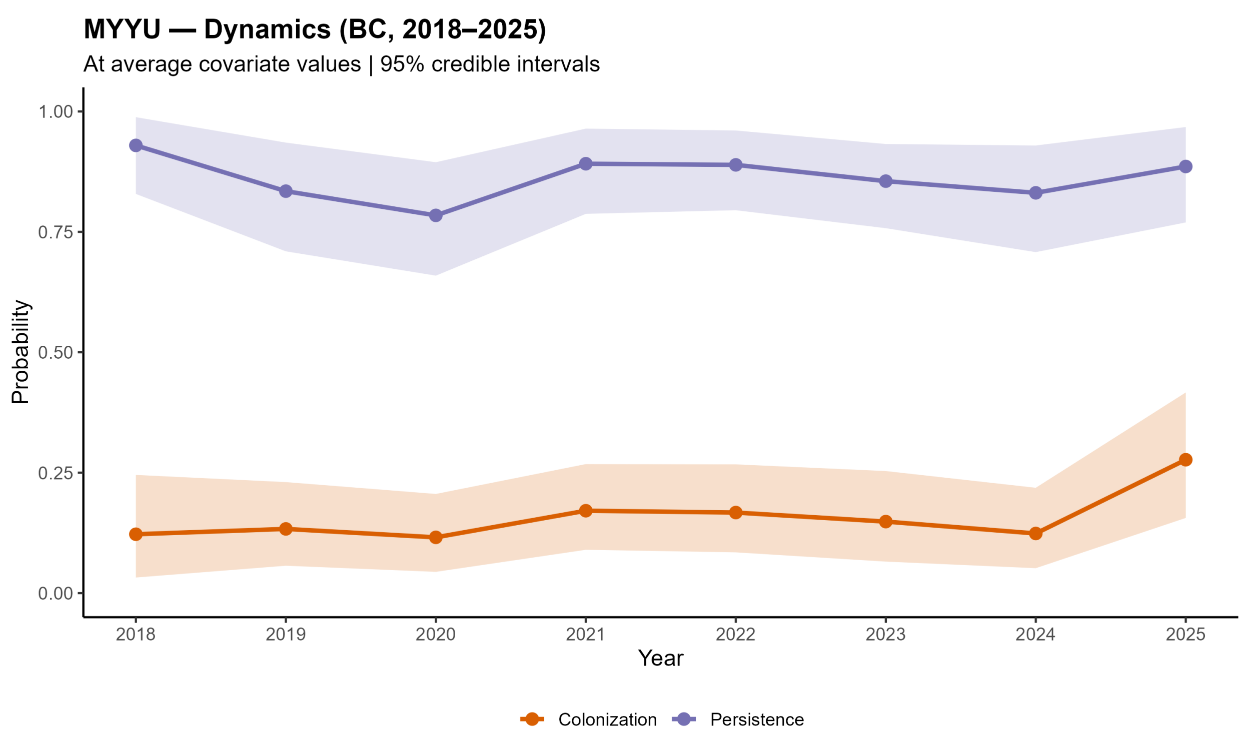


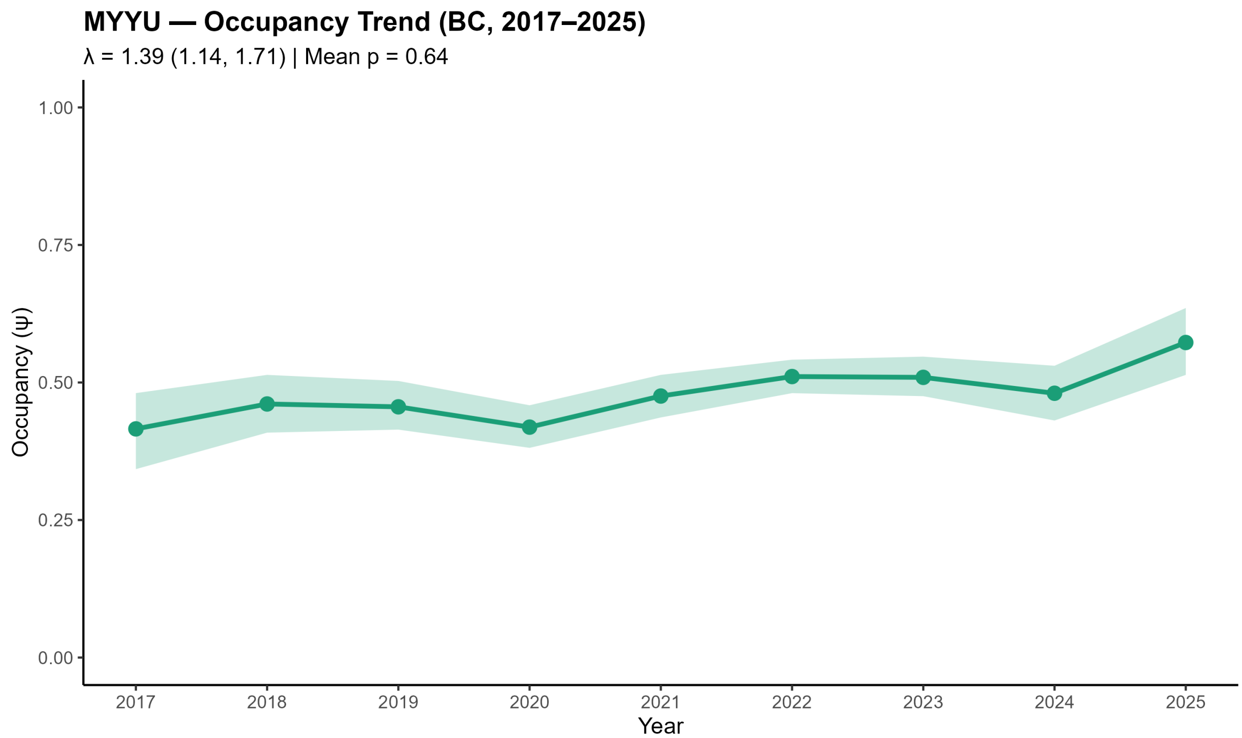
PAHE — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence

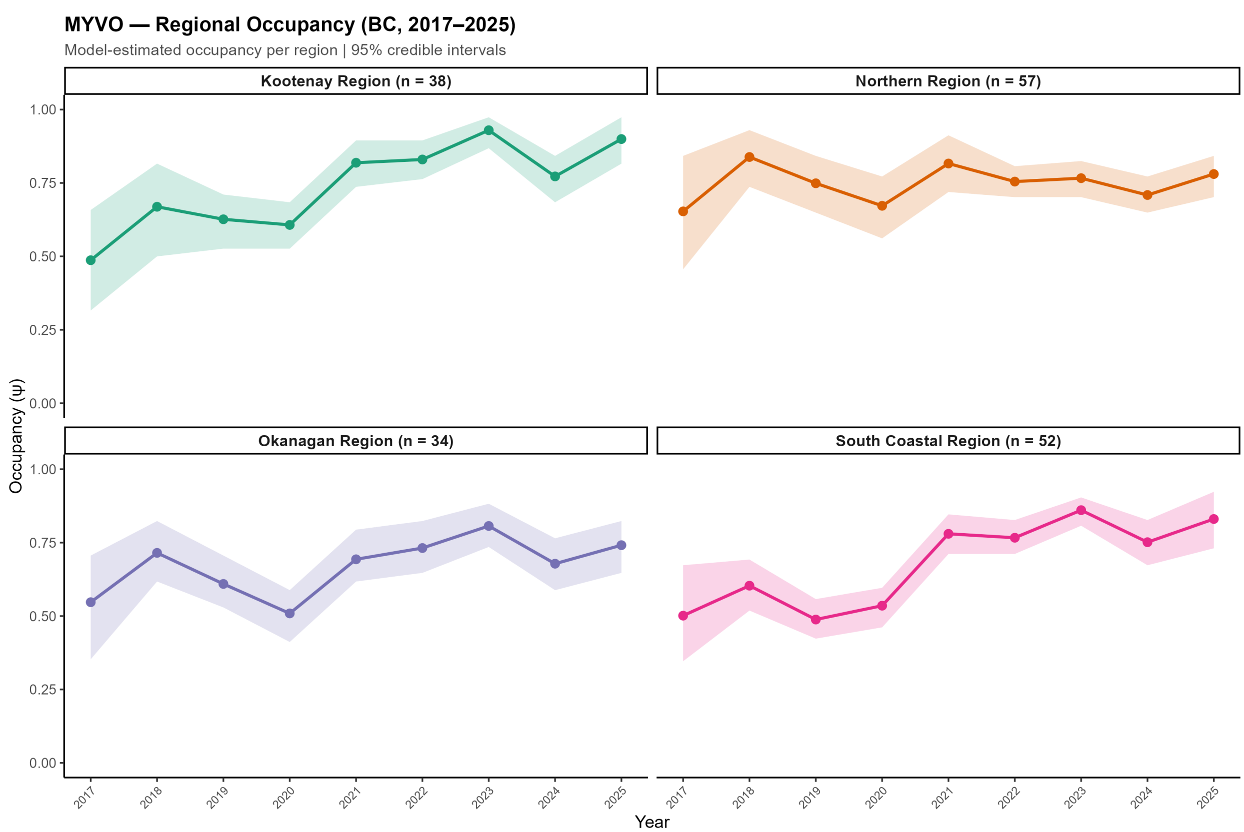

PAHE — Occupancy Trend (BC, 2017–2025)
λ = 1.44 (0.39, 4.30) | Mean p = 0.01
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

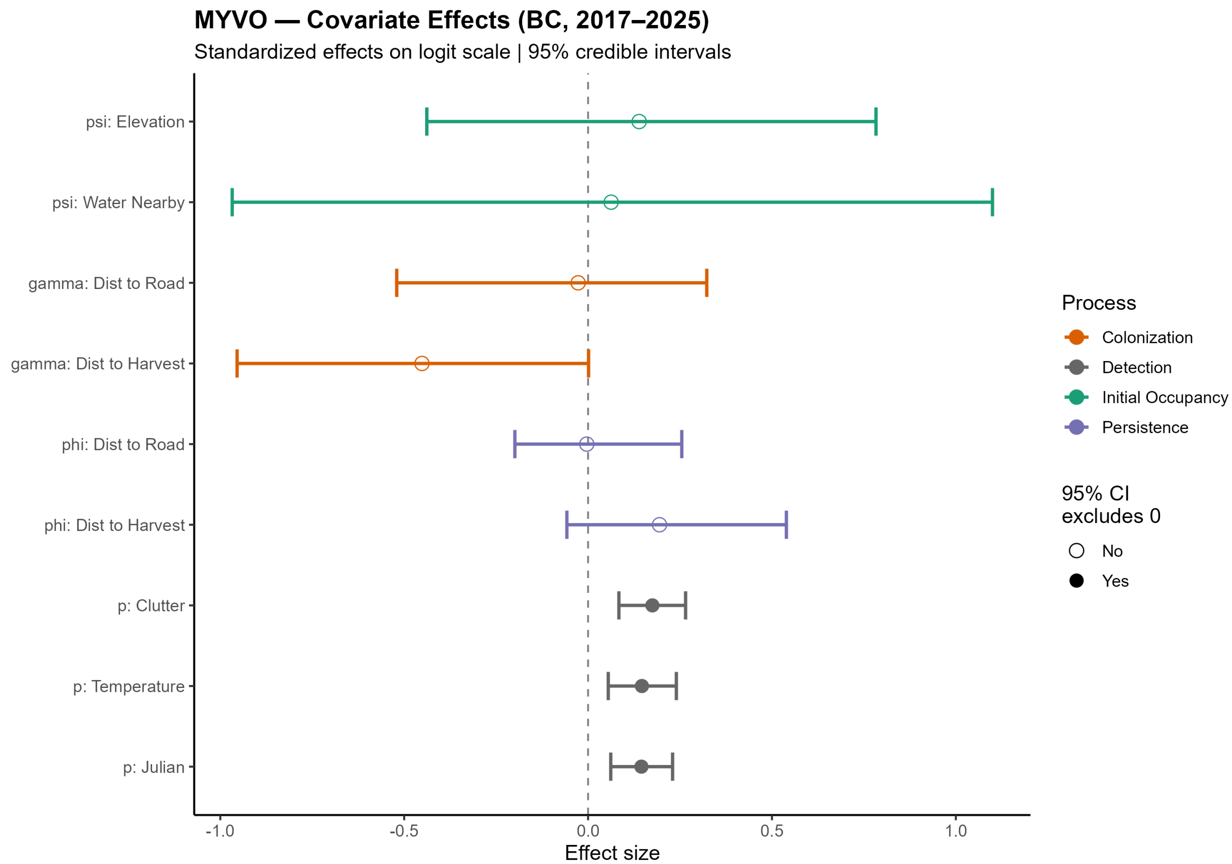

MYYU — Regional Occupancy (BC, 2017–2025)
Model-estimated occupancy per region | 95% credible intervals
Kootenay Region (n = 38)
Northern Region (n = 57)
Okanagan Region (n = 34)
South Coastal Region (n = 52)
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year


MYYU — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-3
-2
-1
0
1
2
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI excludes 0
No
Yes

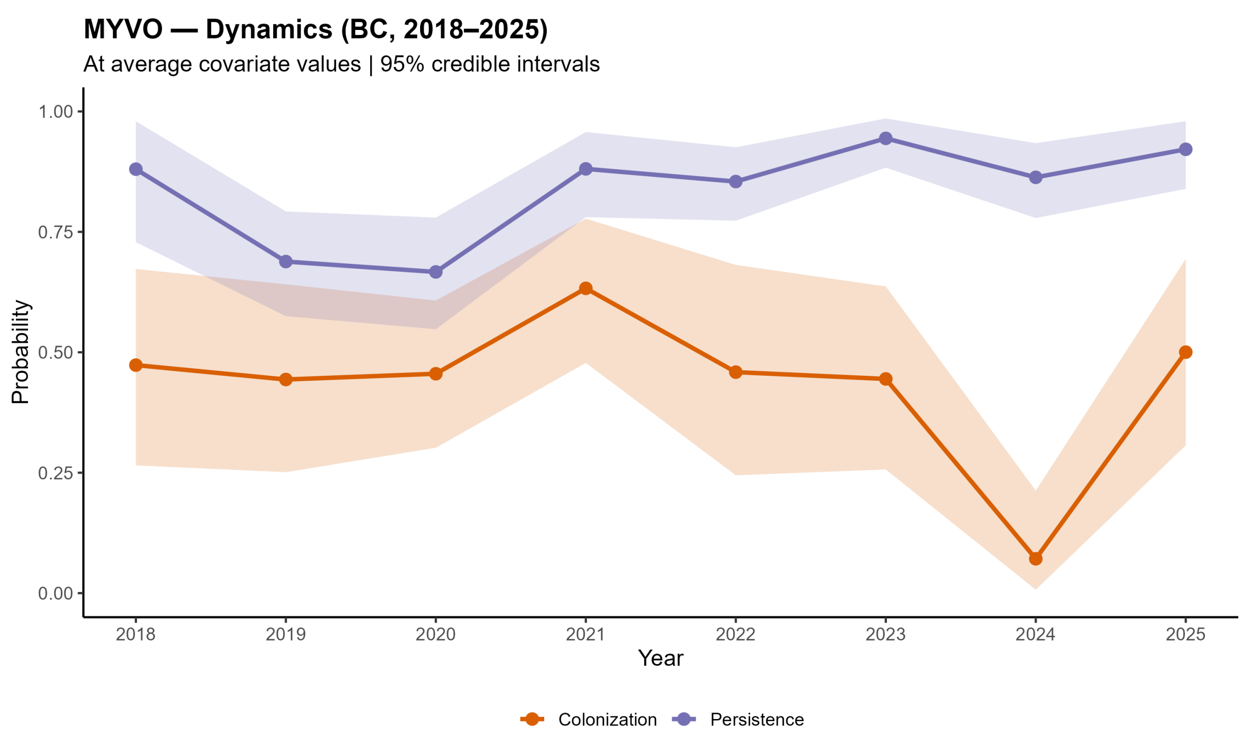

MYYU — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence


MYYU — Occupancy Trend (BC, 2017–2025)
λ = 1.39 (1.14, 1.71) | Mean p = 0.64
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year


MYVO — Regional Occupancy (BC, 2017–2025)
Model-estimated occupancy per region | 95% credible intervals
Kootenay Region (n = 38)
Northern Region (n = 57)
Okanagan Region (n = 34)
South Coastal Region (n = 52)
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year



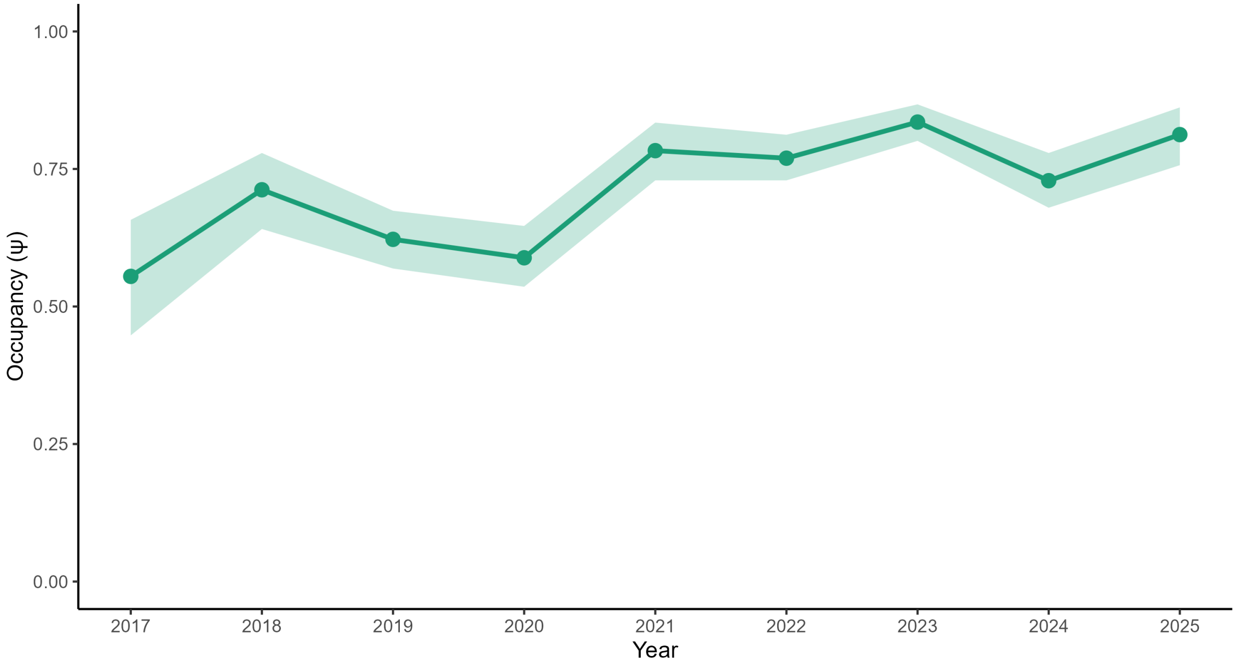
MYVO — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-1.0
-0.5
0.0
0.5
1.0
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI excludes 0
No
Yes



MYVO — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
1.00
0.75
0.50
0.25
0.00
Probability
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence

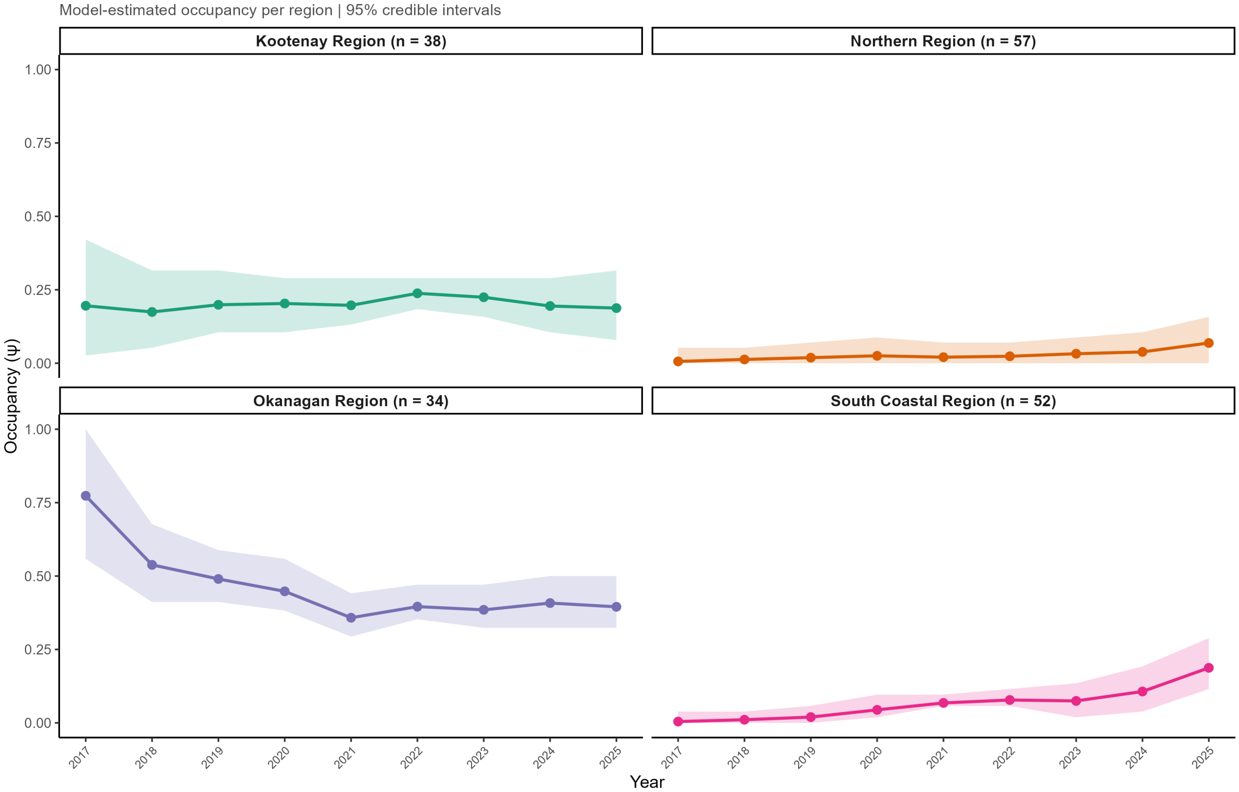
MYVO — Occupancy Trend (BC, 2017–2025)

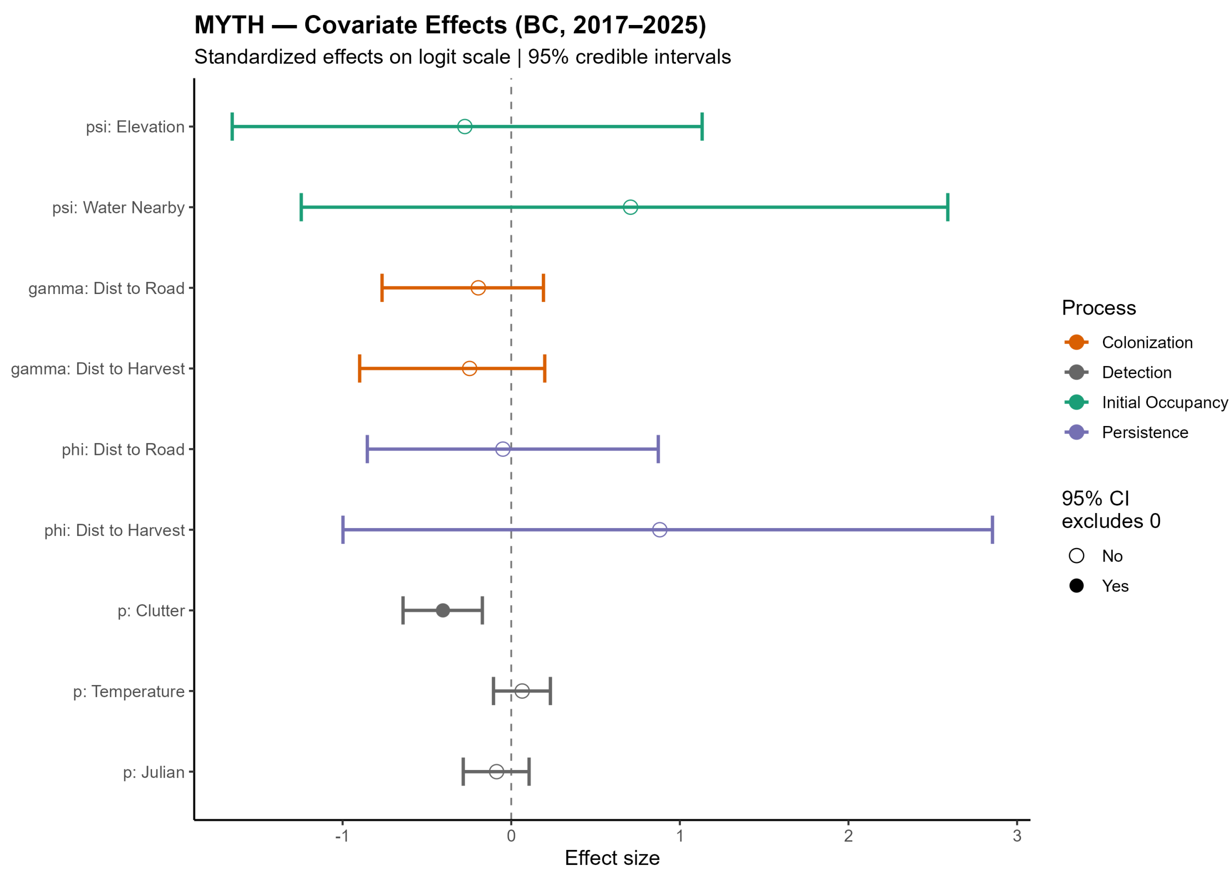
λ = 1.48 (1.22, 1.84) | Mean p = 0.61

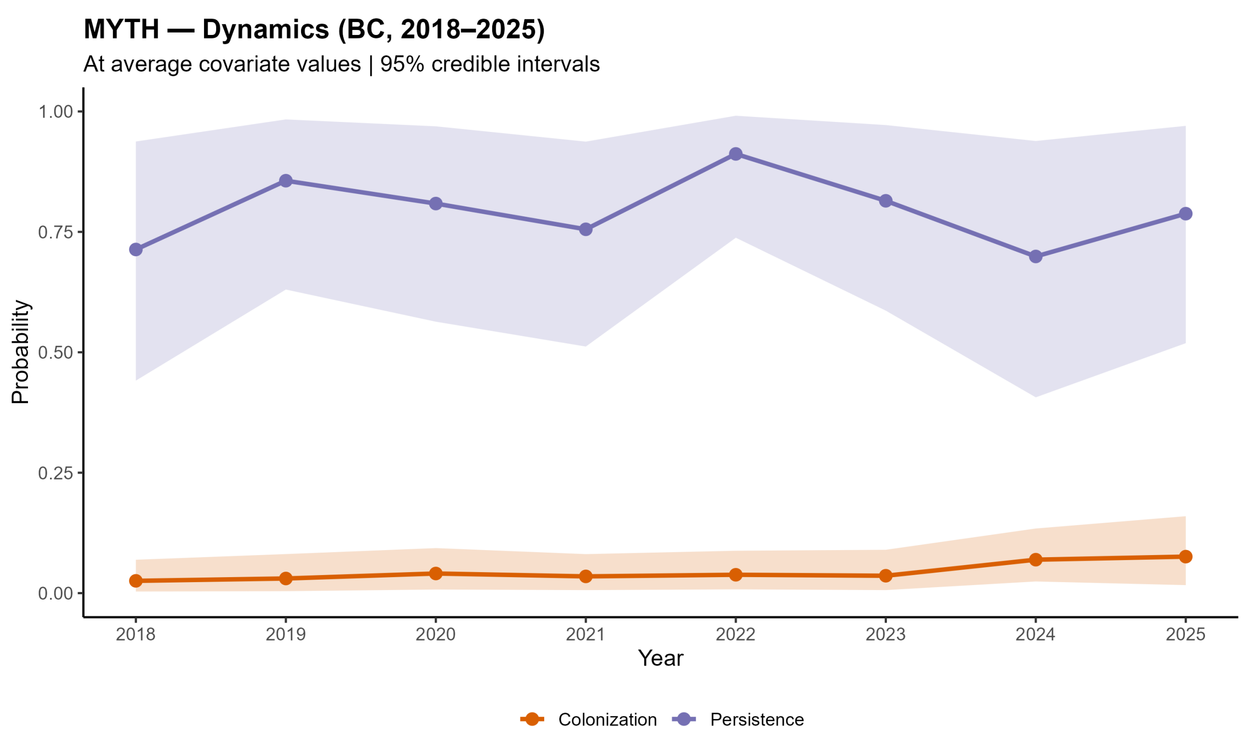


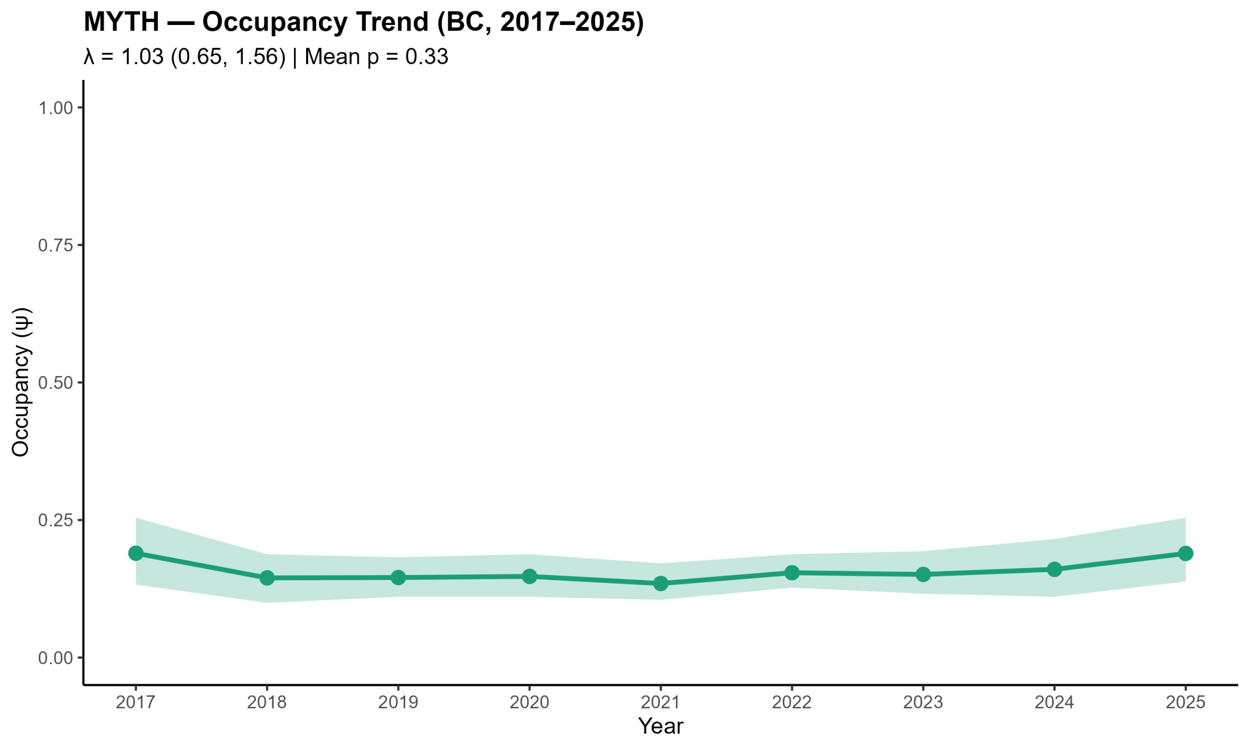
MYTH — Regional Occupancy (BC, 2017–2025)

Model-estimated occupancy per region | 95% credible intervals

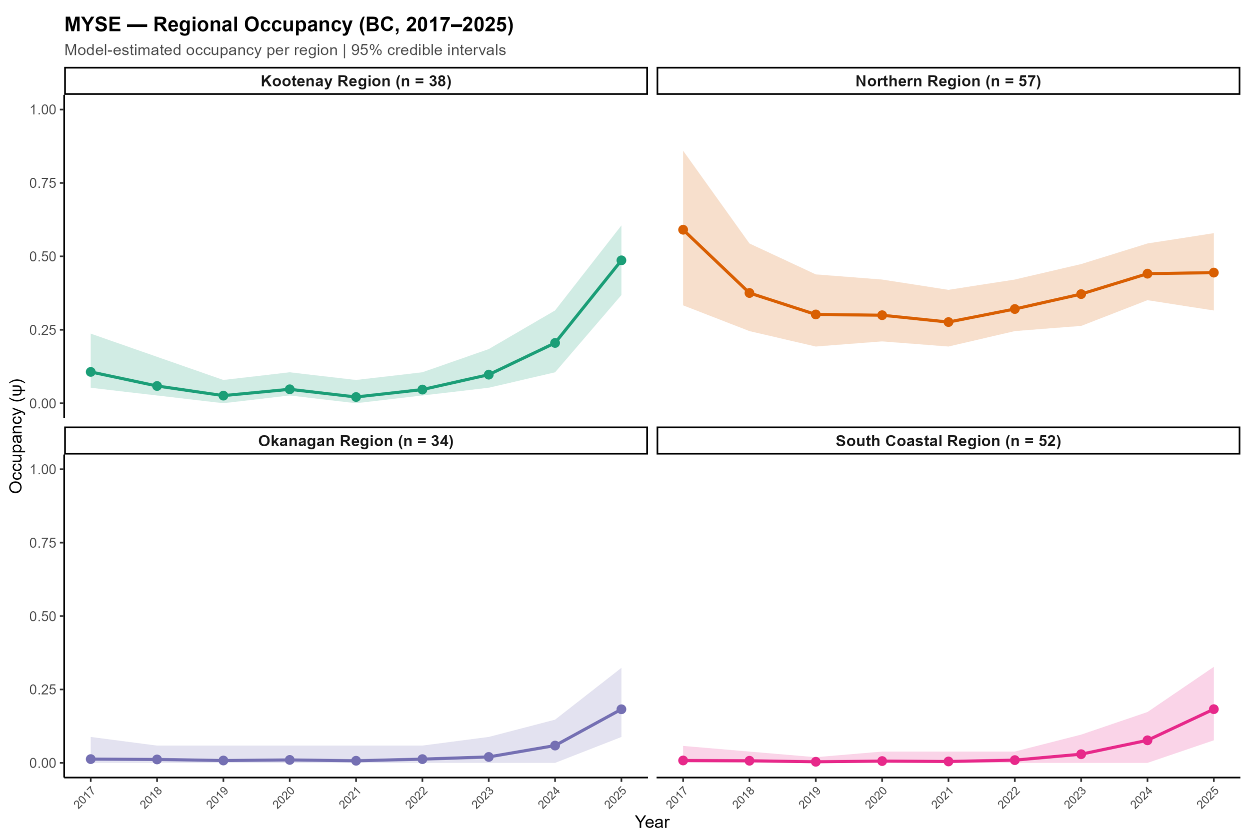



MYTH — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-1
0
1
2
3
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI excludes 0
No
Yes

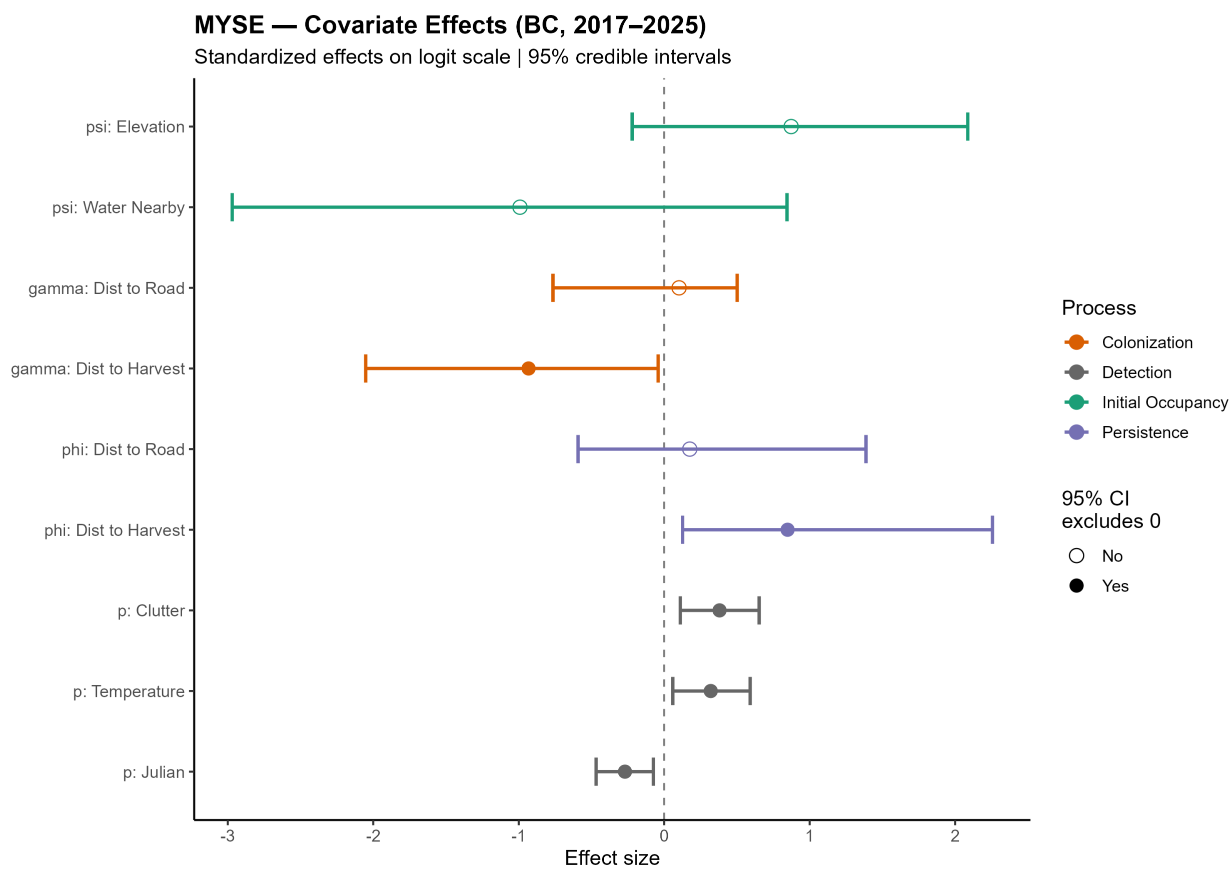

MYTH — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence

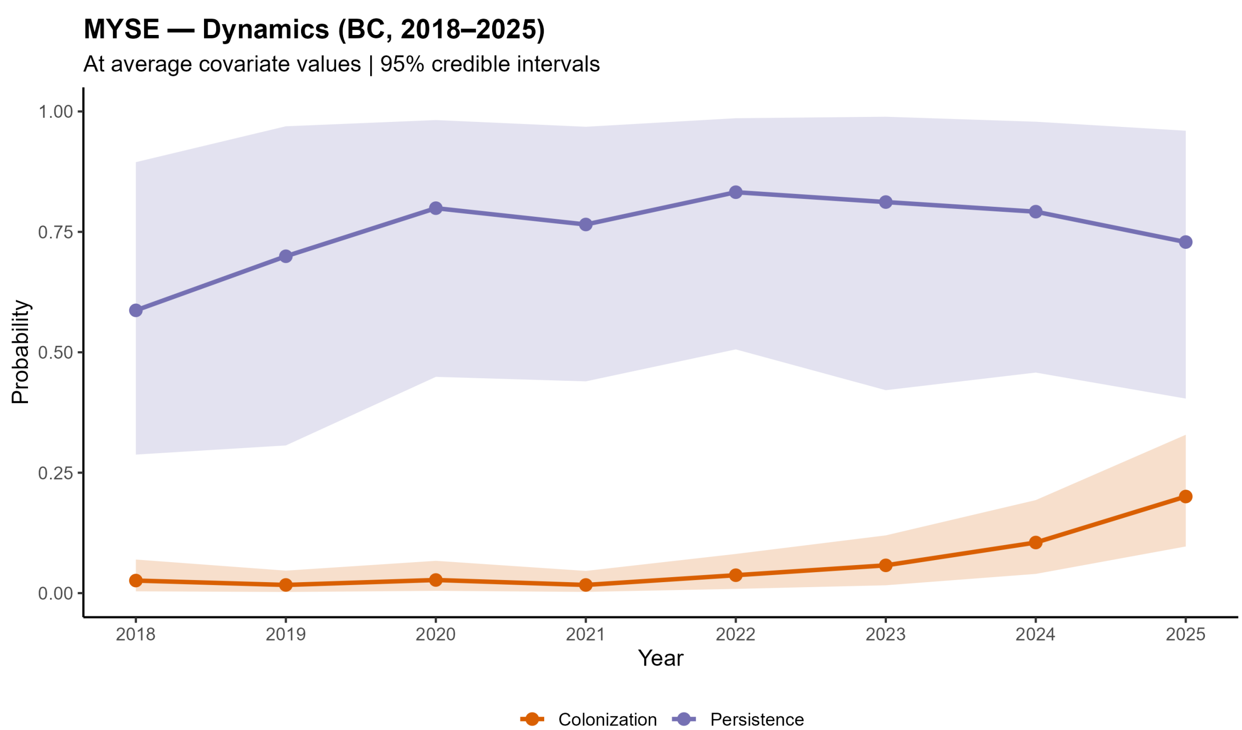


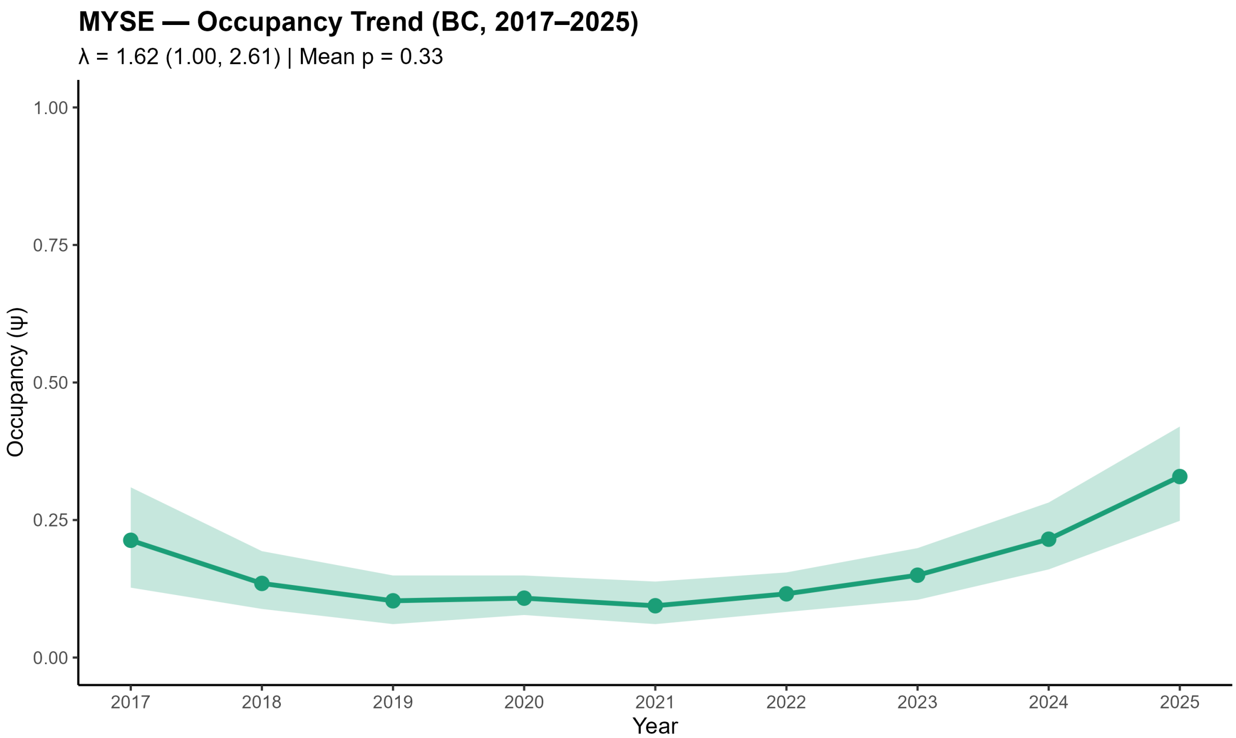
MYTH — Occupancy Trend (BC, 2017–2025)
λ = 1.03 (0.65, 1.56) | Mean p = 0.33
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

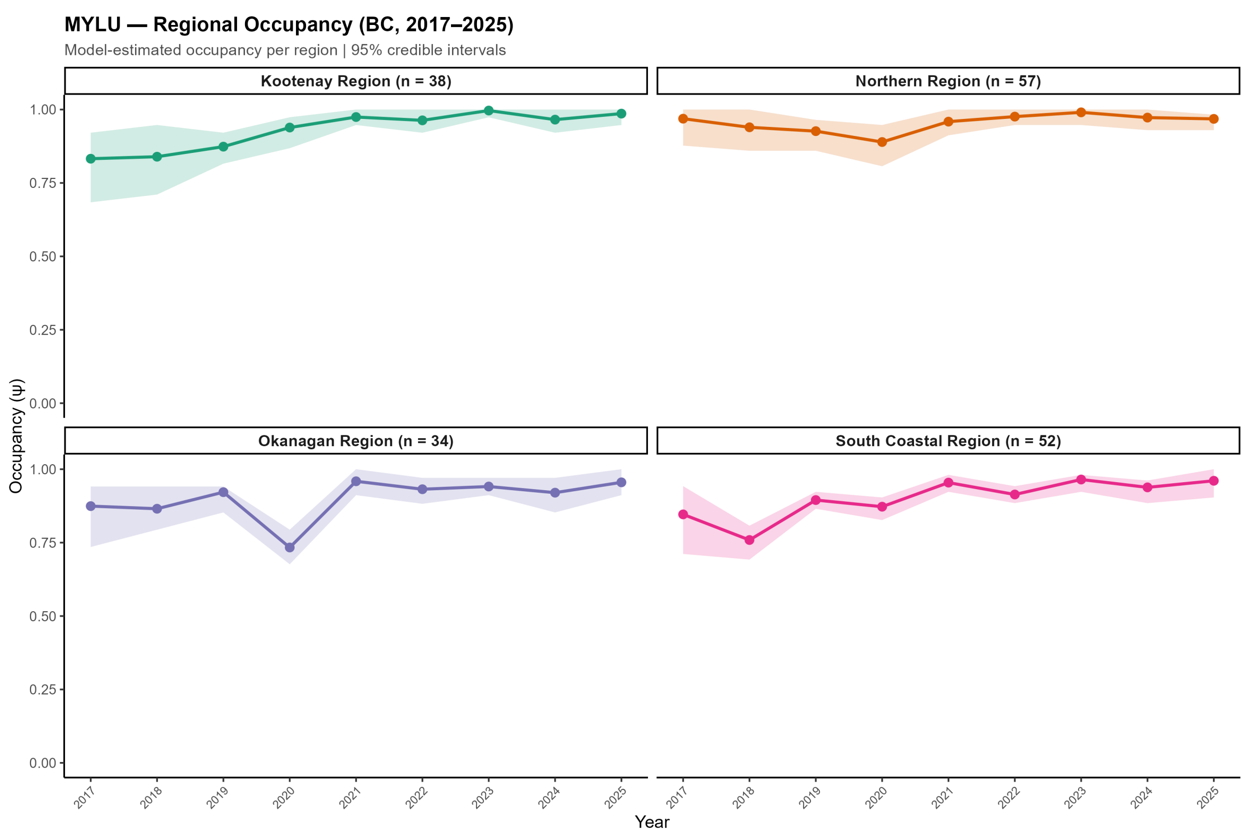


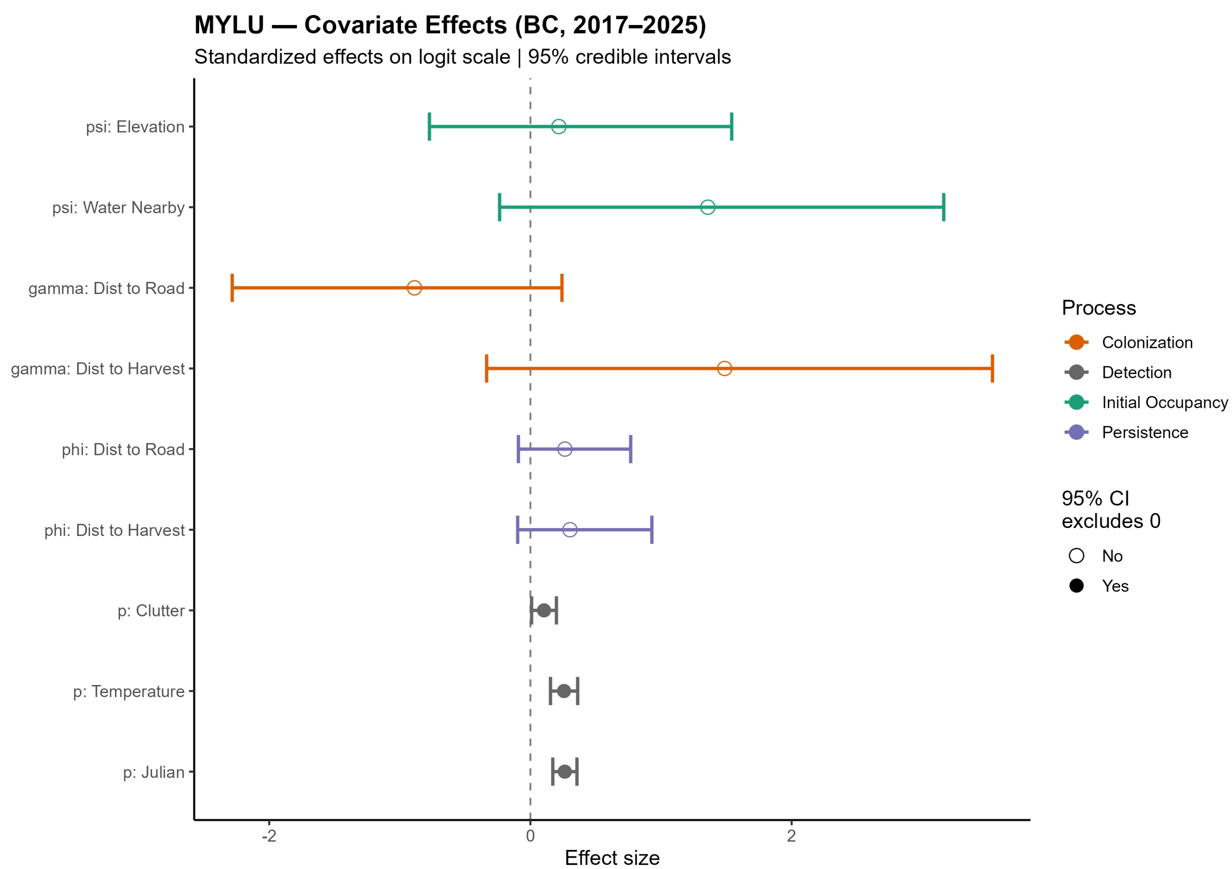
MYSE — Regional Occupancy (BC, 2017–2025)
Model-estimated occupancy per region | 95% credible intervals
Kootenay Region (n = 38)
Northern Region (n = 57)
Okanagan Region (n = 34)
South Coastal Region (n = 52)
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

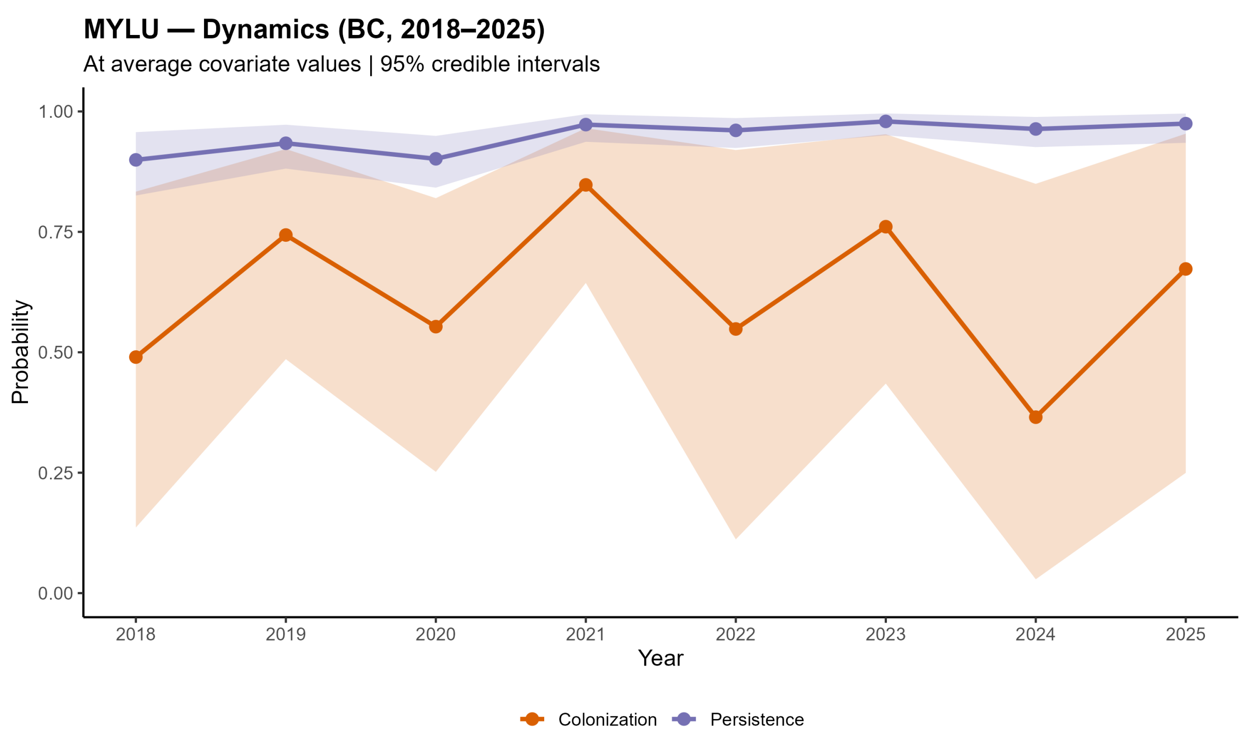

MYSE — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-3
-2
-1
0
1
2
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI
excludes 0
No
Yes


MYSE — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence


MYSE — Occupancy Trend (BC, 2017–2025)
λ = 1.62 (1.00, 2.61) | Mean p = 0.33
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

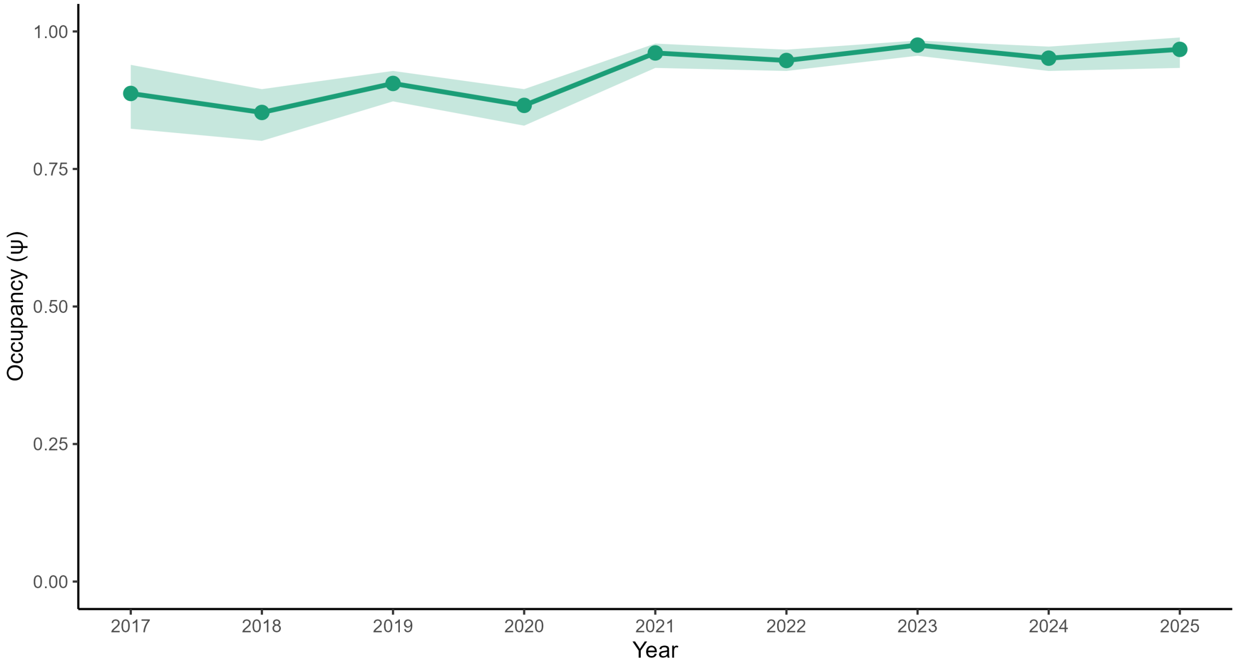

MYLU — Regional Occupancy (BC, 2017–2025)
Model-estimated occupancy per region | 95% credible intervals
Kootenay Region (n = 38)
Northern Region (n = 57)
Okanagan Region (n = 34)
South Coastal Region (n = 52)
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year


MYLU — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-2
0
2
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI excludes 0
No
Yes


MYLU — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence

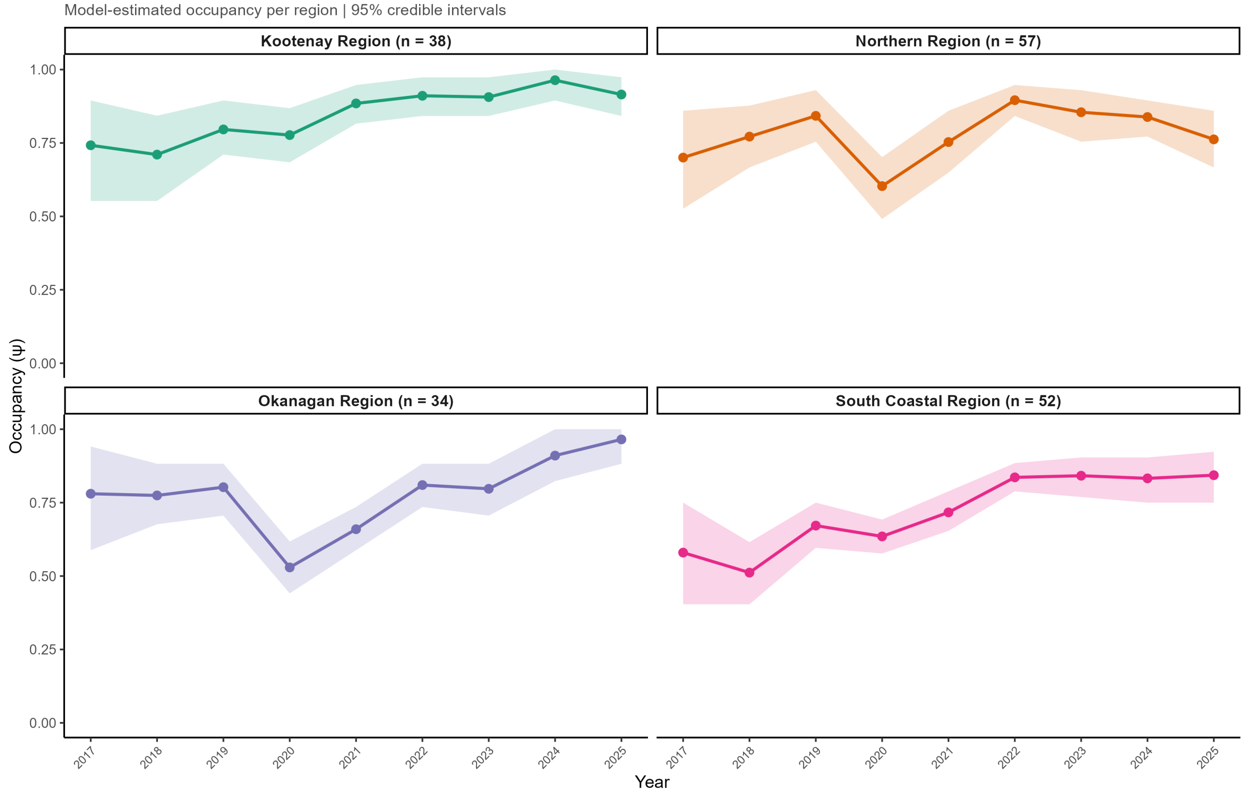
MYLU — Occupancy Trend (BC, 2017–2025)

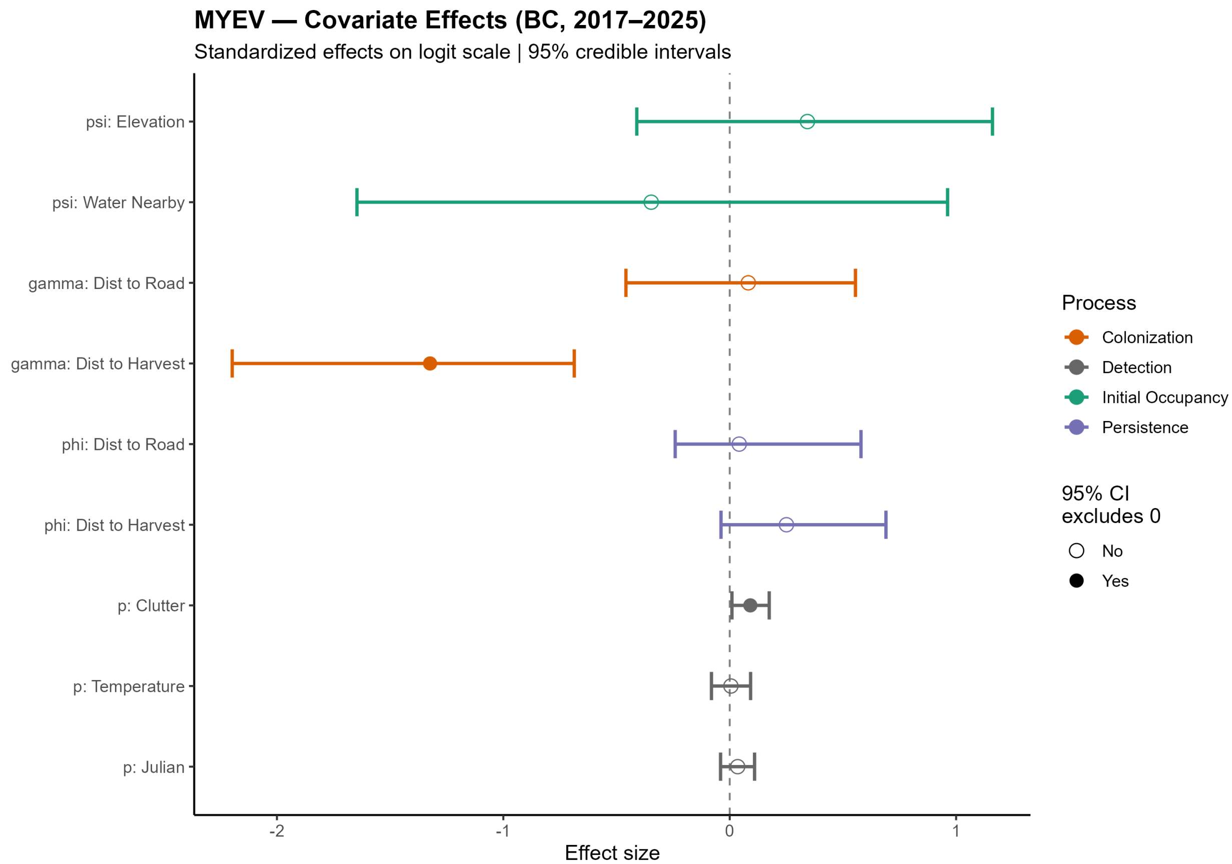
λ = 1.09 (1.02, 1.18) | Mean p = 0.82

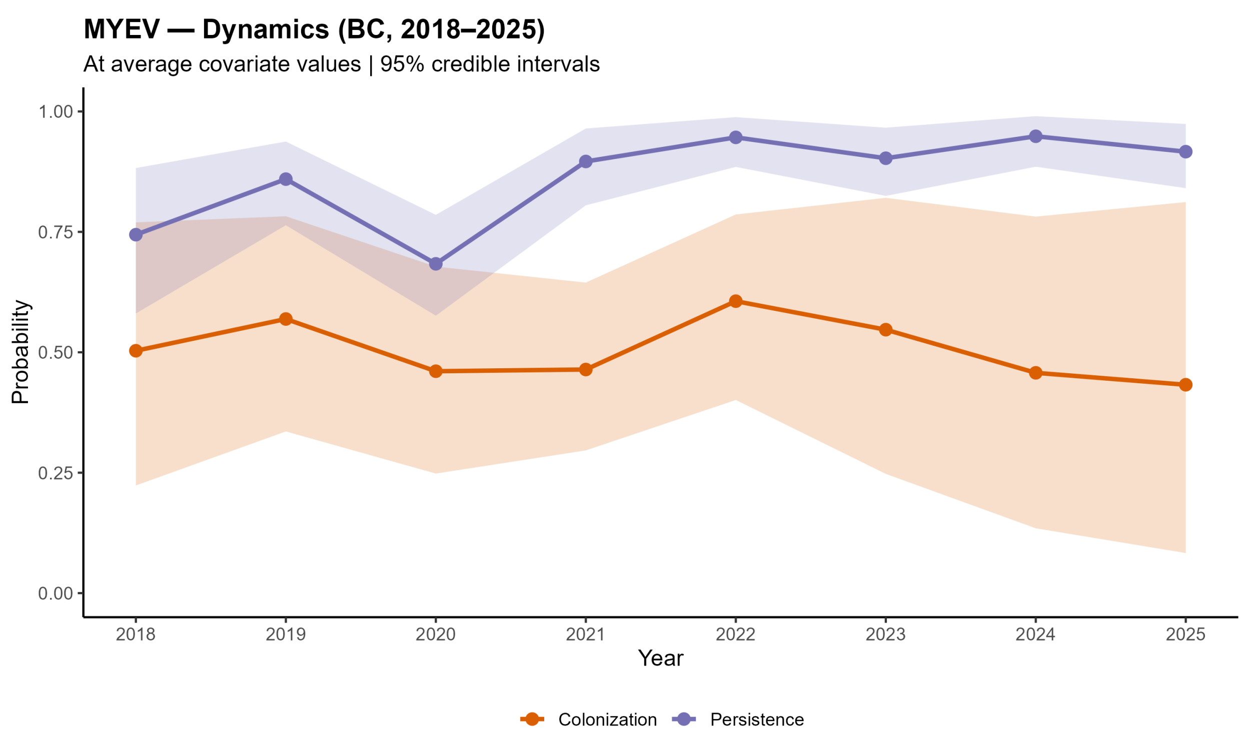


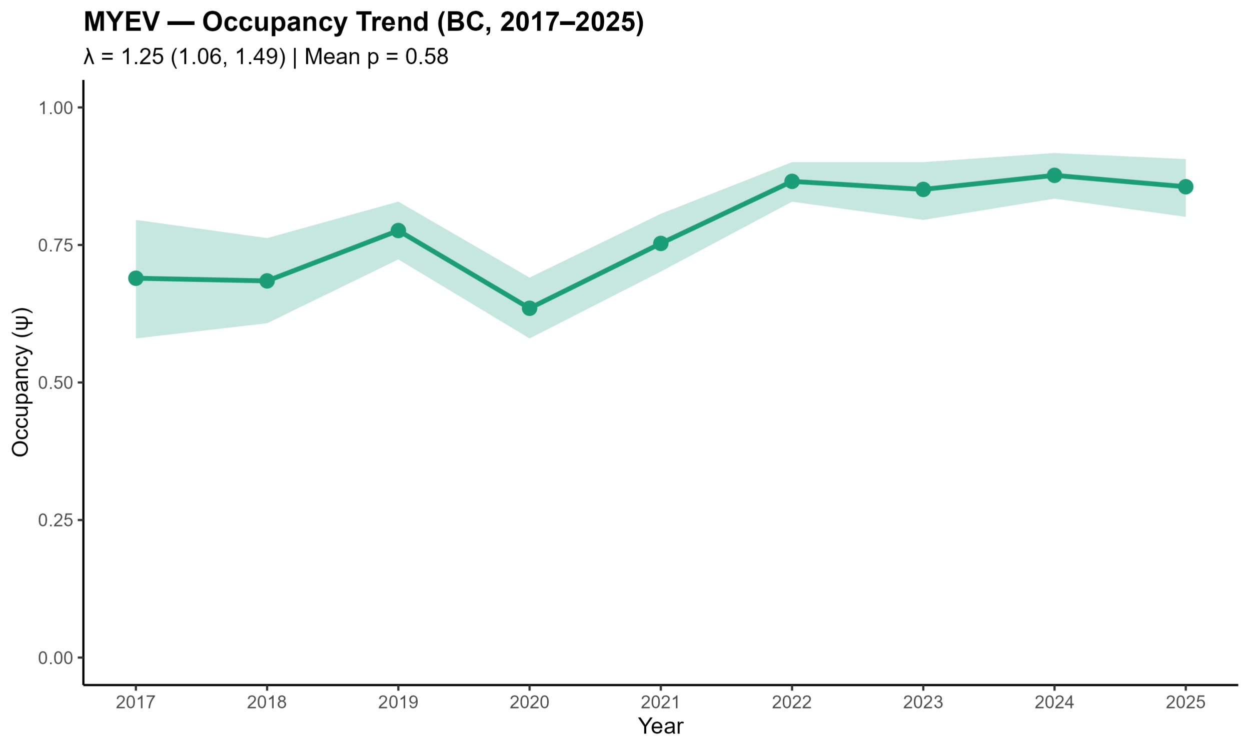
MYEV — Regional Occupancy (BC, 2017–2025)

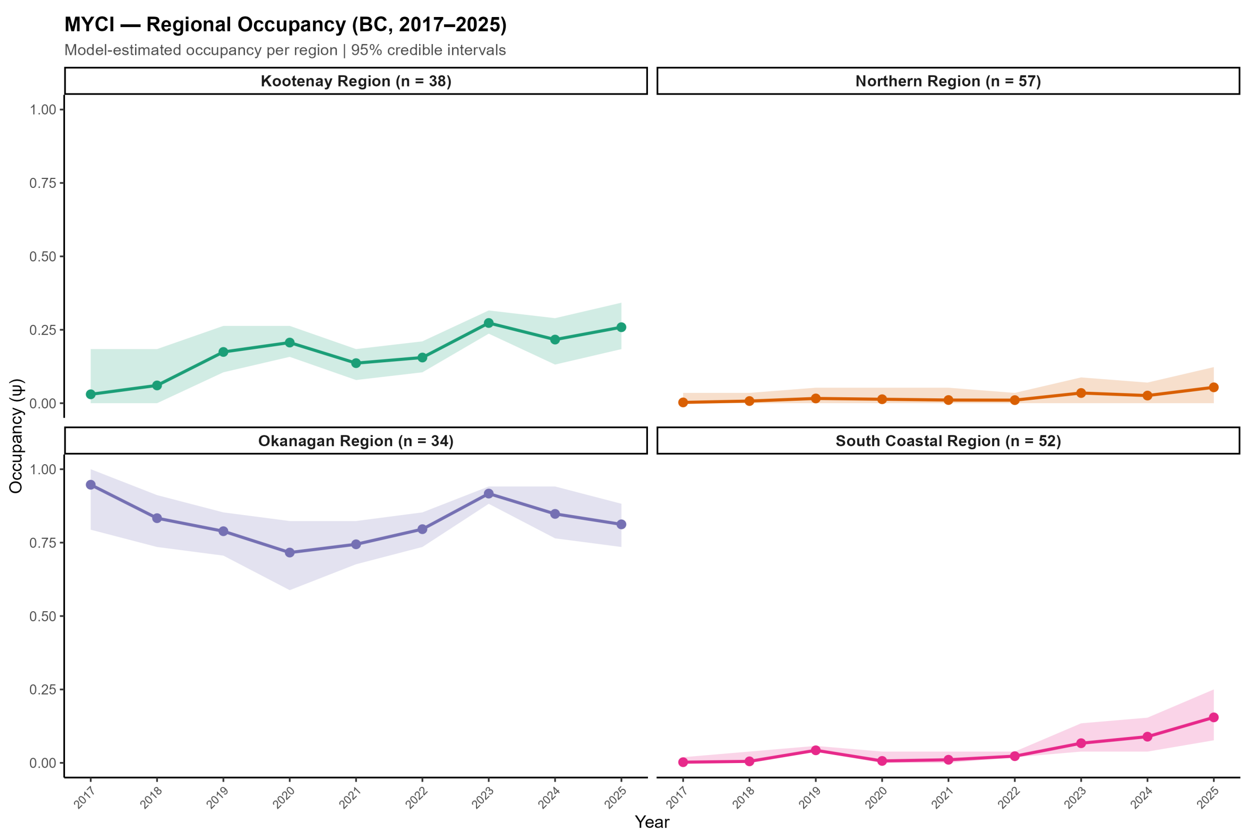
Model-estimated occupancy per region | 95% credible intervals

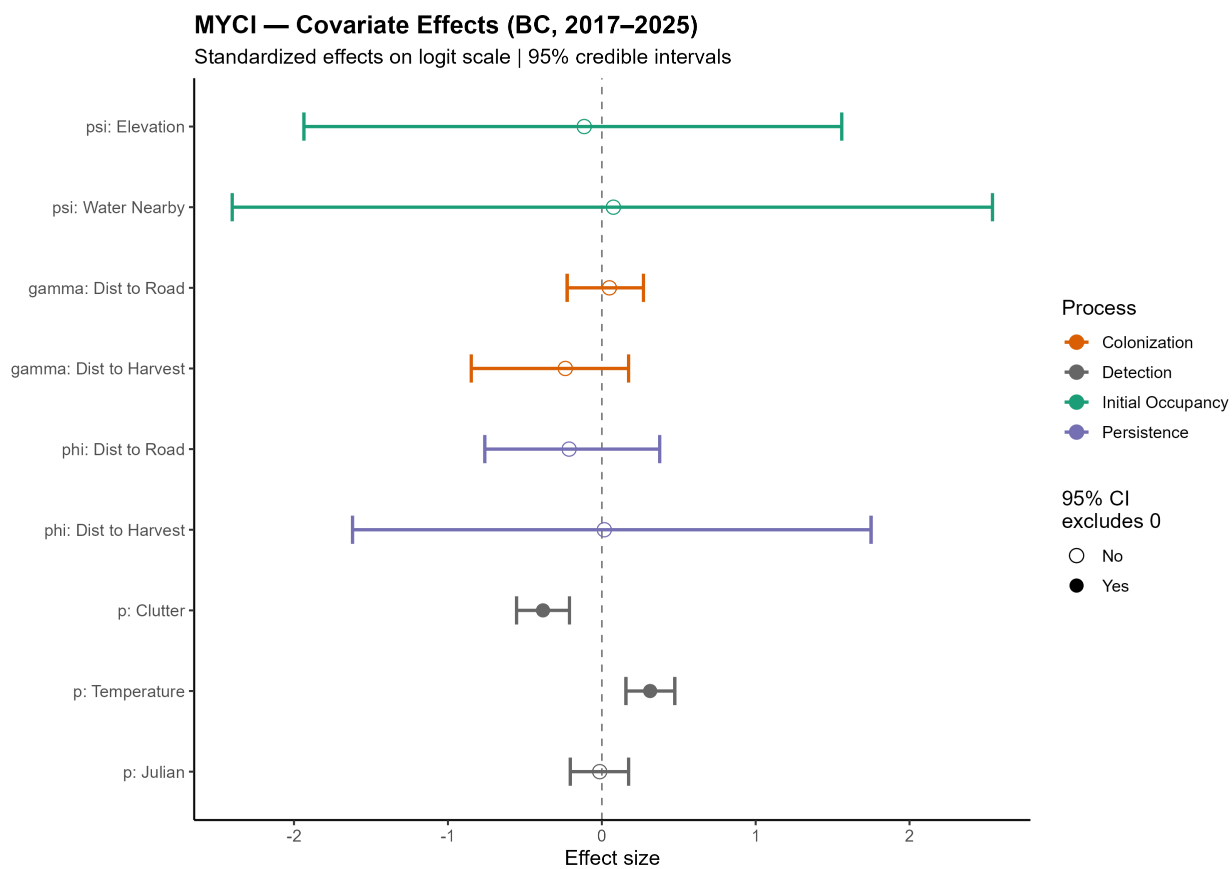


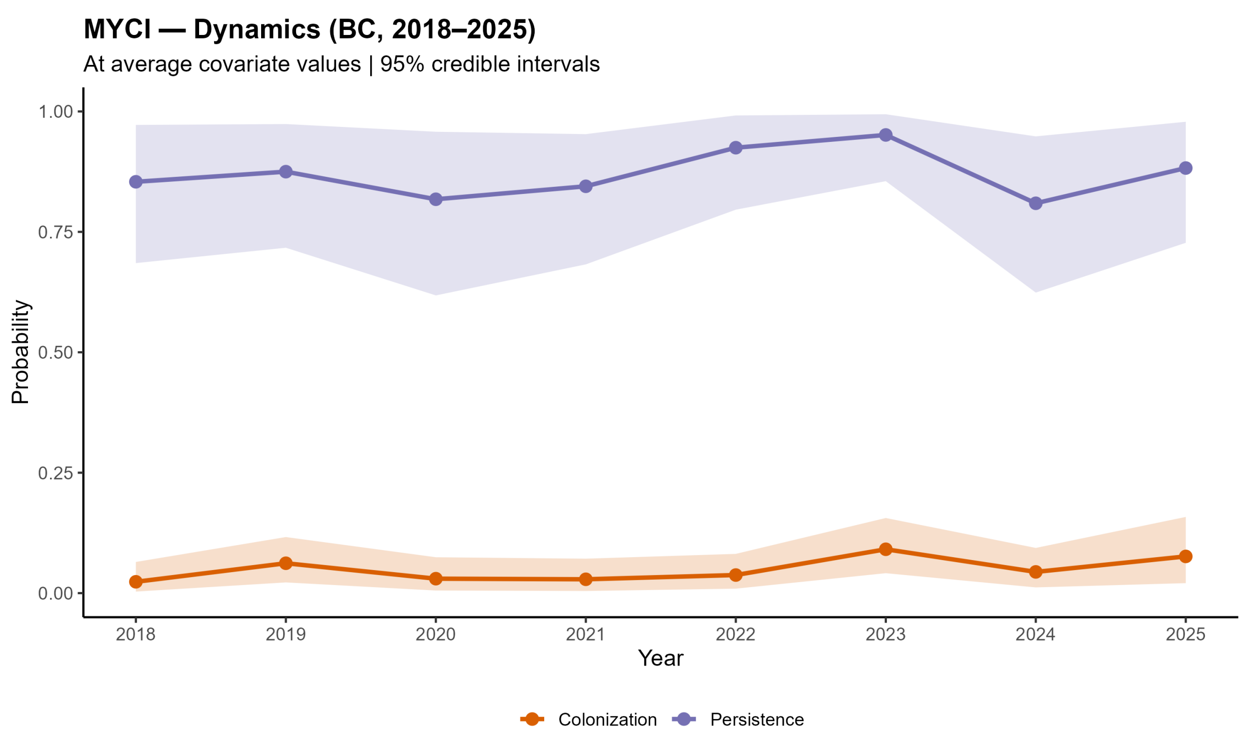

MYEV — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-2
-1
0
1
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI excludes 0
No
Yes

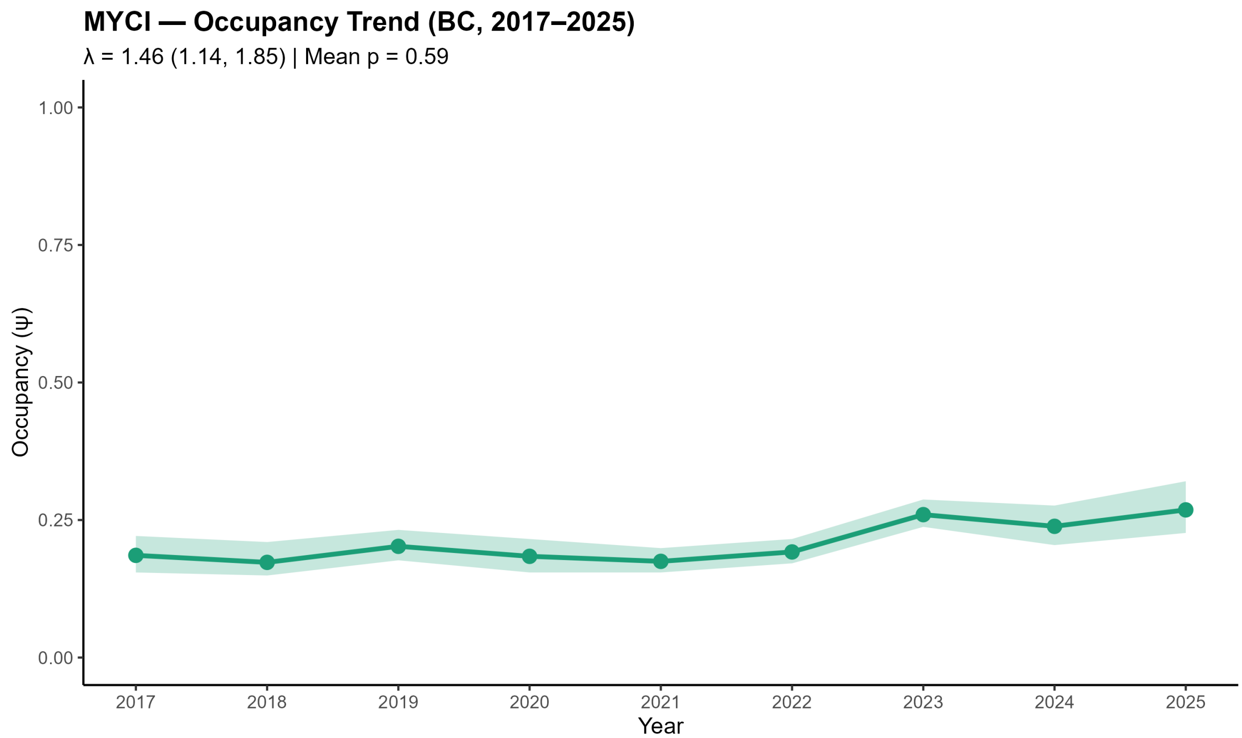

MYEV — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence


MYEV — Occupancy Trend (BC, 2017–2025)
λ = 1.25 (1.06, 1.49) | Mean p = 0.58
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

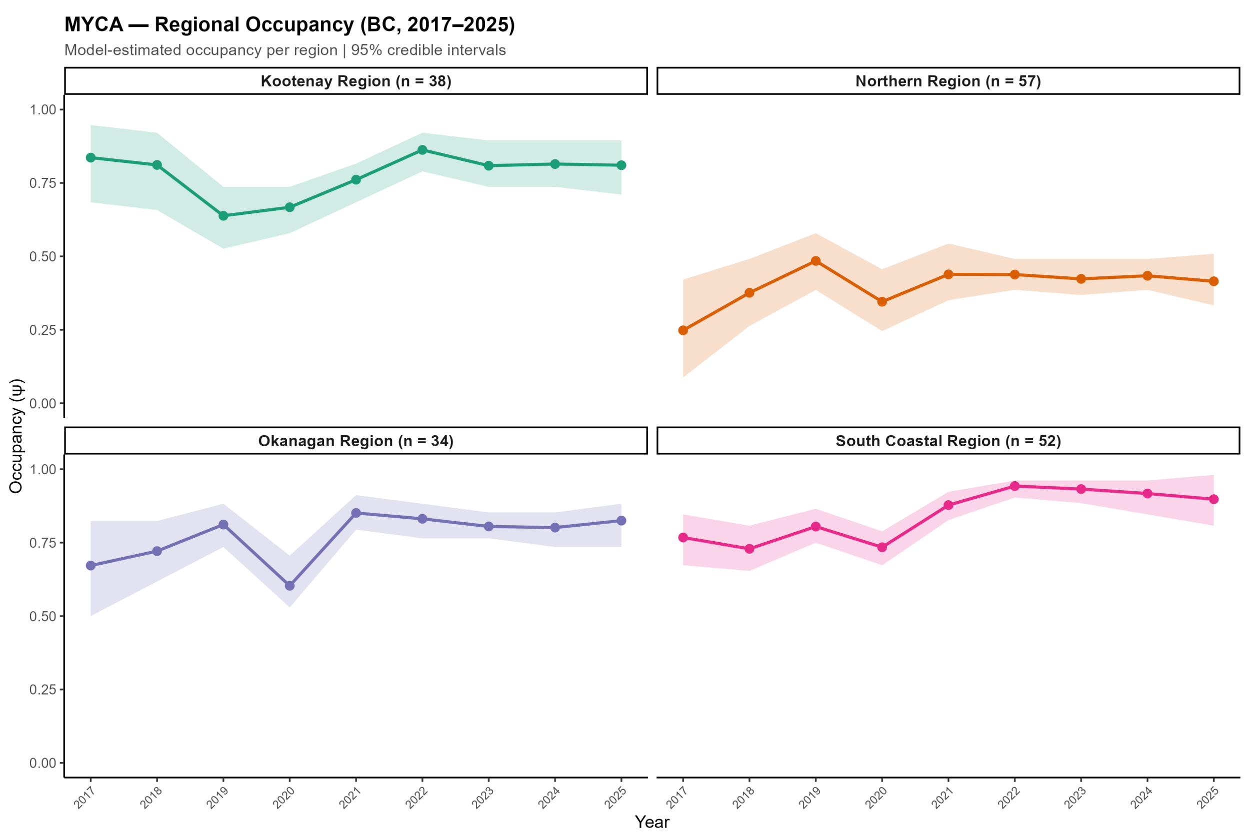

MYCI — Regional Occupancy (BC, 2017–2025)
Model-estimated occupancy per region | 95% credible intervals
Kootenay Region (n = 38)
Northern Region (n = 57)
Okanagan Region (n = 34)
South Coastal Region (n = 52)
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year


MYCI — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-2
-1
0
1
2
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI excludes 0
No
Yes

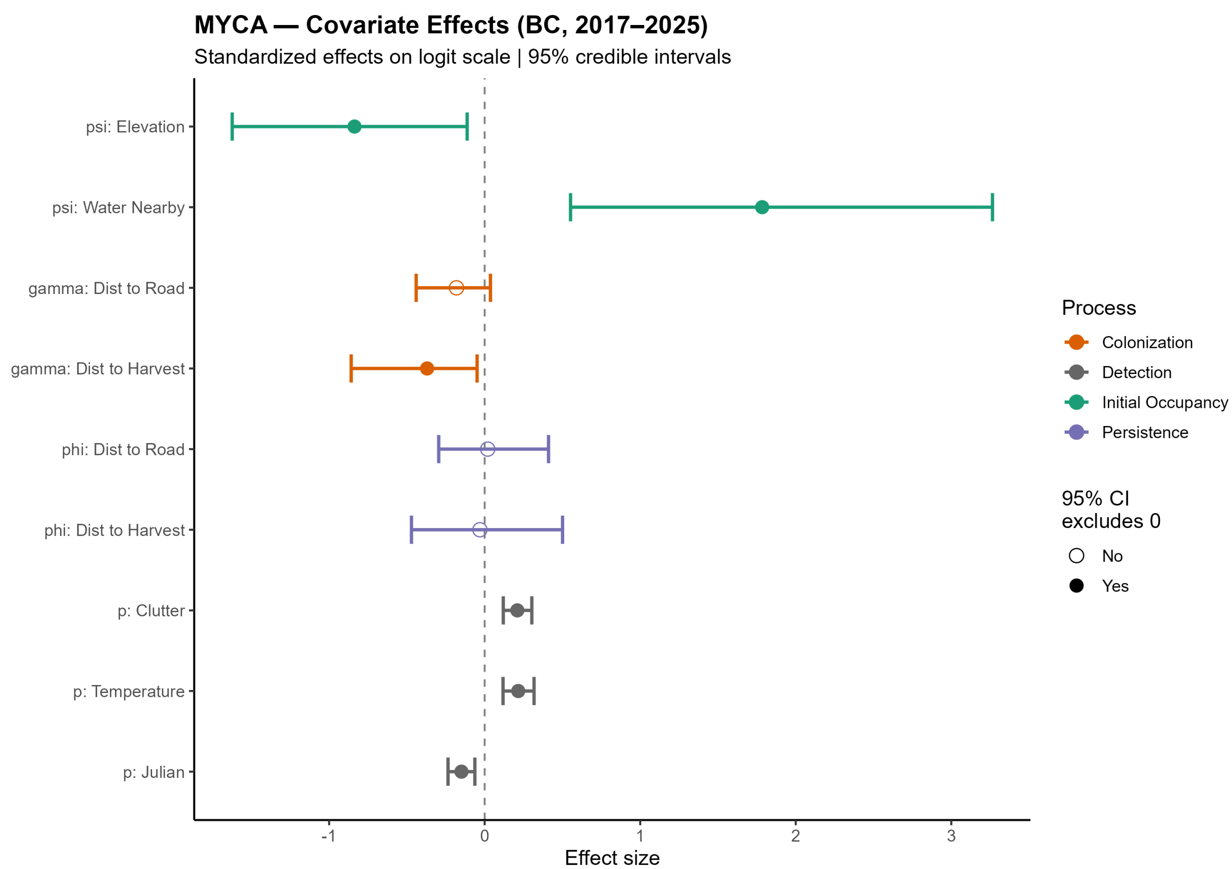

MYCI — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence



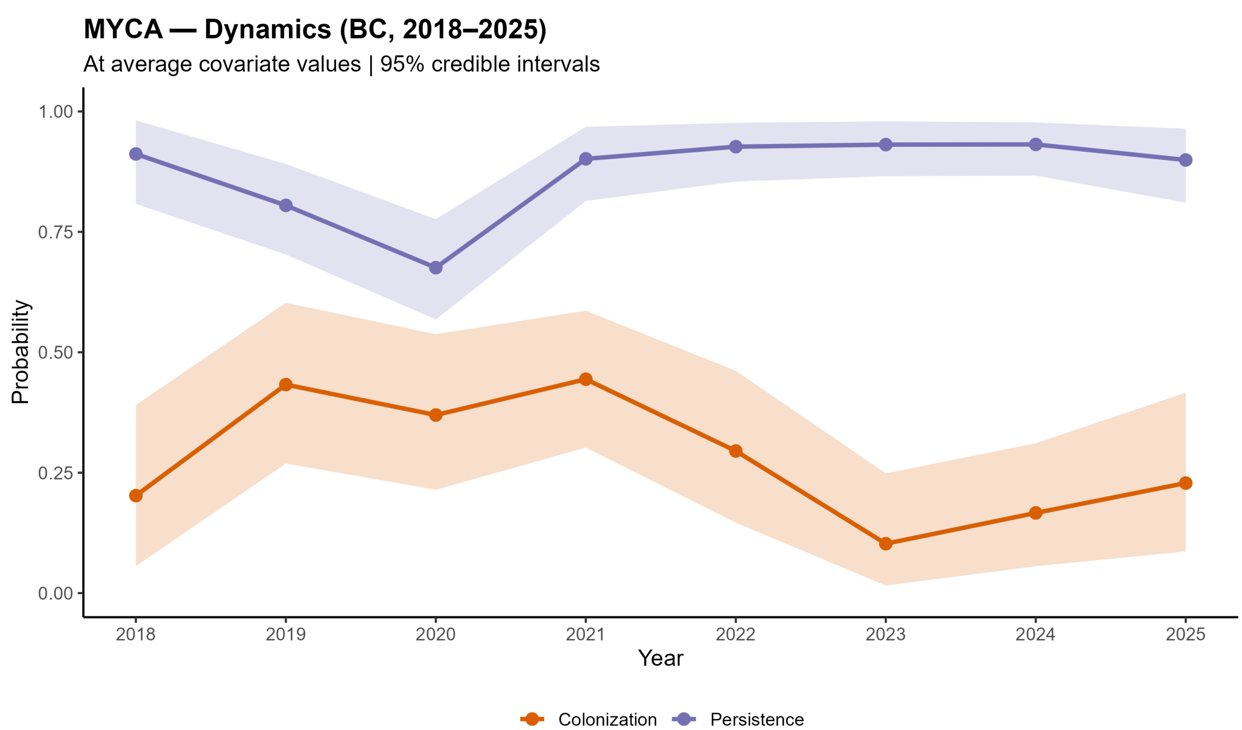
MYCI — Occupancy Trend (BC, 2017–2025)
λ = 1.46 (1.14, 1.85) | Mean p = 0.59
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year



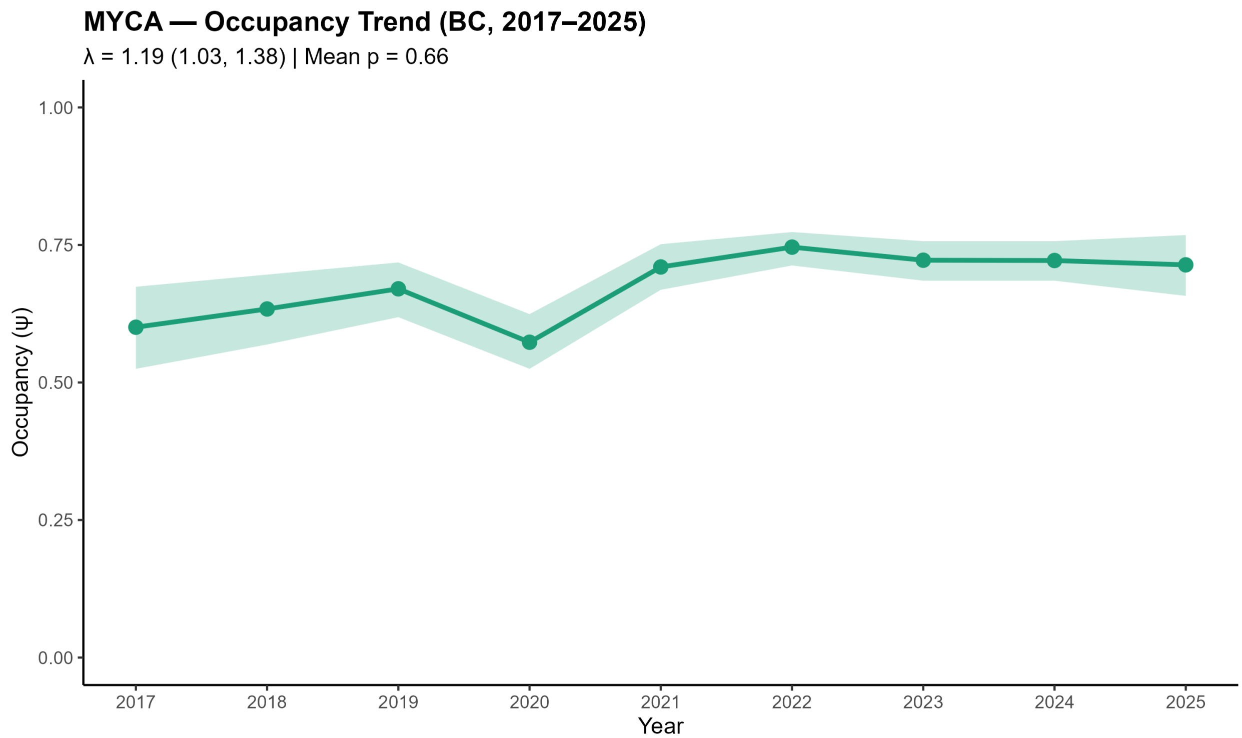
MYCA — Regional Occupancy (BC, 2017–2025)
Model-estimated occupancy per region | 95% credible intervals
Kootenay Region (n = 38)
Northern Region (n = 57)
Okanagan Region (n = 34)
South Coastal Region (n = 52)
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

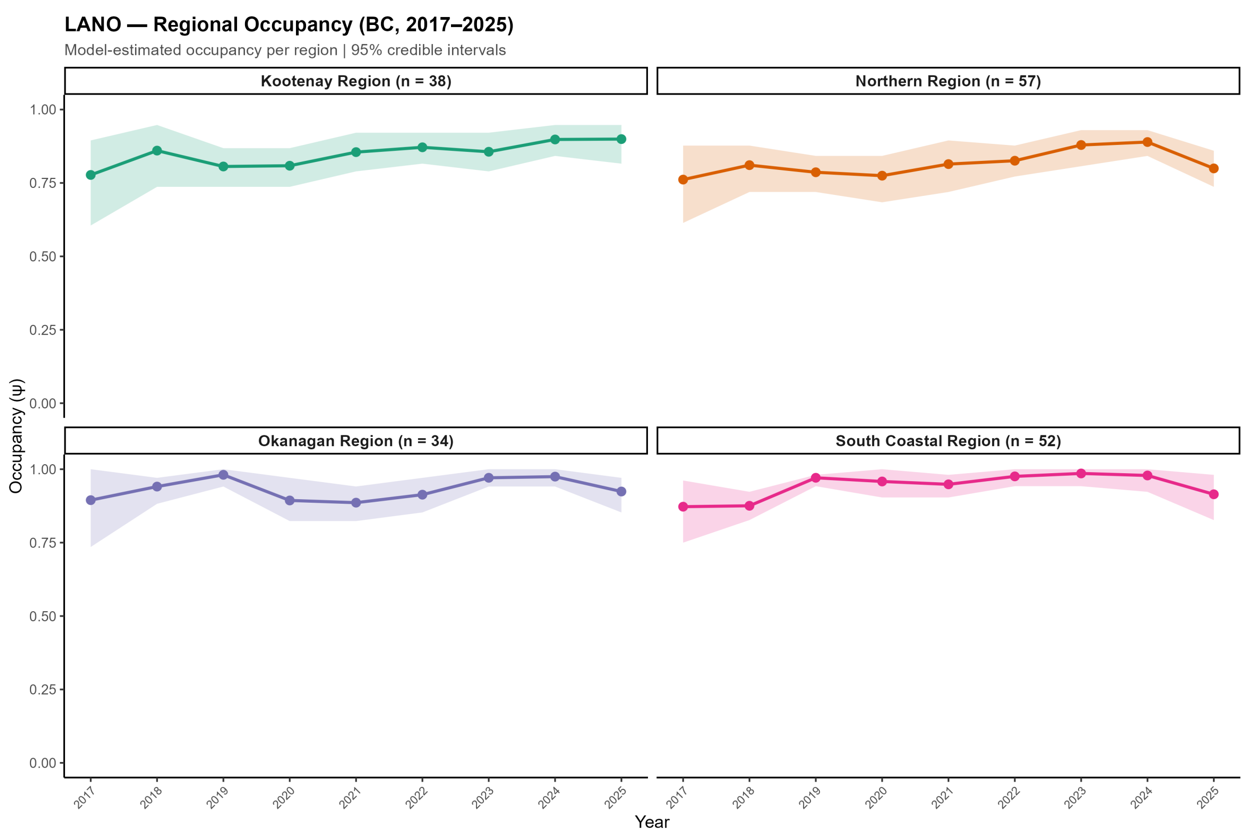


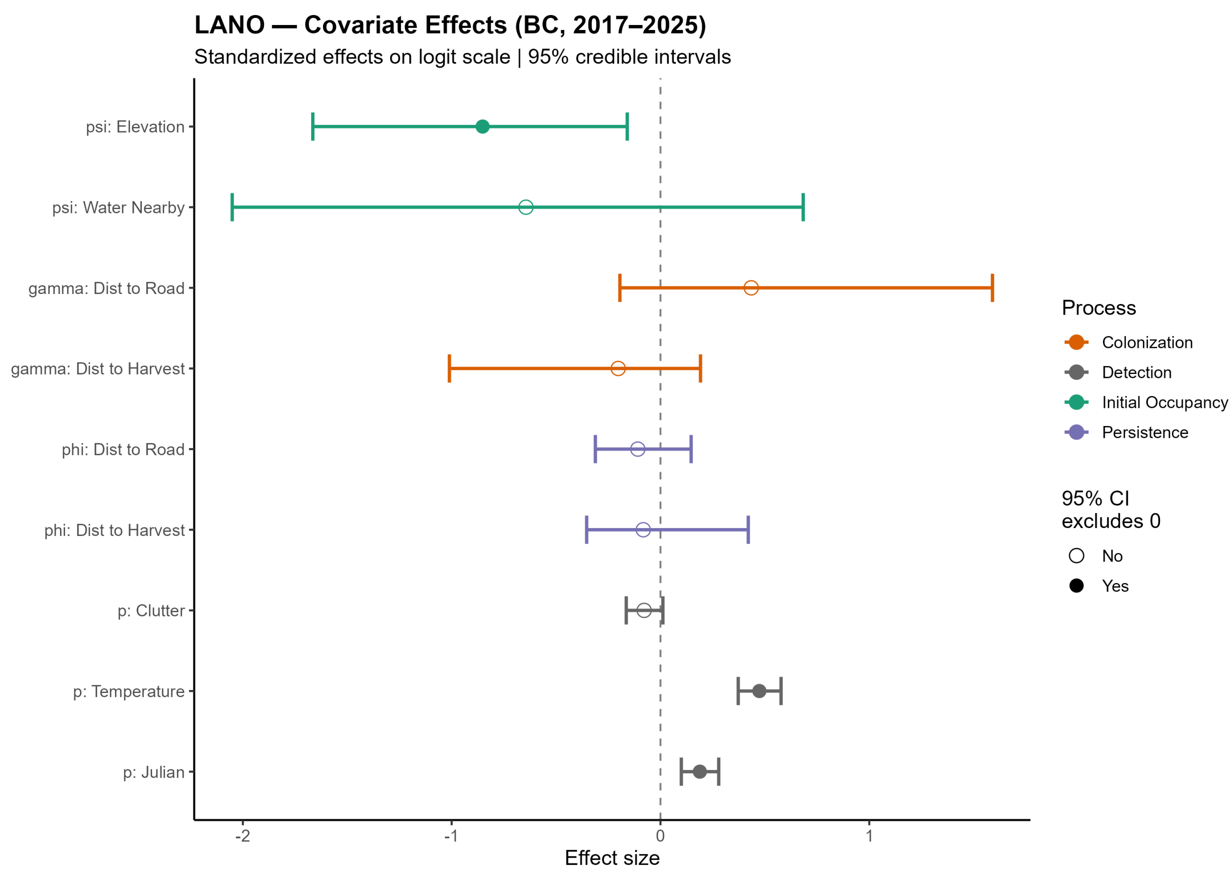
MYCA — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-1
0
1
2
3
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI
excludes 0
No
Yes

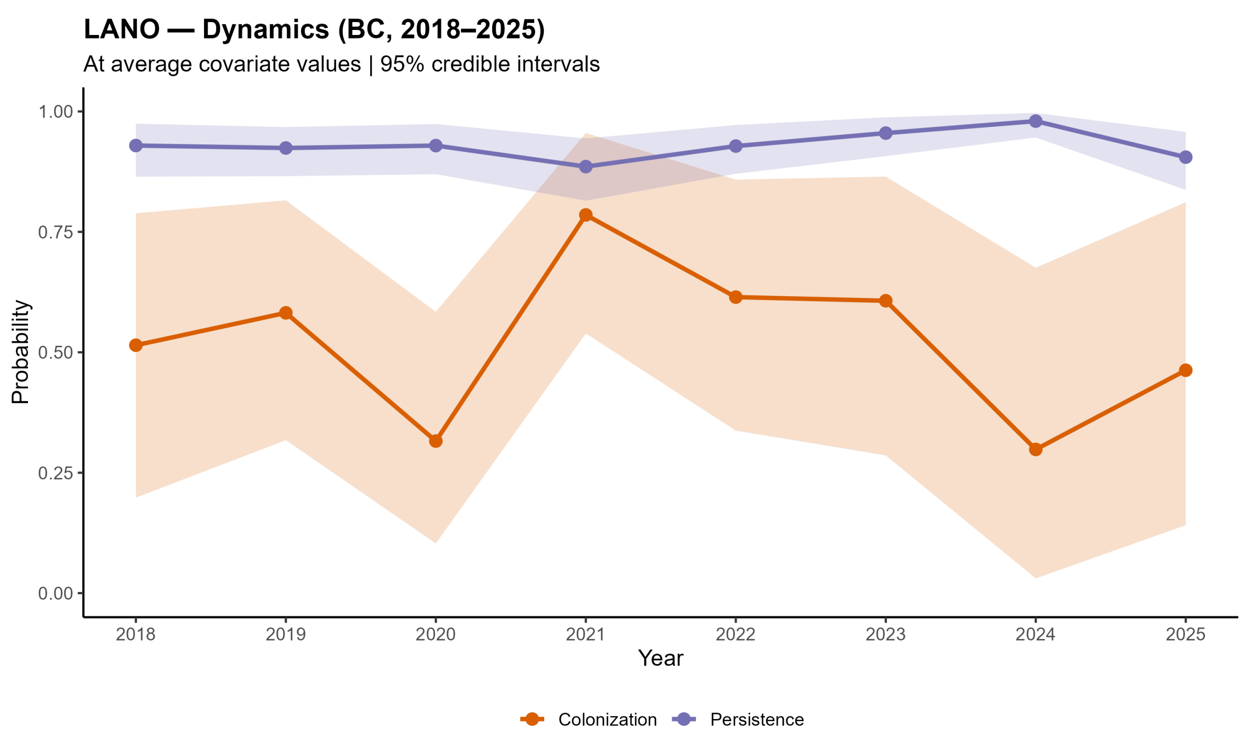


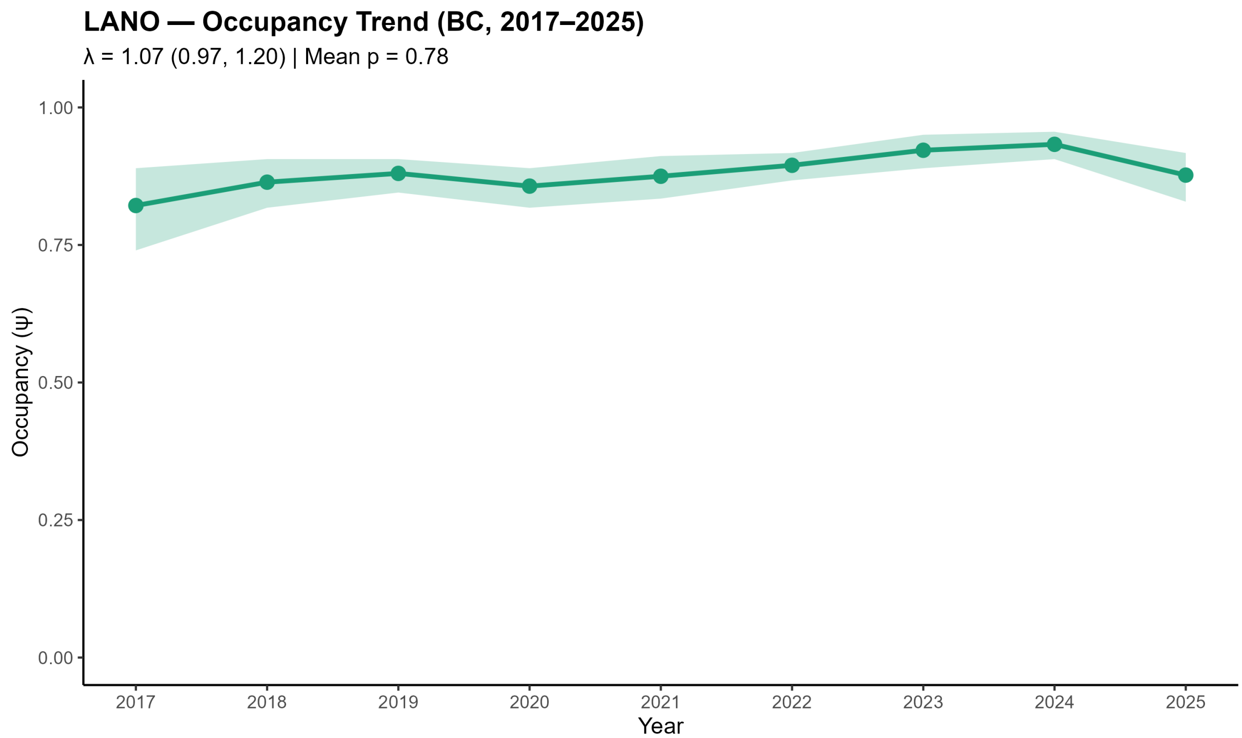
MYCA — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
1.00
0.75
0.50
0.25
0.00
Probability
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence

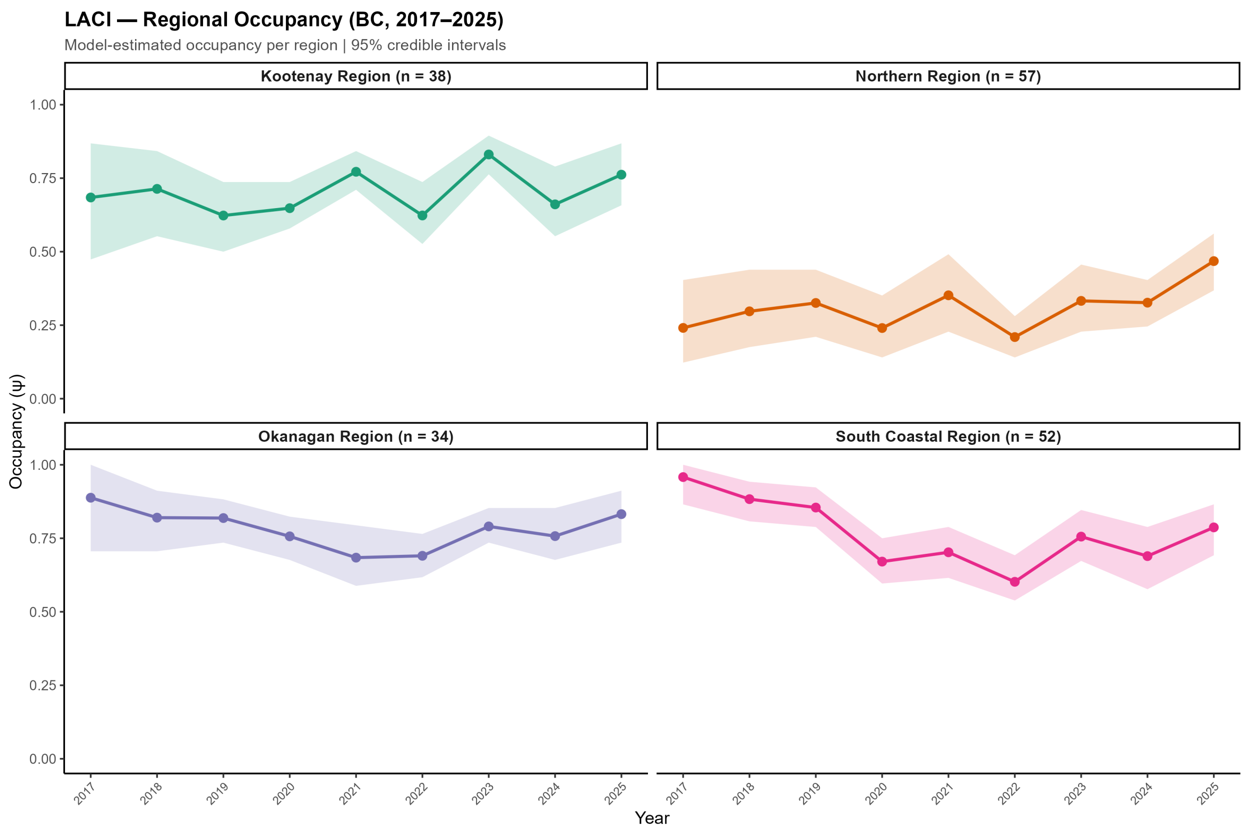

MYCA — Occupancy Trend (BC, 2017–2025)
λ = 1.19 (1.03, 1.38) | Mean p = 0.66
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

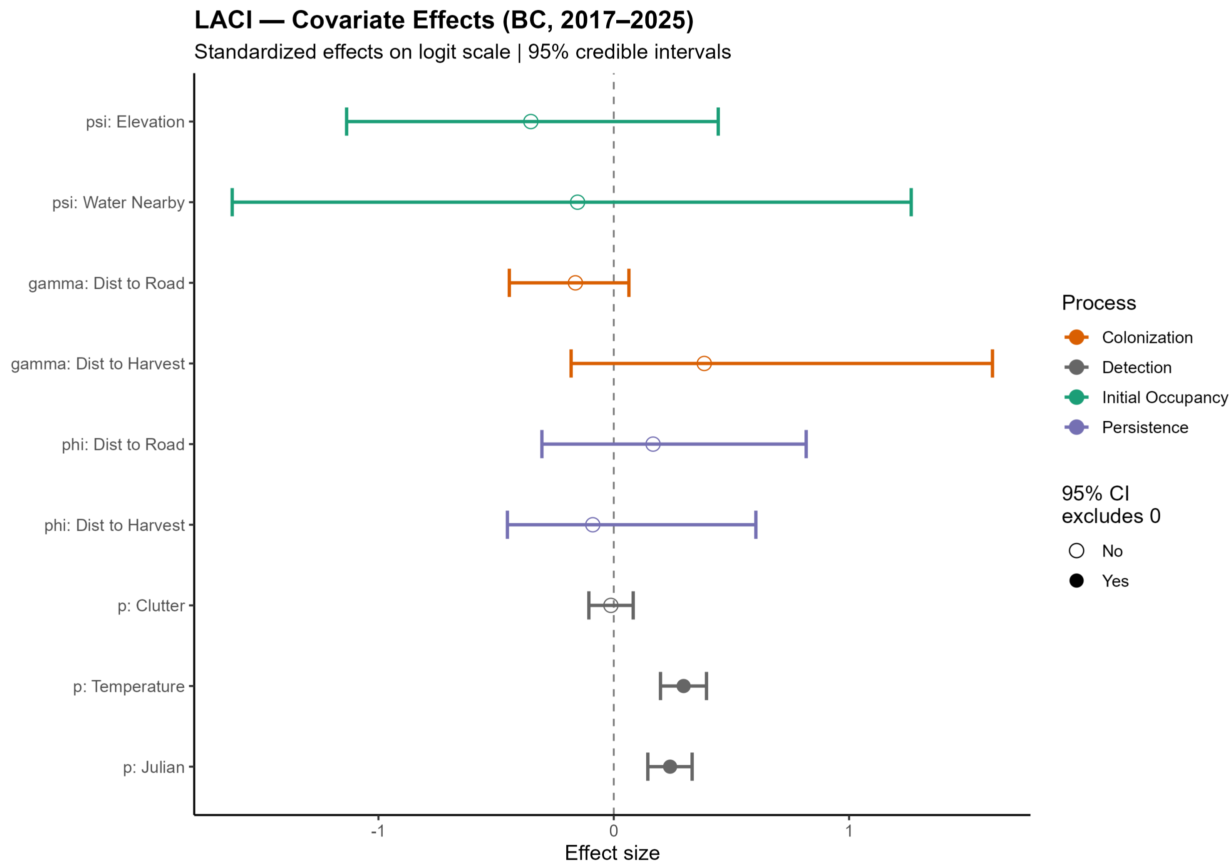

LANO — Regional Occupancy (BC, 2017–2025)
Model-estimated occupancy per region | 95% credible intervals
Kootenay Region (n = 38)
Northern Region (n = 57)
Okanagan Region (n = 34)
South Coastal Region (n = 52)
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year


LANO — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-2
-1
0
1
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI excludes 0
No
Yes

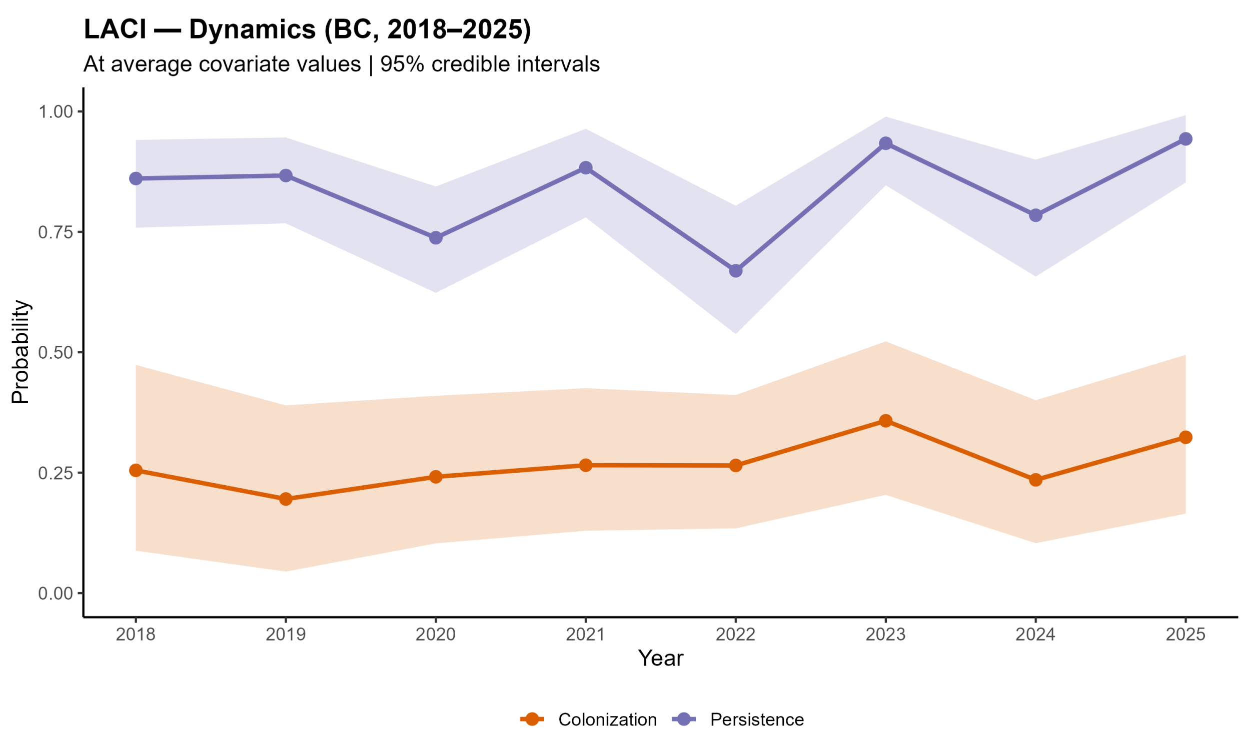

LANO — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence


LANO — Occupancy Trend (BC, 2017–2025)
λ = 1.07 (0.97, 1.20) | Mean p = 0.78
Occupancy (ψ)
Year

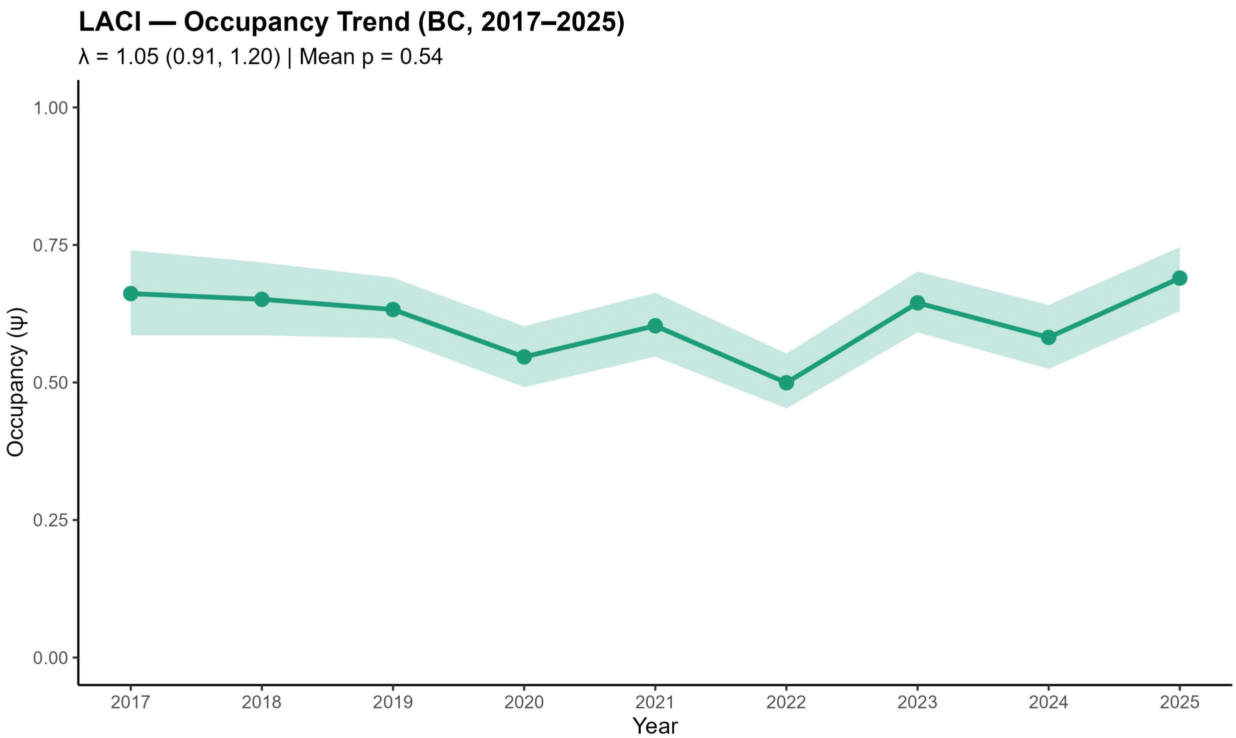

LACI — Regional Occupancy (BC, 2017–2025)
Model-estimated occupancy per region | 95% credible intervals
Kootenay Region (n = 38)
Northern Region (n = 57)
Okanagan Region (n = 34)
South Coastal Region (n = 52)
Occupancy (ψ)
Year

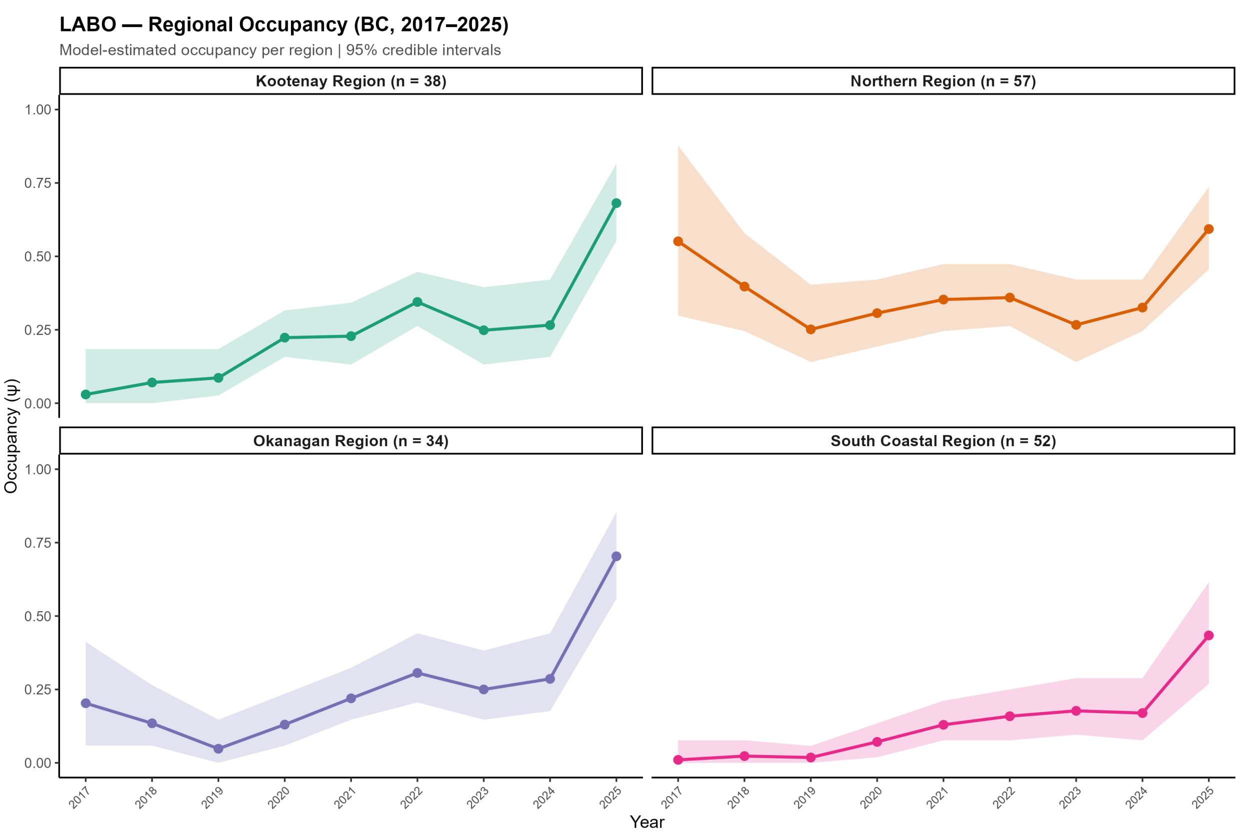


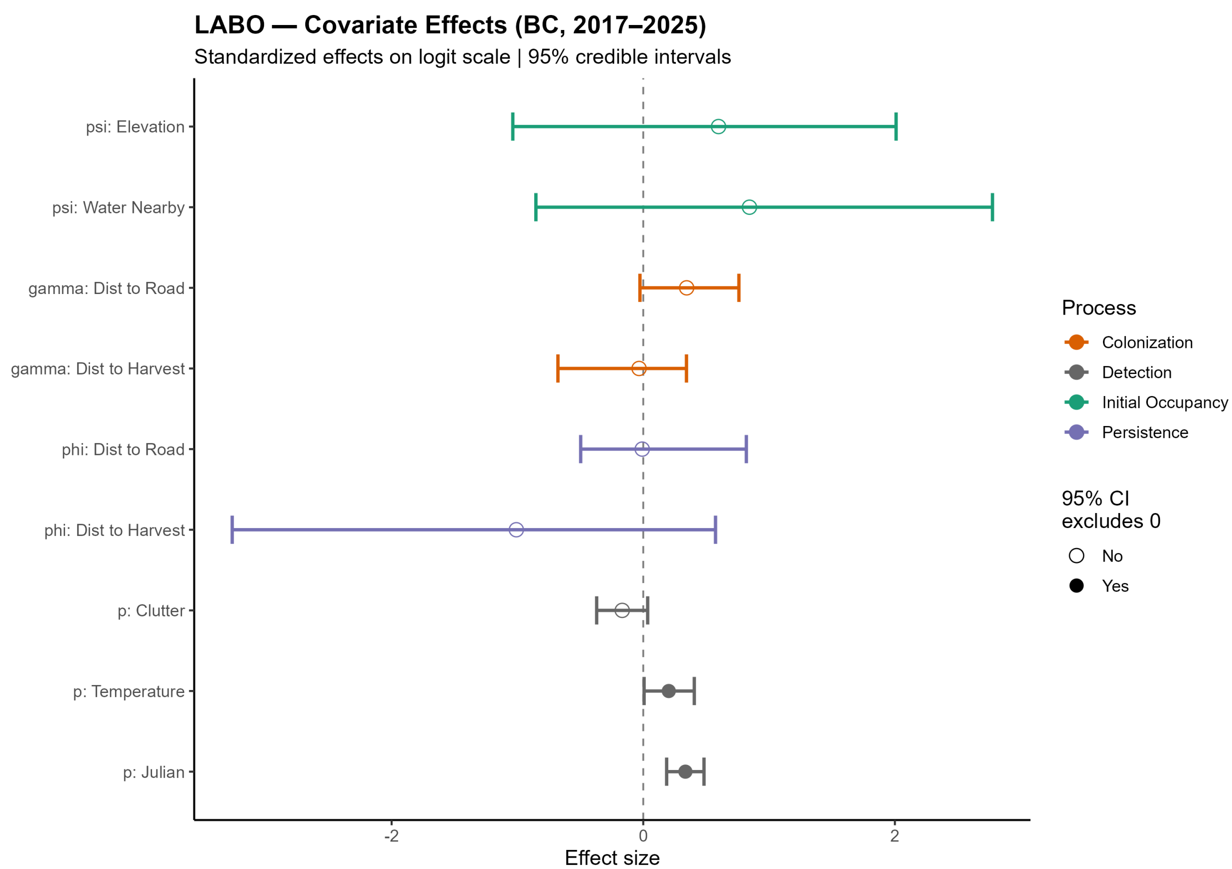
LACI — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-1
0
1
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI excludes 0
No
Yes



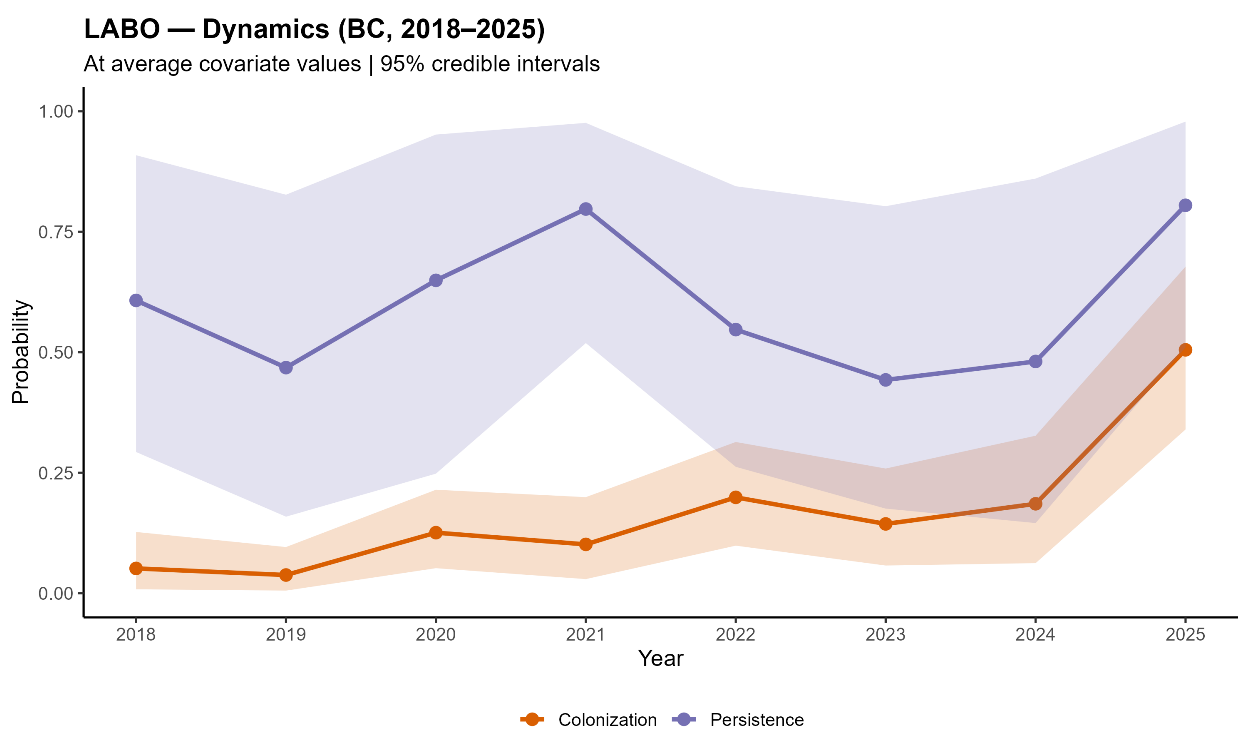
LACI — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence


LACI — Occupancy Trend (BC, 2017–2025)
λ = 1.05 (0.91, 1.20) | Mean p = 0.54
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

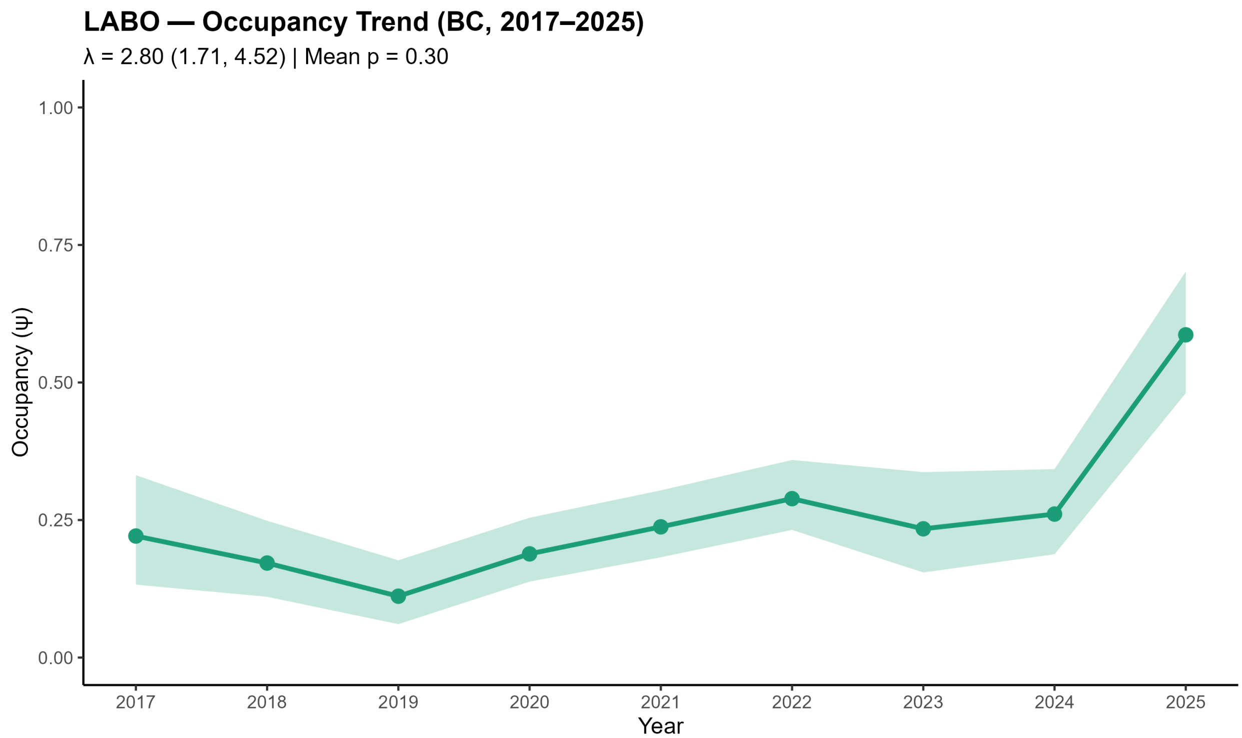

LABO — Regional Occupancy (BC, 2017–2025)
Model-estimated occupancy per region | 95% credible intervals
Kootenay Region (n = 38)
Northern Region (n = 57)
Okanagan Region (n = 34)
South Coastal Region (n = 52)
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

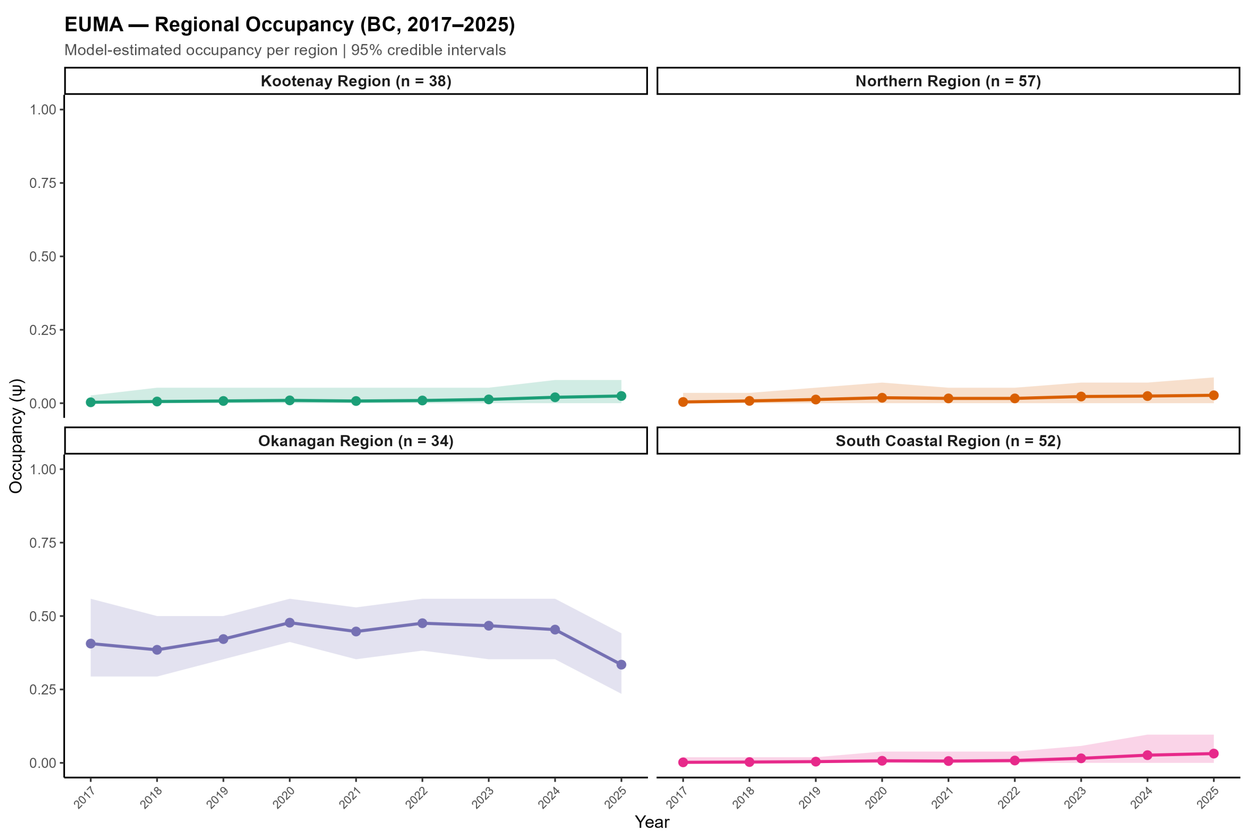


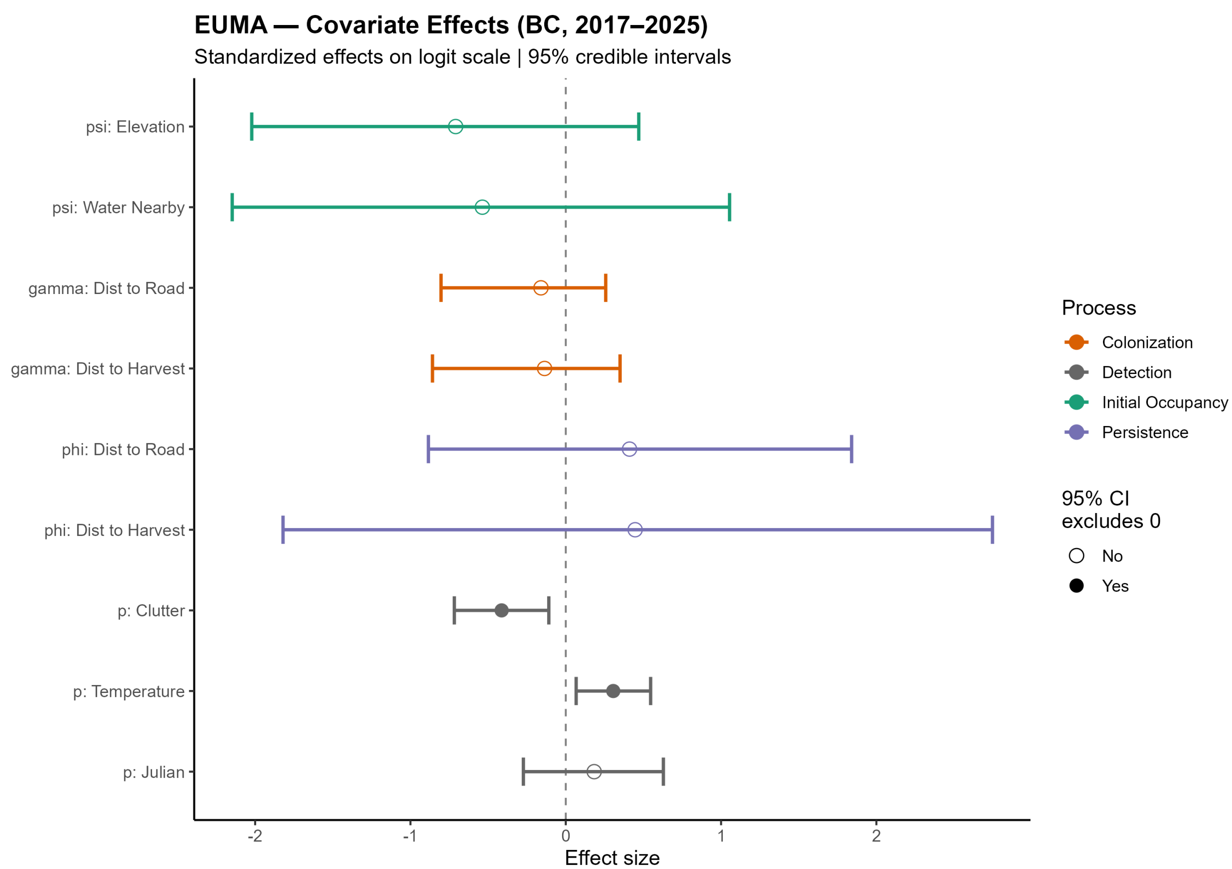
LABO — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-2
0
2
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI excludes 0
No
Yes



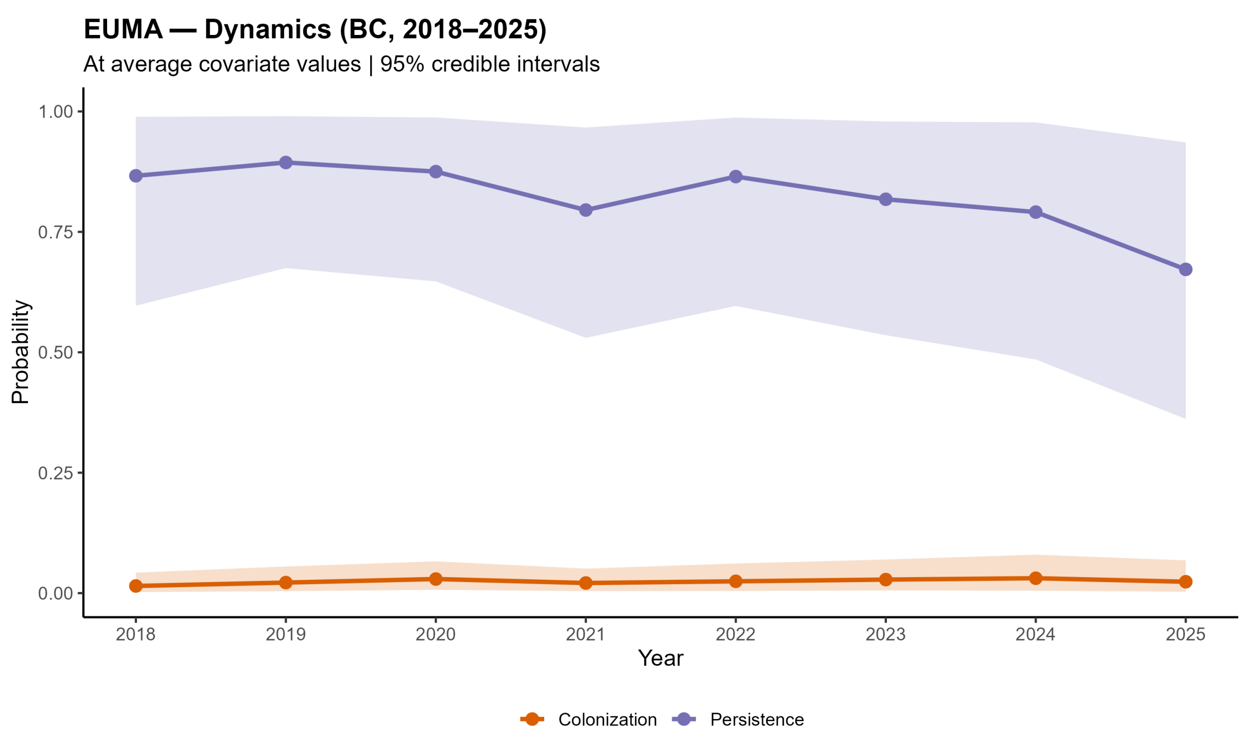
LABO — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence


LABO — Occupancy Trend (BC, 2017–2025)
λ = 2.80 (1.71, 4.52) | Mean p = 0.30
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

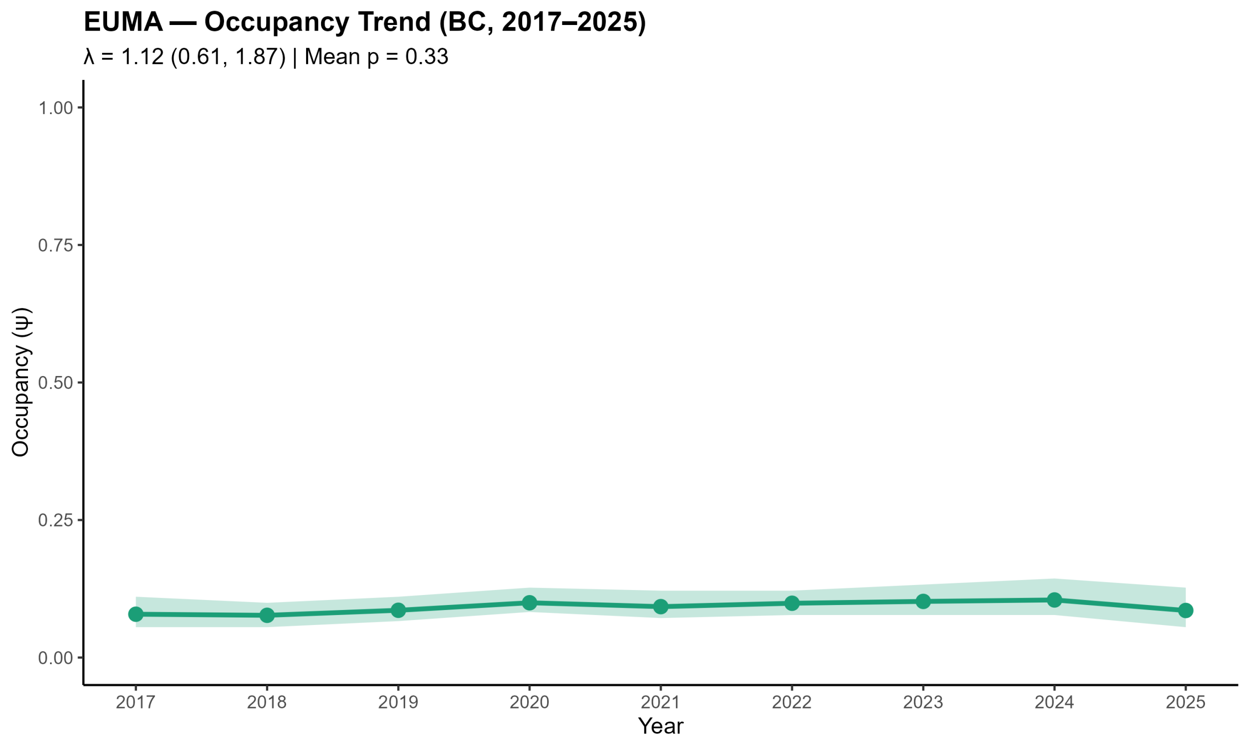

EUMA — Regional Occupancy (BC, 2017–2025)
Model-estimated occupancy per region | 95% credible intervals
Kootenay Region (n = 38)
Northern Region (n = 57)
Okanagan Region (n = 34)
South Coastal Region (n = 52)
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year



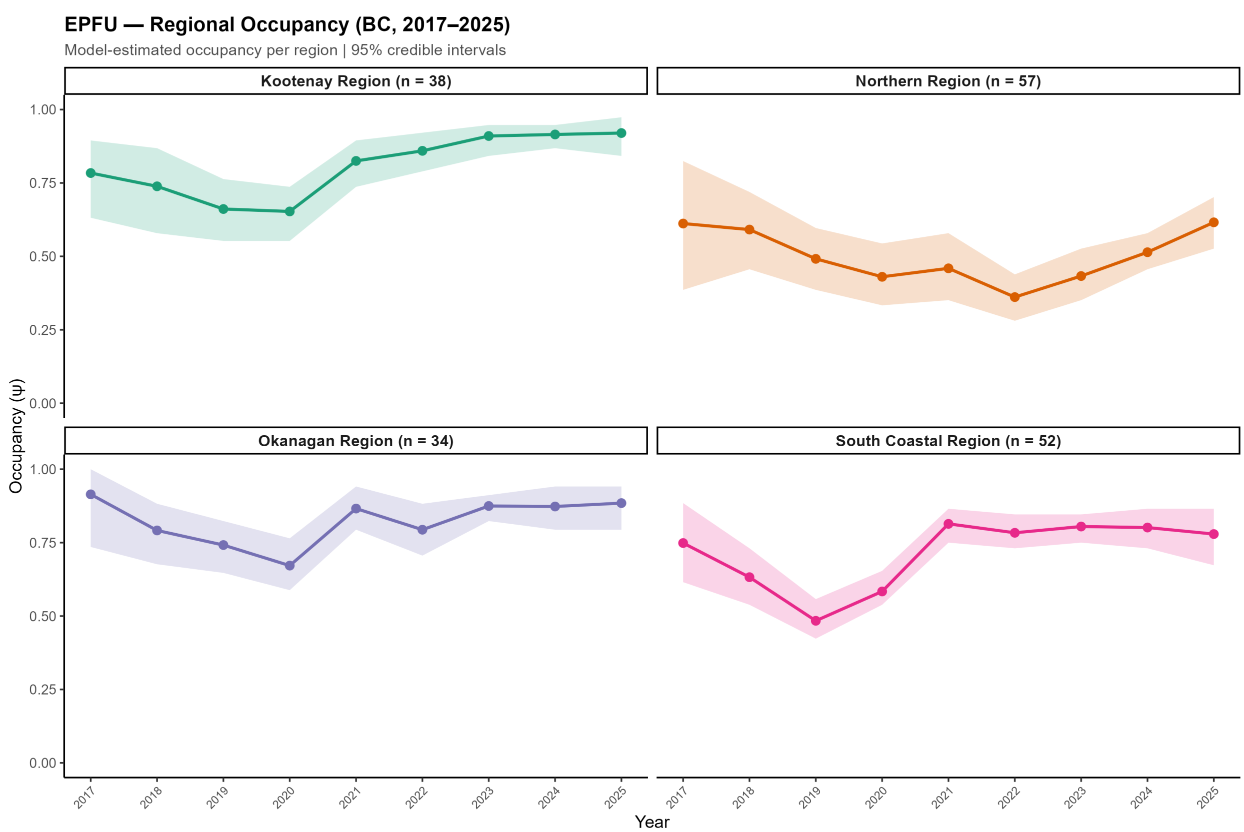
EUMA — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-2
-1
0
1
2
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI excludes 0
No
Yes



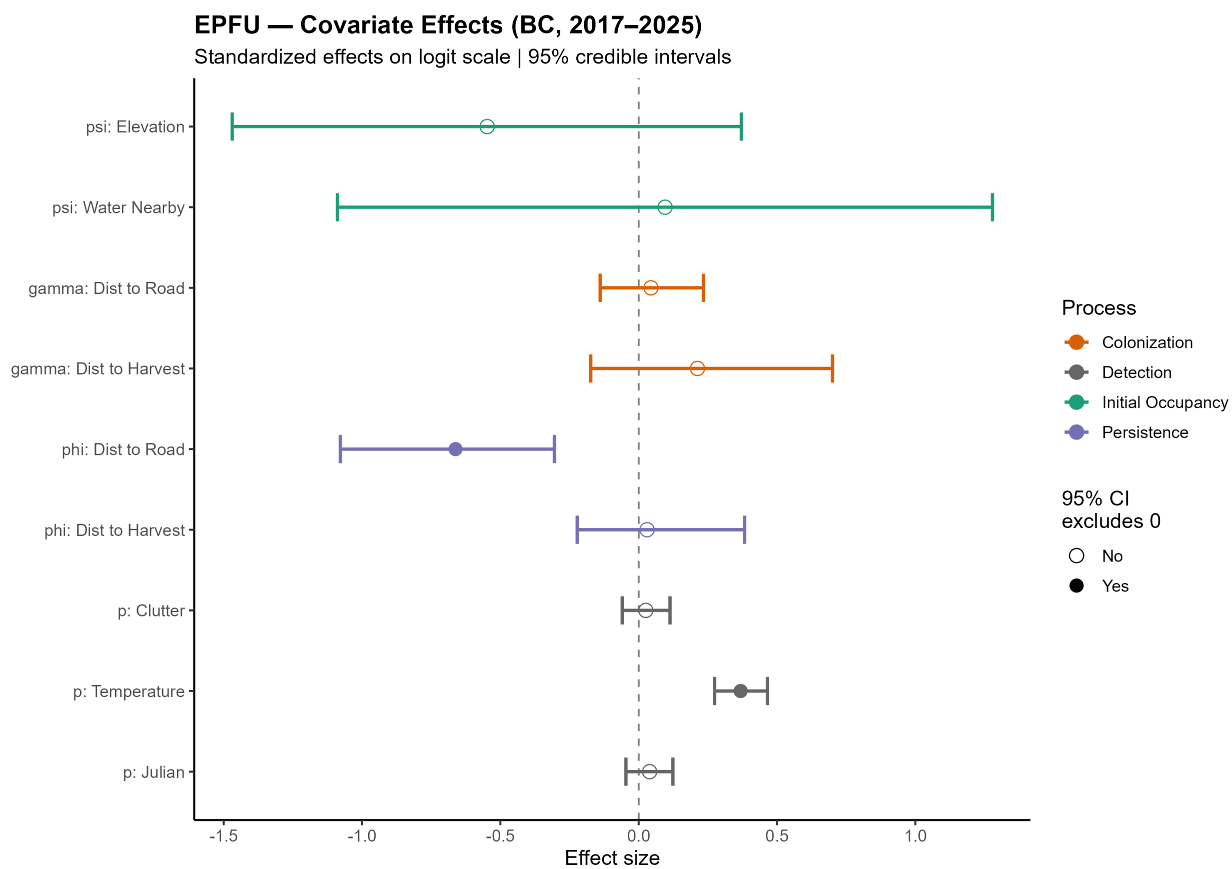
EUMA — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence

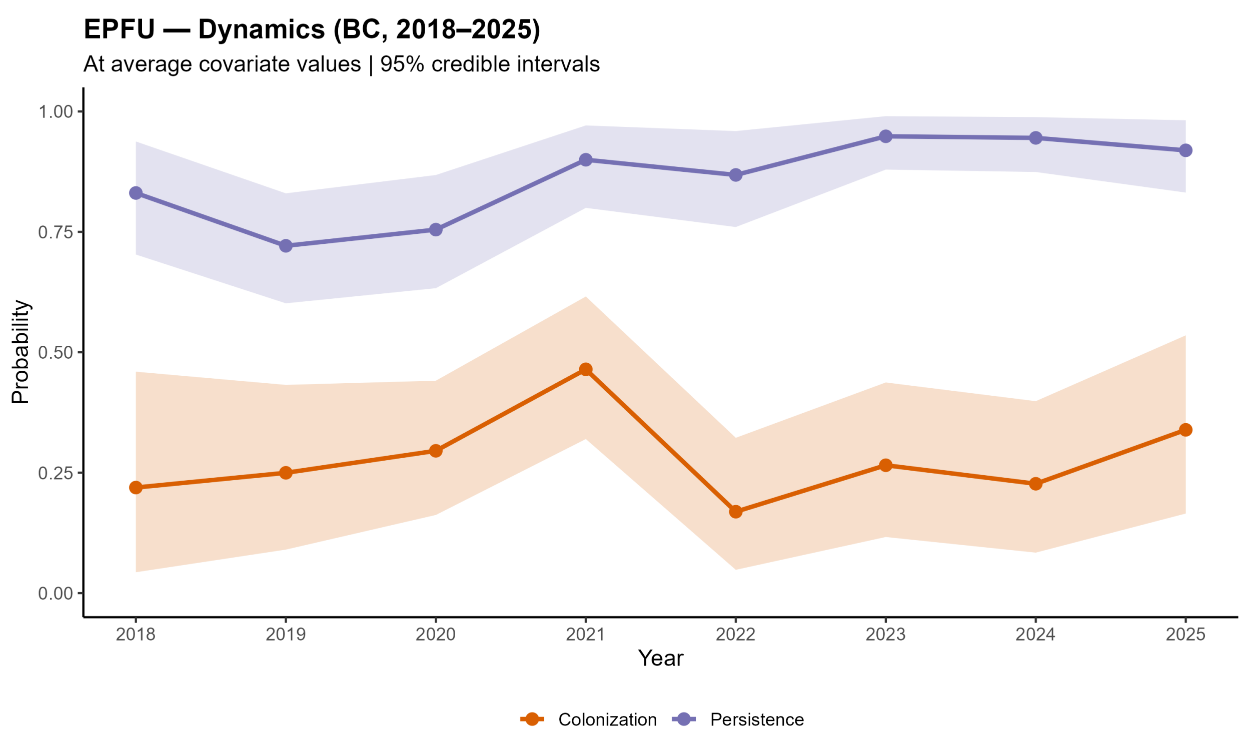


EUMA — Occupancy Trend (BC, 2017–2025)
λ = 1.12 (0.61, 1.87) | Mean p = 0.33
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year



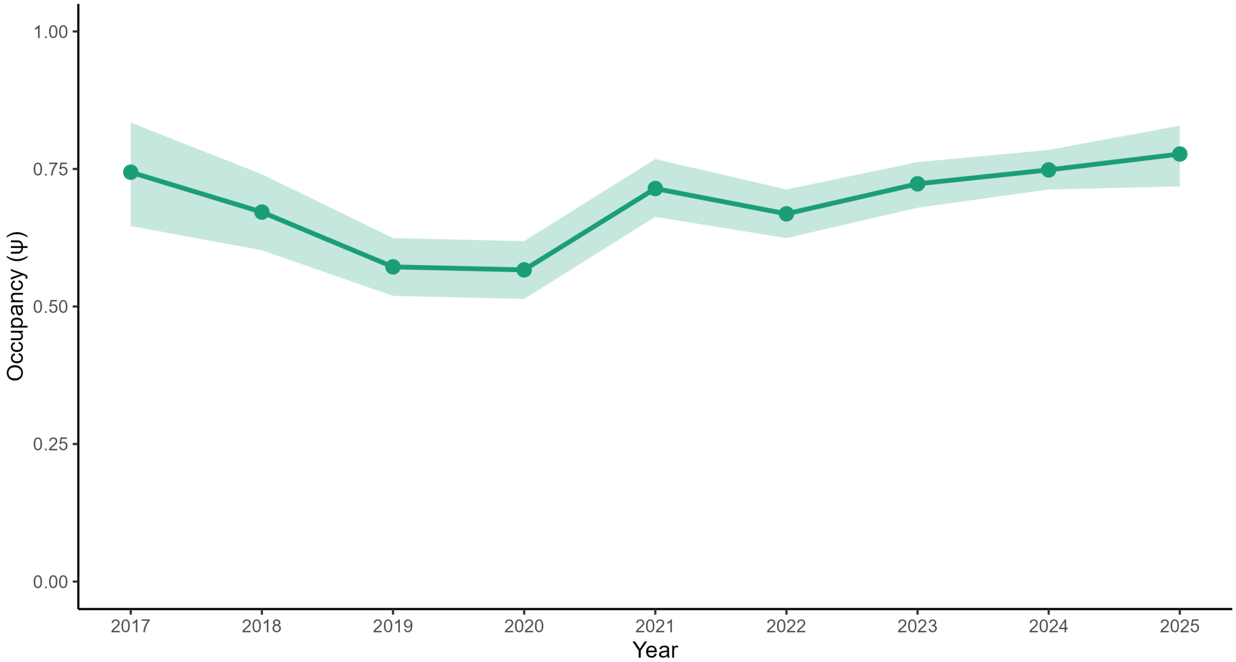
EPFU — Regional Occupancy (BC, 2017–2025)
Model-estimated occupancy per region | 95% credible intervals
Kootenay Region (n = 38)
Northern Region (n = 57)
Okanagan Region (n = 34)
South Coastal Region (n = 52)
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year


EPFU — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-1.5
-1.0
-0.5
0.0
0.5
1.0
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI excludes 0
No
Yes


EPFU — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
Probability
1.00
0.75
0.50
0.25
0.00
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence

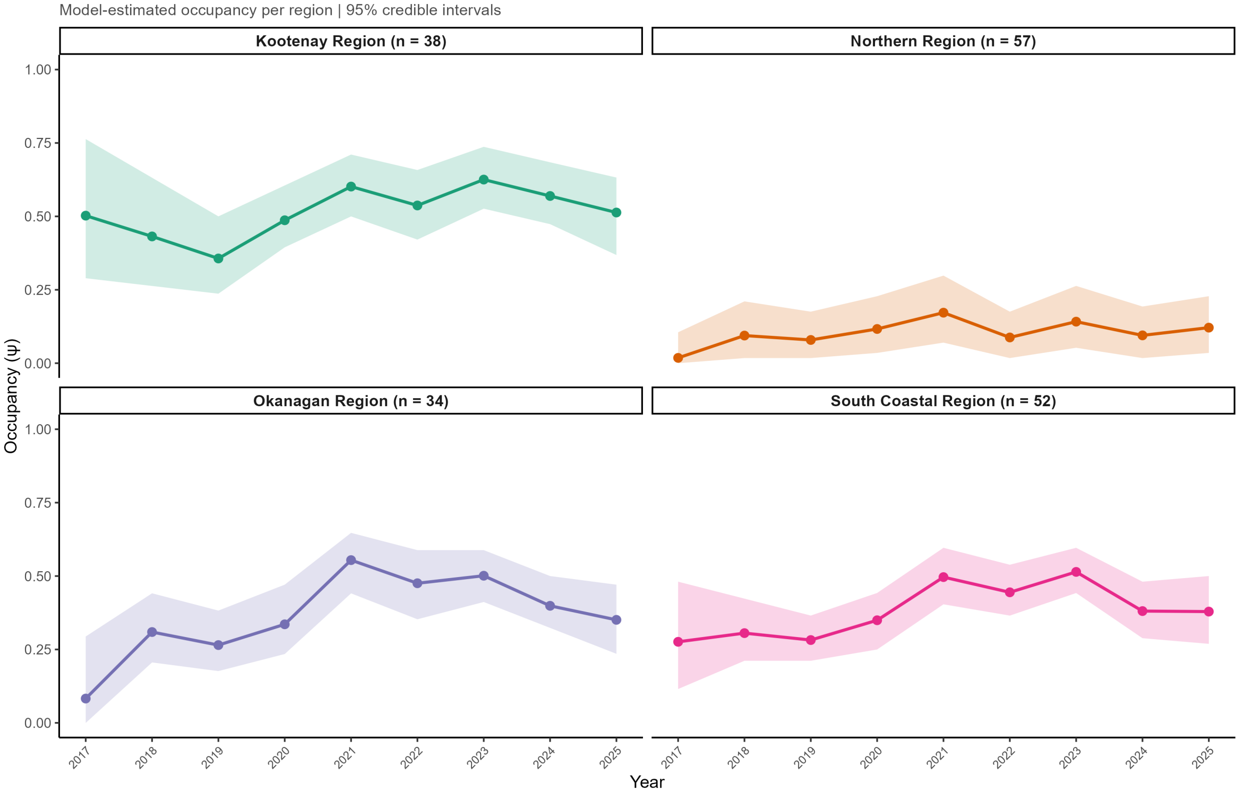
EPFU — Occupancy Trend (BC, 2017–2025)

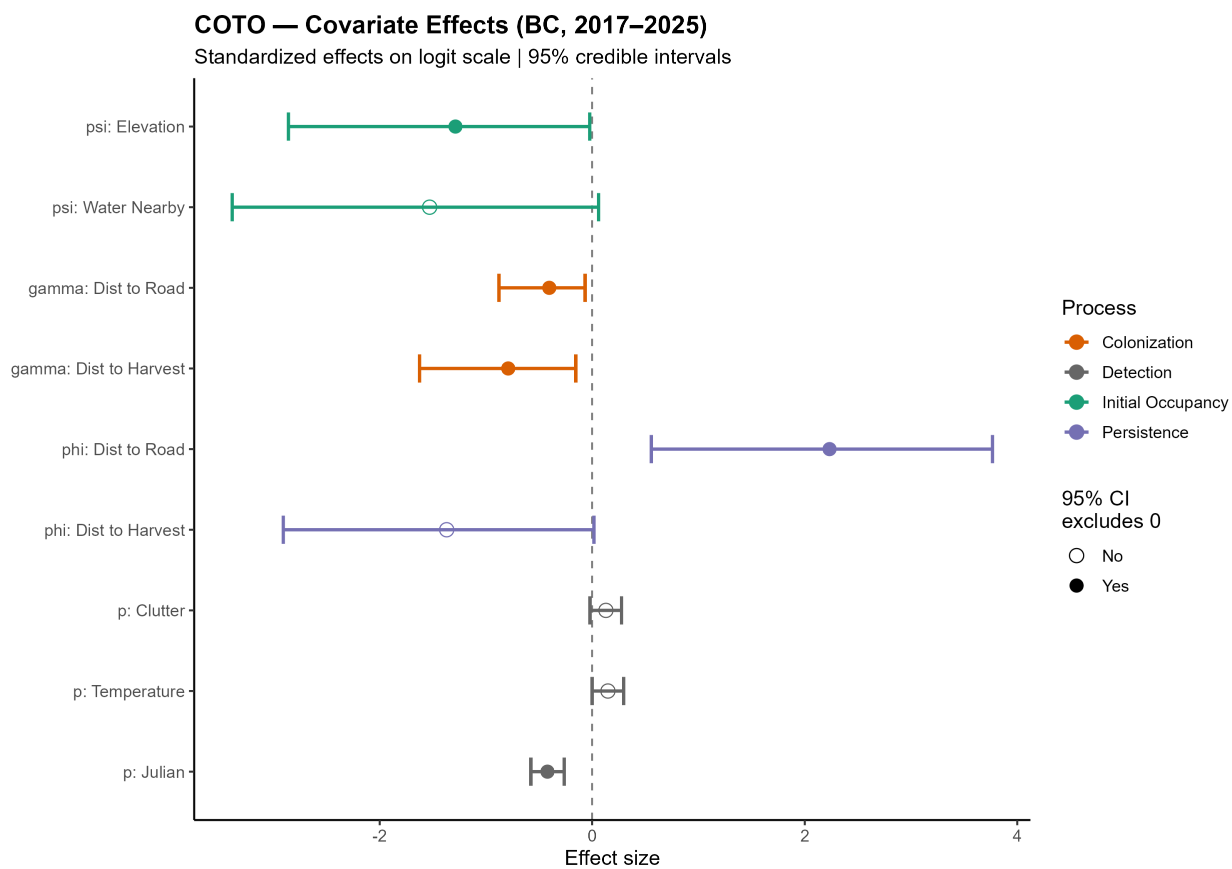
λ = 1.05 (0.91, 1.22) | Mean p = 0.60

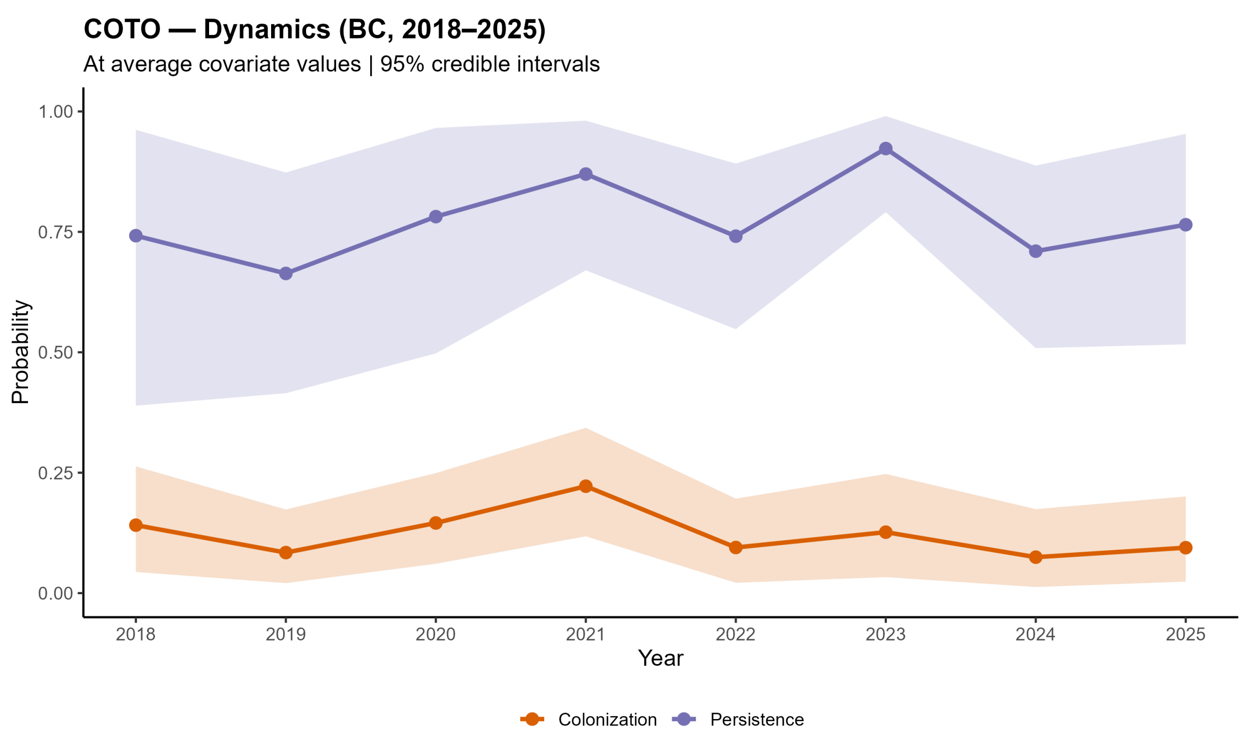


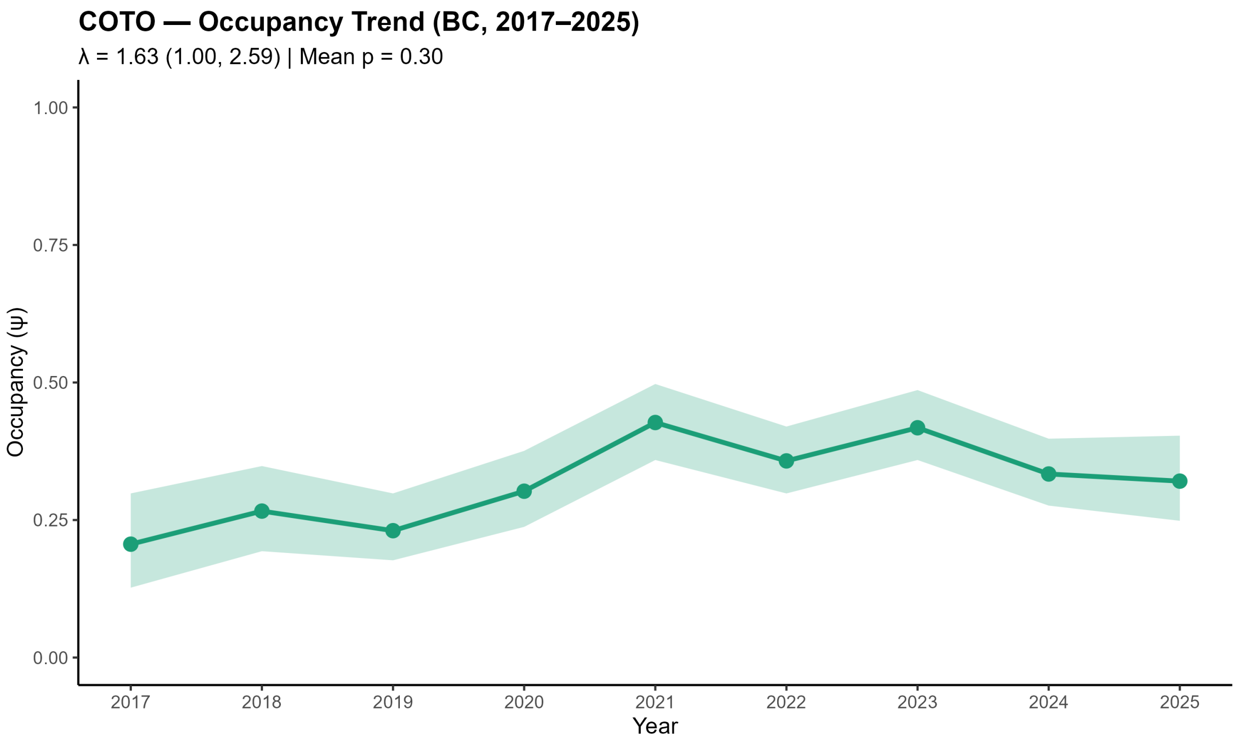
COTO — Regional Occupancy (BC, 2017–2025)

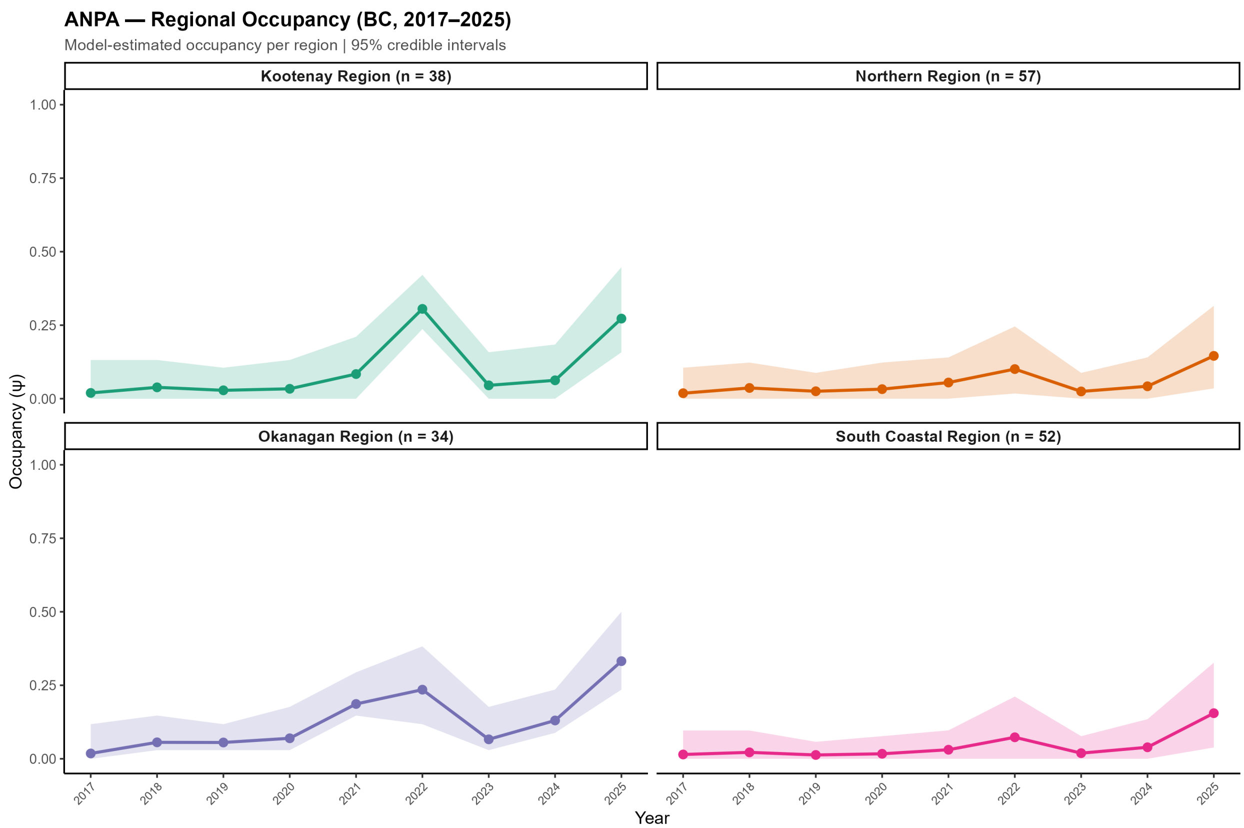
Model-estimated occupancy per region | 95% credible intervals




COTO — Covariate Effects (BC, 2017–2025)
Standardized effects on logit scale | 95% credible intervals
psi: Elevation
psi: Water Nearby
gamma: Dist to Road
gamma: Dist to Harvest
phi: Dist to Road
phi: Dist to Harvest
p: Clutter
p: Temperature
p: Julian
-2
0
2
4
Effect size
Process
Colonization
Detection
Initial Occupancy
Persistence
95% CI
excludes 0
No
Yes

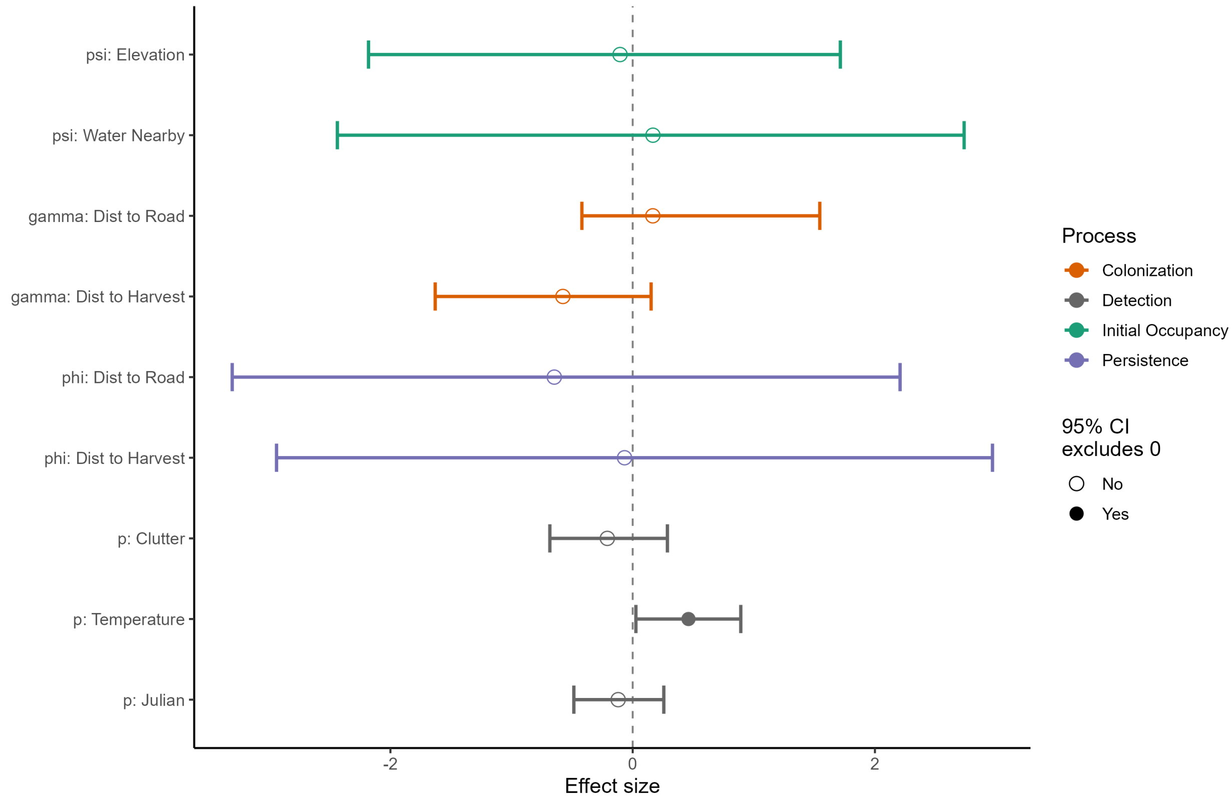

COTO — Dynamics (BC, 2018–2025)
At average covariate values | 95% credible intervals
1.00
0.75
0.50
0.25
0.00
Probability
2018
2019
2020
2021
2022
2023
2024
2025
Year
Colonization
Persistence


COTO — Occupancy Trend (BC, 2017–2025)
λ = 1.63 (1.00, 2.59) | Mean p = 0.30
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year


ANPA — Regional Occupancy (BC, 2017–2025)
Model-estimated occupancy per region | 95% credible intervals
Kootenay Region (n = 38)
Northern Region (n = 57)
Okanagan Region (n = 34)
South Coastal Region (n = 52)
Occupancy (ψ)
1.00
0.75
0.50
0.25
0.00
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

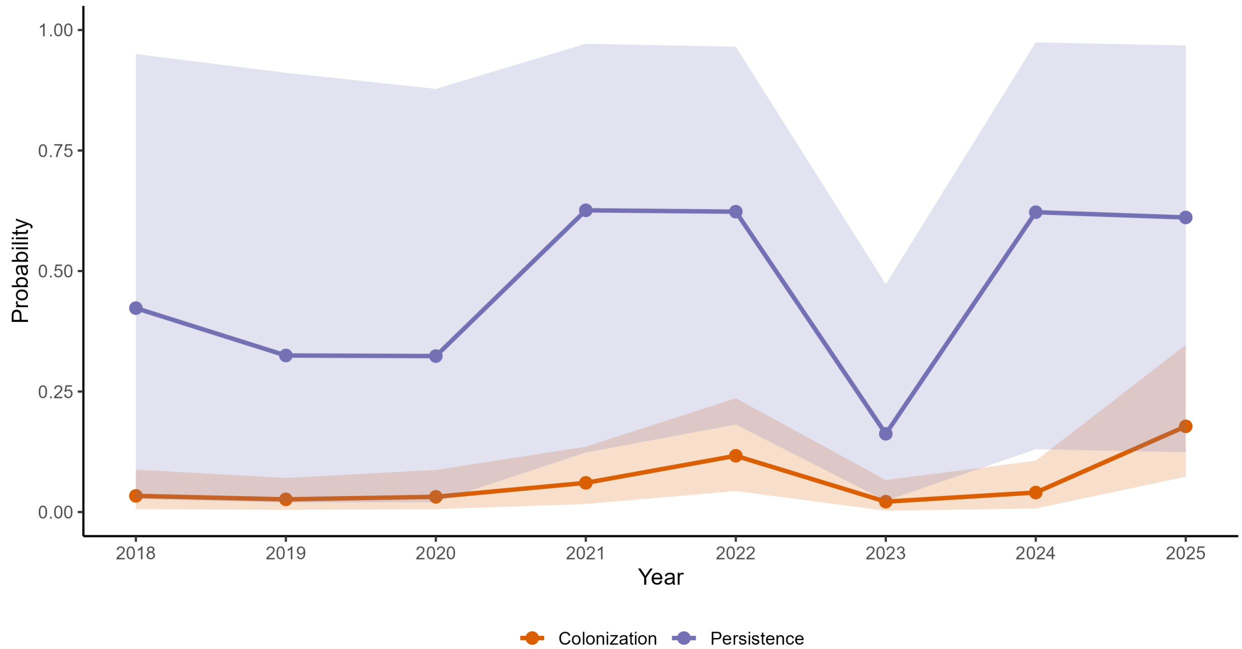
ANPA — Covariate Effects (BC, 2017–2025)

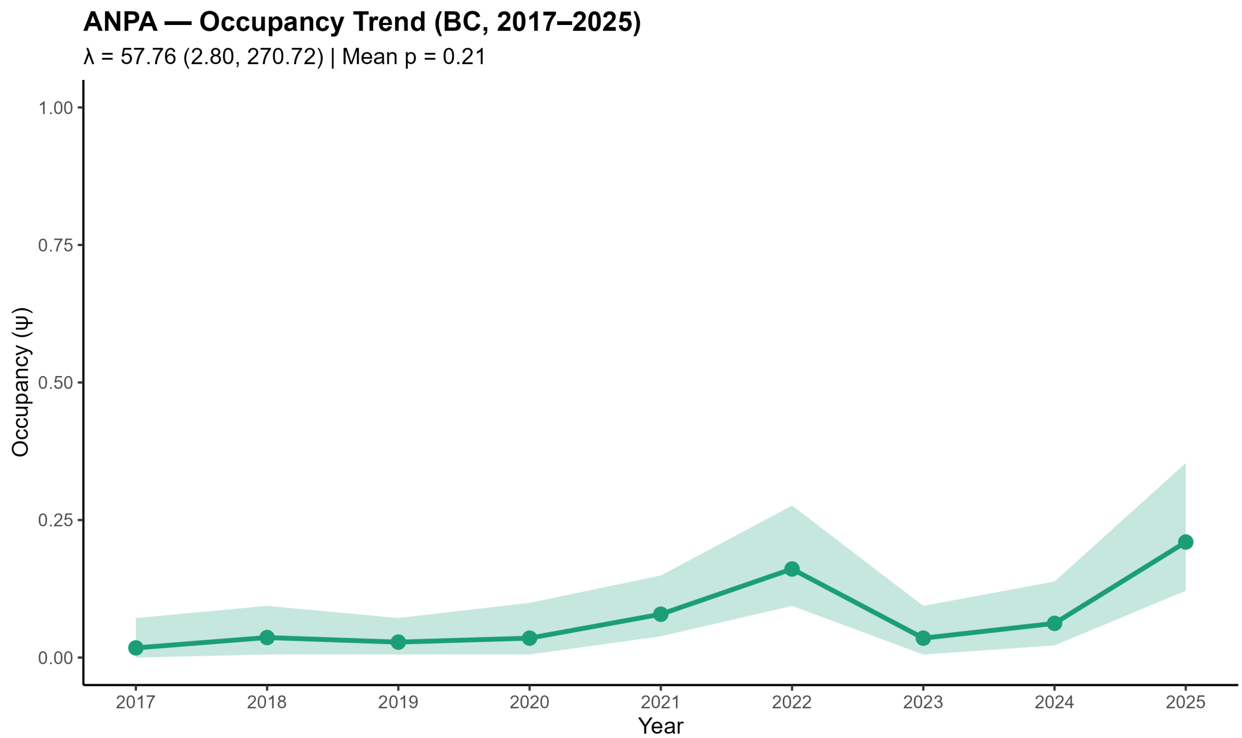
Standardized effects on logit scale | 95% credible intervals



ANPA — Dynamics (BC, 2018–2025)

At average covariate values | 95% credible intervals





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